Large Language Models as Nonlinear Dynamical Systems: The Three Clocks of Depth, Training, and Generation

Maor Farid *maor@getleo.ai*
Leo AI, Inc.
50 Milk Street, Boston, MA 02109, United States

Abstract

Language models are dynamical systems. A transformer is iterated over its layers, over the steps of its training and over the tokens it generates, and on each of these three clocks it shows behavior that nonlinear dynamics has studied for a century: fixed points, bifurcations, synchronization, chaos and noise-driven escape from potential wells. The two communities rarely meet, so this behavior is often rediscovered under new names. We build a bridge between them on three levels. To describe, we condense about 200 papers into the Three-Clock Dictionary, which maps fifteen concepts of nonlinear dynamics onto each clock and marks every correspondence as proved, measured or conjectured, and we show each concept in a mechanical system and in a language model side by side. To explain, we derive reduced-order models: the softmax is a Boltzmann distribution, gradient descent an Euler step on a nonlinear spring, a loss spike a ball thrown out of a stiff valley, attention on the circle a Kuramoto model, and a repetition loop a well that deepens while it is occupied. To predict, we test these models on open models with a laptop CPU. Escape from repetition loops in GPT-2 and Pythia follows the Arrhenius law with the logit gap as the barrier, and the barrier deepens and then saturates, so the temperature needed to leave a loop falls like $1/\ln N$ with the generation length N and rises with model size. Coherent text lives in a narrow window of temperatures, which a small reasoning model shares at its recommended temperature. Training makes each Pythia layer more expansive, and a small transformer trains at a multi-mode edge of stability. Controllers derived from the escape analysis remove loops but do not yet beat the best fixed temperature, and we state twelve open problems where nonlinear dynamics could solve problems of language models.

1 Introduction

1.1 Language models, dynamical systems and the potential between them

A transformer language model is a nonlinear map. It takes a context, a sequence of tokens, and returns a probability distribution over the next token, much as a forced pendulum maps its initial condition and excitation to a history of displacements.[1] This analogy has one important limit. In a mathematical model of a physical or mechanical system, the parameters, such as the length of a pendulum or the stiffness of a frame under earthquake loading, are fixed or prescribed from outside, whereas a learning model changes its own parameters, and it does so through a second dynamical system, the training algorithm. Training is a closed loop in the sense of control theory: it adjusts the parameters until the output reaches a goal, much as one could tune the length of a pendulum by feedback until it resonates with an external drive.

A language model is therefore an iterated map that shows dynamical behavior in high-dimensional spaces, and many of its phenomena can be described, explained and predicted, and some of its problems solved, with the terms, tools and methods of nonlinear dynamics. This is an opportunity, because nonlinear dynamics

[1] Use of AI tools: an AI assistant (Claude, Anthropic) helped search and verify the literature, draft and edit text, and write and run the experiment code, under the author's direction. All references were checked against arXiv or Crossref records and all reported numbers come from the released code. The author takes full responsibility for the content.

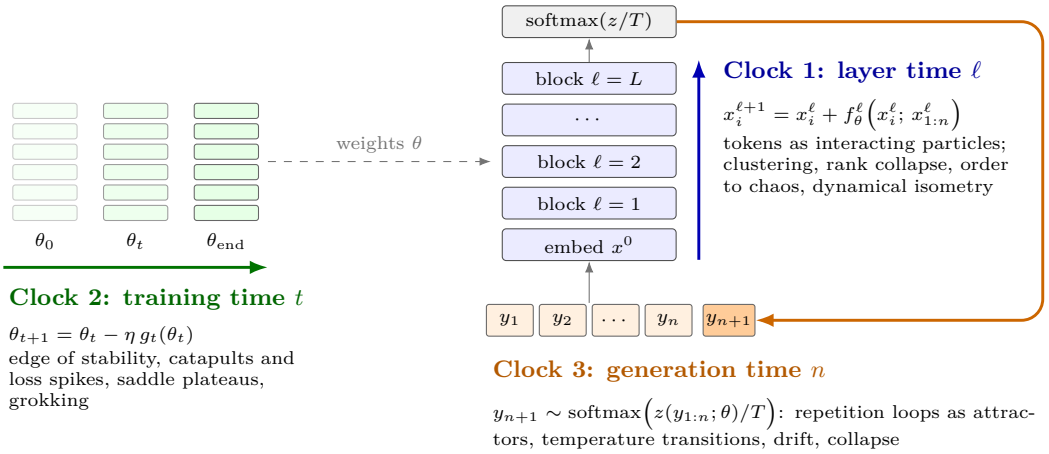


Figure 1: The three clocks of a transformer language model. **Clock 1** (blue) iterates the residual update over layers for a fixed input and fixed weights. **Clock 2** (green) iterates the optimizer over the weights; it hands its current weights to the model. **Clock 3** (orange) iterates sampling: each new token is appended to the context and the whole stack runs again. One tick of clock 3 contains a full run of clock 1, and clock 2 changes the map that both of them iterate. Typical behavior studied on each clock is listed next to it; Table 1 maps it onto dynamical-systems concepts.

has been developed by a large community for more than a century, and its body of knowledge can serve the young field of language-model research.

The two communities, however, are largely separate. Researchers spend their careers deepening their knowledge and experience within one of the fields, and each field has its own literature, its own jargon, its own catalogue of phenomena and its own conferences, so few synergies arise between them.

The cost of this separation is that well-understood dynamics is rediscovered under new names. Nonlinear dynamics has a mature theory for the behaviors that appear in language models, built over more than a century on pendulums, rotating machinery, ships, lasers and networks of coupled oscillators: stability and bifurcations, synchronization, chaos and Lyapunov exponents, and noise-driven escape from potential wells (Strogatz, 2018; Kramers, 1940; Hänggi et al., 1990; Acebrón et al., 2005). These tools transfer directly. Escape from a repetition loop at sampling temperature T is the escape of a particle from a potential well under noise of intensity T, with the gap between the logits of the repeated token and its best competitor in the role of the barrier. The Arrhenius form of the Kramers rate then predicts how the escape probability depends on temperature, and the measured growth of the barrier predicts how long a loop survives (Section 4.1 and Experiment 3). The exchange also runs the other way. Neural networks already forecast spatiotemporally chaotic systems from data (Pathak et al., 2018) and learn coordinates in which nonlinear dynamics becomes linear (Lusch et al., 2017), and a trained transformer is itself a high-dimensional, non-autonomous system that can be run millions of times, observed completely and differentiated exactly.

One goal of this paper is therefore to build a bridge between the two communities and to show where collaboration and joint research can pay off. For this reason we define the terms and methods of both fields that the argument needs, including some basic ones, so that members of each community can follow the other side. Appendix A collects these definitions in one place, and Appendix B lists the notation.

A bridge between two fields can carry three kinds of traffic. The first is description, in which a behavior seen in one field receives a model, a name and a set of measurable quantities from the other, so that a repetition loop becomes a periodic orbit and a step-to-step oscillation of the loss becomes a period-2 orbit of a map. The second is explanation, in which the tools of the older field account for why the behavior occurs and predict how it changes with a parameter, as Kramers' theory predicts how the rate of escape from a well changes with the noise. The third is solution, in which the explanation points to an intervention that fixes a problem the newer field had not solved. Most of this paper works at the first two levels.

1.2 A language model as three coupled dynamical systems

A language model is used as an iterated map in three modes:

1. the forward pass, which carries the representation of a token through the layers ℓ;

2. training, which updates the weights one optimizer step t at a time;

3. generation, which chains forward passes to generate text one token n at a time.

The forward pass iterates over the depth of the model. Each residual block adds an update to the hidden state of every token, so a stack of L blocks is a discrete-time dynamical system with the layer index as time (E, 2017; Haber & Ruthotto, 2017; Lu et al., 2017; Chen et al., 2018), and attention couples the tokens into a system of interacting particles (Lu et al., 2019; Sander et al., 2021b; Geshkovski et al., 2023b). Attention acts like a force field: it attracts and repels the representations of the tokens as they flow through the layers of the model.

Training iterates over optimizer steps. The state is the parameter vector θ, and gradient descent, $\theta_{t+1} = \theta_t - \eta\nabla\mathcal{L}(\theta_t)$, is a map on parameter space. It is the explicit Euler discretization of the gradient flow $\dot{\theta} = -\nabla\mathcal{L}(\theta)$, which describes an overdamped marble sliding down the loss landscape toward a minimum. Like any discretized flow, this map has dynamics of its own: at the step sizes used in practice it sits at the boundary of linear stability (Cohen et al., 2021), it produces loss spikes (Chowdhery et al., 2022; Wortsman et al., 2023), and it goes through abrupt transitions such as grokking (Power et al., 2022) and the formation of induction heads (Olsson et al., 2022).

Generation iterates over the generated tokens. An autoregressive model feeds its own samples back as input, so decoding is a stochastic dynamical system on the space of contexts, and its best known pathology, the repetition loop (Holtzman et al., 2019; Xu et al., 2022), has recently been analyzed as an attractor (Wang et al., 2025; Jones, 2026).

Each mode advances its own time index. We therefore call the modes the *three clocks* of a language model: clock 1 is the forward pass, with layer time ℓ; clock 2 is training, with training time t; and clock 3 is generation, with generation time n (Figure 1). The clocks are coupled, since the parameters produced by training define the map that the forward pass iterates, and every tick of generation calls a full forward pass. They also run at very different rates, since the weights do not move during generation and each weight update averages over many forward passes. Separated time scales of this kind are the standard setting of slow-fast dynamical systems (Strogatz, 2018), and they explain why three literatures could grow up around one object with few citations between them.

1.3 This work

In the current work we build this bridge in both directions. We write the three clocks in the language of nonlinear dynamics, record which correspondences are proved, measured or only conjectured, and use the tools of the field to explain behavior that the machine learning literature documents but does not yet explain.

At the first level, the paper shows the description directly: each concept is drawn in a mechanical system and in a language model in the same representation, such as the same time history, spectrum or escape plot. At the second level, reduced-order models explain each correspondence, in the way an engineer replaces a sloshing tank or a rolling ship by an equivalent oscillator. The softmax turns out to be a Boltzmann distribution, gradient descent an Euler step on a nonlinear spring, a loss spike a ball thrown out of a stiff valley, attention on the circle a Kuramoto model, and a repetition loop a well that deepens while it is occupied. At the third level we tested three controllers derived from the escape analysis on GPT-2. They remove exact loops and find the temperature window without a sweep, but none produces more coherent text than the best fixed temperature, because the trap is stored in the context while the controllers act on the noise (**OP 12**). Solutions at this level remain the main open task, and the open problems of Section 6 list the places where we expect them. Our contributions are the following.

1. **A unified framework** that writes the three clocks as coupled maps and organizes about 200 papers from machine learning theory, interpretability, optimization, statistical physics and applied mathematics around shared dynamical concepts, each shown in a physical system and a language model side by side.

2. **The Three-Clock Dictionary** (Table 1), which maps fifteen concepts from nonlinear dynamics onto each clock with the evidence level of every entry; each empty cell defines a research question.

3. **Reduced-order models and theory**, including an exact Arrhenius law for escape from repetition loops and conditions under which a loop locks.

4. **Experiments on open models** that run on a laptop CPU, one per clock, and **twelve open problems**, each stated with a first test.

Organization. Section 2 defines the three clocks as maps and reads each one as a dynamical system. Section 3 opens with the Three-Clock Dictionary and then goes through its main concepts, from stability and basins to synchronization, noise-driven escape and chaos, each shown in a mechanical system and in a language model side by side. Section 4 explains why the pairs behave alike by reducing each behavior of the language model to a physical model with the same governing term, and proves when a repetition loop locks. Section 5 tests three of these correspondences in new experiments, one per clock: whether training changes how the forward pass stretches perturbations, how gradient descent on a transformer moves from a period-2 orbit to irregular oscillation as the step size grows, and whether escape from repetition loops follows the Arrhenius law with a barrier that deepens over time. Section 6 states twelve open problems, and the appendices give the glossary, the notation, the experimental details and further work on each clock.

Related work and scope. Several papers develop one of these views in depth, and we rely on them throughout: attention as an interacting particle system over depth (Geshkovski et al., 2023b; Castin et al., 2025), gradient descent at the edge of stability (Cohen et al., 2021), and stochastic and chaotic accounts of generation (Krishnamurthy et al., 2026; Jones, 2026). To our knowledge, no earlier work treats the three clocks side by side in one vocabulary, with an explicit record of which correspondences are proved, which are measured and which are conjectures. We use the language of dynamical systems in its technical sense wherever a result supports it, and we flag analogies as analogies; for example, we call a loss spike a bifurcation only when a parameter can be shown to cross a stability threshold.

Practical takeaways.

- *Generation.* In GPT-2 and Pythia, coherent text appears only in a narrow band of temperatures between repetition and incoherence, and for some prompts in none. A loop is not locked by a fixed temperature: the temperature needed to leave it within a given budget falls like $1/\ln N$ with the length N of the generation, and larger models need a higher temperature to leave the same kind of loop (Section 5.3). In Qwen3-0.6B on GSM8K, accuracy within a fixed budget peaks at the vendor's $T = 0.6$, and below it the model rethinks the same step until the budget runs out.
- *Anti-repetition.* Raising the temperature when a loop starts, rolling the text back to the start of the loop, or steering the temperature by feedback all remove exact loops, but none beats the best fixed temperature on coherent text, because the attractor is stored in the context, not in the sampler (**OP 12**).
- *Training.* At the edge of stability several Hessian eigenvalues sit 5% to 14% above $2/\eta$ while the curvature along the actual step equals $2/\eta$, so $\eta\lambda_{\max} > 2$ alone does not signal divergence. A power iteration can underestimate $\lambda_{\max}$ there, because the top eigenvalues form a cluster that reorders at every step (Section 5.2).
- *Depth.* Training makes each layer stretch perturbations more, while the stretched directions of consecutive layers become less aligned (Section 5.1).
- *Tools.* Per-layer Lyapunov exponents, learning-rate bifurcation diagrams and exact loop-escape probabilities from a single greedy pass all run on a laptop CPU, and the code is released.

2 The three clocks

Section 1 described three ways in which a language model is iterated. This section writes them as maps, so that each can be read as a dynamical system with a state, a time and a rule of evolution. It then explains how the three maps are nested and why they run at different speeds, and it ends with the places where the dynamical reading needs care. The rest of the paper uses these definitions throughout.

Consider a decoder-only transformer (Vaswani et al., 2017) with L residual blocks, model width d and vocabulary $\mathcal{V}$. For a context of n tokens, write $x^\ell = (x_1^\ell, \dots, x_n^\ell) \in \mathbb{R}^{n\times d}$ for the residual stream after block ℓ, and θ for all parameters. The model defines three maps. Appendix B lists all symbols used in the paper.

Definition 1 (Clock 1: layer time)**.** For fixed θ and fixed input tokens,

$$x^{\ell+1} = x^\ell + f_\theta^\ell(x^\ell), \qquad \ell = 0, \dots, L-1, \tag{1}$$

where f_θ^ℓ contains attention, which couples token i to tokens $j \le i$, and a token-wise MLP. The logits are $z = W_U\,\mathrm{LN}(x_n^L)$.

Clock 1 read as a dynamical system. Equation equation 1 is a forward Euler step, with unit step size, of an ordinary differential equation $\dot{x} = f(x, \ell)$ for n particles in $\mathbb{R}^d$. The tokens are the particles, their hidden states are the positions, and attention is the force through which they interact. The system is non-autonomous, since each block has its own weights, and it runs for a fixed and small number of steps. This observation was made early for residual networks (He et al., 2015) by E (2017), Haber & Ruthotto (2017) and Lu et al. (2017), and it led to continuous-depth models (Chen et al., 2018), to architectures designed from the stability theory of ODEs and PDEs (Haber & Ruthotto, 2017; Ruthotto & Haber, 2018; Chang et al., 2019), to invertible second-order variants (Sander et al., 2021a), and to a view of training as optimal control (E et al., 2018). Whether a trained residual network actually converges to an ODE as depth grows depends on the initialization and on how the residual branch is scaled (Sander et al., 2022). In transformers, several stabilization methods amount to a choice of this step size, either by scaling the residual branch at initialization (Bachlechner et al., 2020; Wang et al., 2022) or by moving the layer normalization inside the branch (Xiong et al., 2020; Liu et al., 2020).

Definition 2 (Clock 2: training time)**.** For a loss $\mathcal{L}$ and a stream of minibatches $\mathcal{B}_t$,

$$\theta_{t+1} = \theta_t - \eta_t\, P_t\, \nabla_\theta \mathcal{L}(\theta_t; \mathcal{B}_t), \qquad t = 0, 1, \dots \tag{2}$$

where P_t is a preconditioner (identity for gradient descent, a diagonal matrix built from gradient moments for Adam) and $\mathcal{L}(\cdot; \mathcal{B}_t)$ is the loss on minibatch $\mathcal{B}_t$. With full batches the map is deterministic; with minibatches it is a random dynamical system.

Clock 2 read as a dynamical system. With P_t the identity, equation 2 is the explicit Euler scheme for the gradient flow $\dot{\theta} = -\nabla\mathcal{L}(\theta)$, the motion of an overdamped particle in the potential $\mathcal{L}$, with the learning rate η as time step. The state is the parameter vector, with millions to billions of entries. The data stream is the sequence of minibatches presented to the optimizer, one per step. Because each minibatch is drawn at random, the map is driven by noise, and because the data, the schedule η_t and the preconditioner change over training, it is non-autonomous.

Definition 3 (Clock 3: generation time)**.** For fixed θ, a sampling temperature $T > 0$ and a context $s_n = (y_1, \dots, y_n)$,

$$y_{n+1} \sim \pi_T(\cdot \mid s_n) = \mathrm{softmax}\big(z(s_n; \theta)/T\big), \qquad s_{n+1} = (s_n, y_{n+1}). \tag{3}$$

This is a Markov chain on the growing space of contexts. Truncation rules such as top-k or nucleus sampling change π_T but not the structure.

Clock 3 read as a dynamical system. Each step chooses one of $|\mathcal{V}|$ discrete states, and the chosen token becomes part of the input of the next step, so the output is fed back as input. Definition 3 makes generation a Markov chain whose state is the whole context. The state space grows with n, so the usual

results on stationary distributions do not apply directly, but the questions of dynamical systems still make sense: which sets of contexts trap the chain, how long the chain stays in them, and how these answers depend on the temperature T and on truncation. Krishnamurthy et al. (2026) formulate language models as high-dimensional nonlinear autoregressive models and analyze networks of interacting models with mean-field equations. A deterministic version of clock 3, obtained by feeding back the expected embedding of the next-token distribution instead of a sample, is a continuous map that can be analyzed with standard tools (Jones, 2026). Temperature divides the logits by T, so π_T in equation 3 is a Gibbs distribution with energy $-z$ at temperature T. Greedy decoding is the limit $T \to 0$, and T sets how often the chain leaves the most likely path, so it serves as both the noise intensity and the natural bifurcation parameter of clock 3. Its effect on task accuracy (Renze & Guven, 2024) and on novelty and coherence (Peeperkorn et al., 2024) has been measured directly.

Nested clocks and separated time scales. The three maps are nested. Evaluating $z(s_n; \theta)$ in equation 3 means running equation 1 to the end, and computing the gradient in equation 2 means running equation 1 on teacher-forced contexts, taken from the training text and not from the model's own samples. The weights produced by equation 2 define the maps in equation 1 and equation 3. The clocks also run at very different speeds. Generation never changes θ, and one training step changes it by an amount proportional to the small learning rate η, so the forward map changes only by $O(\eta)$ per step, an amount bounded by a constant times η. Counted in optimizer steps t, the weights therefore evolve on the slow time ηt, while each forward pass and each generation runs to completion at frozen weights. In the notation of the method of multiple scales, with $\varepsilon = \eta$, the fast time is $\tau_0 = t$ and the weights depend on the slow time $\tau_1 = \varepsilon t$, and Berthier (2026) carry this expansion to three time scales, $\tau_2 = \varepsilon^2 t$ included, for gradient descent at the edge of stability. In the language of slow-fast systems, clock 2 is the slow variable and clocks 1 and 3 are fast dynamics evaluated at a frozen slow state (Strogatz, 2018).

Two well-studied effects break this separation. With in-context learning the fast clock imitates the slow one: for in-context regression, a linear self-attention layer can implement one step of gradient descent on examples given in the prompt, so a stack of layers runs an optimizer whose iterations are the layers (von Oswald et al., 2022; Akyürek et al., 2022; Garg et al., 2022). Trained linear transformers learn preconditioned versions of this algorithm (Ahn et al., 2023), and the gradient-flow dynamics that produce them can be solved in closed form (Zhang et al., 2023b). In self-training the fast clock drives the slow one: text produced by clock 3 is used as training data for clock 2 (Shumailov et al., 2023; Bertrand et al., 2023), which we discuss in Appendix E.

Where the analogies are weak. First, clock 1 runs for a fixed and small number of steps (between 6 and about 100 layers), so asymptotic notions such as attractors and Lyapunov exponents must be replaced by finite-time versions. Second, clock 2 is driven by a data stream whose statistics are not stationary in general, and by learning-rate schedules, which makes it a non-autonomous system. Third, the state of clock 3 is the whole context, so its dimension grows with time and the chain never returns to an earlier state. Statements about loops and attractors on clock 3 are therefore about the projection of the state onto recent tokens. Each of these differences limits which theorems can be imported from the classical theory without change.

3 Shared behavior: a dictionary between the clocks and nonlinear dynamics

With the clocks written as maps, the next question is which behaviors of nonlinear systems they show. Table 1 answers it for fifteen concepts at once, and it serves as the map of this section. Each row is a concept from nonlinear dynamics and each column is a clock. A cell names the observed behavior that the concept describes on that clock and marks its evidence level: P when the correspondence is proved for a model of the system, M when it has been measured in trained networks, and C when it is a conjecture. A question mark means that we found no work that tests the concept on that clock.

Reading the table. Fixed points, stability and Lyapunov exponents have been used on all three clocks, although with different meanings: on clock 1 they describe a single finite trajectory, on clock 2 the long-run behavior of an optimizer, and on clock 3 a random process. Phase transitions and escape appear on clocks 2 and 3 but not on clock 1, where the number of steps is fixed and small, so there is no long-time limit in

Concept	Clock 1: layer time ℓ	Clock 2: training time t	Clock 3: generation time n
Fixed point	Tokens collapse to a cluster; rank collapse P (Geshkovski et al., 2023b; Dong et al., 2021). Massive activations nearly input independent M (Sun et al., 2024)	Convergence to a minimum; neural collapse at the end of training M (Papyan et al., 2020)	Period-one loops (one token repeated) as the simplest case of degenerate repetition M (Holtzman et al., 2019)
Linear stability, Jacobian spectrum	Dynamical isometry; layer Jacobians of trained models P/M (Pennington et al., 2017; Fernando & Guitchounts, 2026)	Sharpness vs. $2/\eta$ M (Cohen et al., 2021)	Derivative of the feedback map factorizes through $1/T$ M (Jones, 2026)
Finite-time Lyapunov exponent	Ridges in input space separate classes M (Storm et al., 2023); per-layer profile over training M (Sec. 5.1)	Sensitivity of runs to minibatch noise early in training M (Frankle et al., 2019)	Divergence of nearby generations M (Jaca et al., 2026; Jones, 2026)
Edge of chaos, criticality	Order-chaos transition at initialization sets trainable depth P (Poole et al., 2016; Schoenholz et al., 2016)	Edge of stability M (Cohen et al., 2021)	Critical temperature with power-law text statistics M (Nakaishi et al., 2024)
Limit cycle (periodic orbit)	?	Period-2 orbits of GD P (Chen & Bruna, 2022); SGD converges to cycles P (Chaudhari & Soatto, 2017)	Repetition loops M (Xu et al., 2022); paraphrase 2-cycles M (Wang et al., 2025); loops converge in hidden-state space M (Exp. 3)
Period doubling, route to chaos	?	Larger step sizes give period doubling and chaos in minimal models P (Zhu et al., 2022; Kong & Tao, 2020); period-2 orbit, then irregular oscillation, in a transformer M (Exp. 2)	Feigenbaum scaling in embedding feedback M (Jones, 2026)
Fractal boundary	?	Trainable vs. divergent hyperparameters M (Sohl-Dickstein, 2024); interleaved trainable and divergent step sizes M (Exp. 2); fractal attractors of optimizers M (Tuci et al., 2026)	?
Basin of attraction	Class regions bounded by FTLE ridges M (Storm et al., 2023)	Which minimum is reached depends on step size and batch size M (Jastrzębski et al., 2017)	Prompts that do or do not fall into loops C
Metastability, slow manifold	Multi-cluster states last exponentially long P (Geshkovski et al., 2024)	Plateaus P (Saxe et al., 2013); slow phase of pretraining M (Zhang et al., 2026a; Wen et al., 2024)	Long coherent stretches before drift C
Saddle and ghost	?	Saddles dominate M (Dauphin et al., 2014); abrupt learning from a ghost P (Dinc et al., 2025)	?
Phase transition	?	Grokking P/M (Rubin et al., 2023; Power et al., 2022); induction heads M (Olsson et al., 2022)	Low-T repetitive vs. high-T incoherent phase M (Nakaishi et al., 2024; Arnold et al., 2024); narrow coherent window between them, also in a reasoning model M (Exp. 3)
Noise-driven (Kramers) escape	?	Escape of SGD from minima P (Xie et al., 2020; Mori et al., 2021)	Escape from repetition loops, logit gap as barrier P/M (Prop. 1, Exp. 3); locking temperature P (Prop. 3), horizon-dependent in practice M (Exp. 3)
Synchronization, consensus	Clustering as consensus; Kuramoto form P (Geshkovski et al., 2023b; Li et al., 2026); synchronization by noise P (Agazzi et al., 2026)	?	Collective dynamics of interacting models P (Krishnamurthy et al., 2026)
Koopman, linearization	Local linear description of each layer M (Fernando & Guitchounts, 2026)	Koopman-based training and pruning M (Dogra & Redman, 2020; Redman et al., 2021); lazy regime P (Jacot et al., 2018)	?
Conservation law, symmetry	Gradient-flow structure of attention P (Sander et al., 2021b)	Symmetries give conserved quantities that finite steps break P (Kunin et al., 2020)	?

Table 1: The Three-Clock Dictionary. Each cell gives the behavior that a dynamical-systems concept describes on one clock, with its evidence level (P: proved in a model; M: measured in trained networks; C: conjecture) and representative references. Question marks are cells for which we found no study; they define the open problems of Section 6.

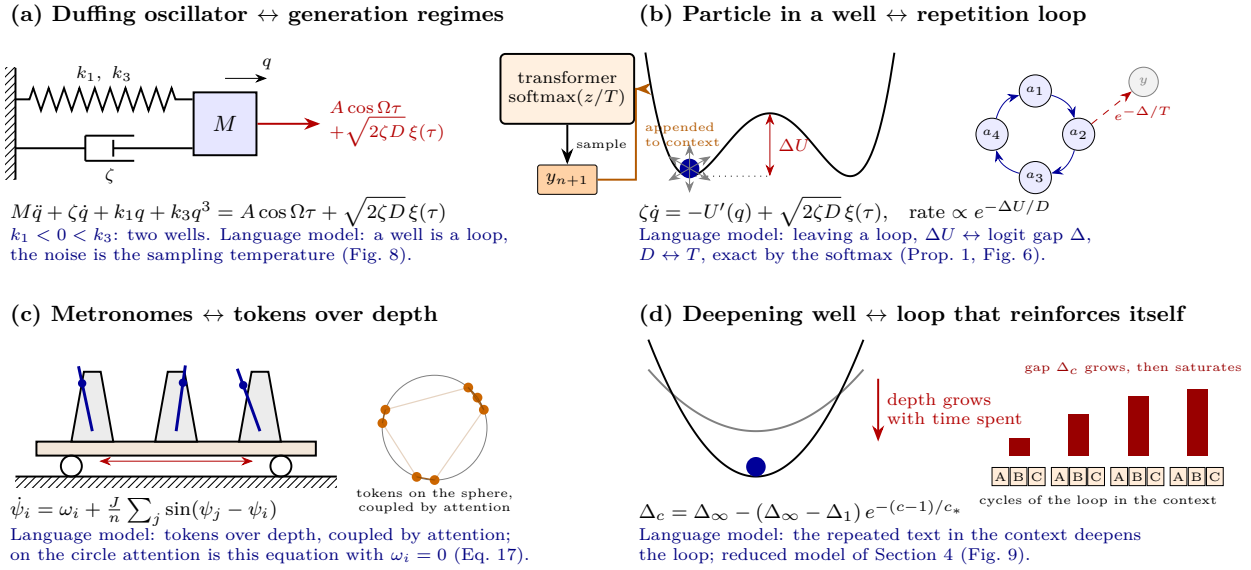


Figure 2: Four physical systems and their language-model twins. In each panel the mechanical system (left, with its governing equation) and the language model (right) look nothing alike, yet they share the term of the governing equation that produces the behavior; Section 4 derives each correspondence. (a) A mass M on a spring with linear stiffness k_1 and cubic stiffness k_3, friction coefficient ζ, harmonic forcing of amplitude A and frequency Ω, and white noise ξ of intensity D; with $k_1<0$ the potential $\frac{1}{2}k_1q^2+\frac{1}{4}k_3q^4$ has two wells. (b) An overdamped particle in a potential well under thermal noise of intensity D, whose escape rate over the barrier ΔU is given by Kramers' formula equation 6. (c) Metronomes on a board that can roll, which synchronize through the small motion of the board; their phases follow the Kuramoto model equation 4. (d) A well whose depth grows with the time the particle has spent in it and then saturates.

which a transition could be sharp. Even so, the metastability result of Geshkovski et al. (2024) shows that finite-time transients on clock 1 can be analyzed with care. Clock 2 is the best covered column, because optimization theory has always been a theory of iterated maps. Clock 1 is covered mostly by results at initialization and by models with simplified attention. Clock 3 is the youngest column, and most of its entries date from 2024 or later.

The rest of this section goes through the rows that carry most of the evidence. Each subsection defines one concept, shows it in a mechanical system and in a language model side by side, and summarizes what is known on each clock. The order follows the way the concepts build on each other: the stability of a fixed point and its loss through period doubling (Section 3.1), the basins that decide which state is reached and the slow transients near unstable states (Section 3.2), the synchronization of coupled units (Section 3.3), escape from a well under noise (Section 3.4), the ways to tell periodic, noisy and chaotic motion apart in a measured signal (Section 3.5), and the growth of small perturbations (Section 3.6). Figure 2 shows the four physical systems that recur in this comparison next to their language-model counterparts. Section 4 then explains why each pair behaves alike.

3.1 Stability, period doubling and the edge of stability

We start with the simplest question one can ask of an iterated map: does a fixed point attract nearby states, and what happens when it stops doing so.

Maps, fixed points and stability. A discrete-time system is a map $u_{t+1}=F(u_t)$. A fixed point $u^*=F(u^*)$ is linearly stable if every eigenvalue of the Jacobian $\mathrm{D}F(u^*)$ lies inside the unit circle. The textbook case is the explicit Euler scheme for a stiff ordinary differential equation: integrating $\dot{u}=-\lambda u$ with step η gives $u_{t+1}=(1-\eta\lambda)u_t$, which converges only if $\eta\lambda<2$ and otherwise oscillates with growing

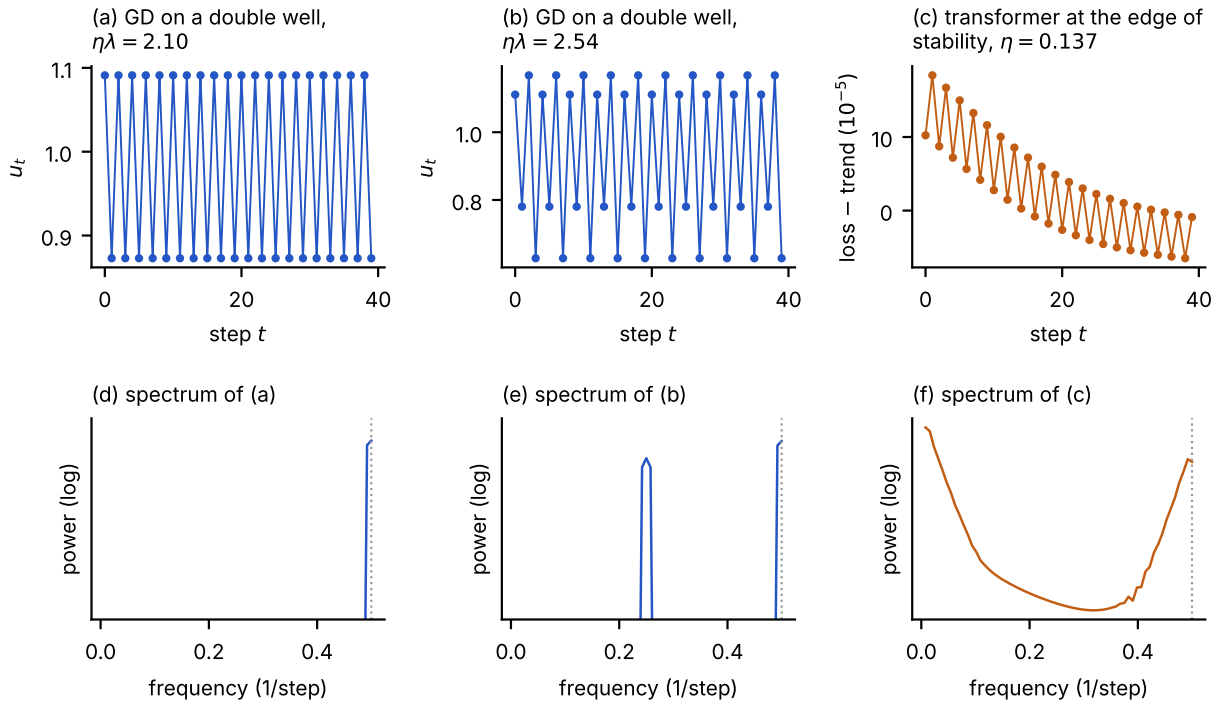


Figure 3: Training as a dynamical system: period two at the edge of stability. (a, b) Gradient descent on the double well $\mathcal{L}(u) = (u^2 - 1)^2/4$, whose minimum has curvature $\lambda = 2$, with $\eta\lambda = 2.10$ and 2.54: the last 40 iterates lie on a period-2 and a period-4 orbit. (d, e) Their power spectra, with peaks at frequencies 1/2 and 1/4. (c) The one-block transformer of Experiment 2 trained by full-batch gradient descent at $\eta = 0.137$, at the edge of stability: training loss over the last 40 steps after removing a slow polynomial trend, in units of 10^{-5}. The loss alternates from step to step. (f) Spectrum of the detrended loss over the last 400 steps: the same peak at 1/2, on top of a low-frequency band from the slow modulation of the oscillation (Section 5.2).

amplitude. Gradient descent with step size η on a quadratic with Hessian eigenvalue λ is the same map, and this one-line calculation is the origin of the threshold $2/\eta$ in the edge-of-stability literature discussed below.

Periodic orbits and bifurcations. A period-k orbit is a fixed point of F^k, and under a change of parameter such orbits can appear, disappear or exchange stability. In a period-doubling bifurcation an eigenvalue crosses -1 and a period-2 orbit is born. The logistic map $u_{t+1} = \mu u_t(1 - u_t)$, introduced as a model of insect populations, passes through a cascade of such doublings to chaos as μ grows (May, 1976), and the ratio of successive bifurcation intervals tends to a universal constant, 4.669..., for every smooth one-humped map (Feigenbaum, 1978). In a saddle-node bifurcation a pair of fixed points collides and disappears. Just past it the trajectory still slows down where the pair used to be (a "ghost"), which produces long plateaus followed by fast jumps (Strogatz, 2018; Dinc et al., 2025). Gradient descent on a double-well loss $\mathcal{L}(u) = (u^2 - 1)^2/4$ already shows both behaviors as η grows. The minimum loses stability at $\eta = 1$, a doubling cascade follows, and near $\eta \approx 1.6$ the chaotic iterate hops between the two wells.

On the clocks. Classical optimization theory treats gradient descent as a discretization of gradient flow, which is valid when the step size is small compared with the inverse curvature. Cohen et al. (2021) showed that neural network training does not stay in that regime. Under full-batch gradient descent the sharpness, meaning the largest Hessian eigenvalue $\lambda_{\max}$, first rises (progressive sharpening) until it reaches $2/\eta$, and then hovers near that value while the loss keeps decreasing in a non-monotone way. By the linear stability argument above, the iteration then sits on the boundary where the top direction loses stability. Adaptive methods show the same behavior with respect to the preconditioned sharpness (Cohen et al., 2022).

A continuous-time description of the whole edge-of-stability regime is now available. The central flow of Cohen et al. (2024) is an ODE for the time-averaged trajectory that keeps the sharpness at $2/\eta$ and predicts the loss curve of real networks, and Berthier (2026) derive it with the method of multiple scales, in three time scales. Litman (2026) shows that a weighted average of the curvature along the steps is forced to $2/\eta$. Bifurcation theory gives the local picture, with the onset of oscillation treated as a flip bifurcation near a manifold of minimizers (Appendix E). Section 5.2 adds a measurement: in a small transformer the edge is multi-mode, with several eigenvalues above $2/\eta$, while the curvature along the step equals $2/\eta$.

At larger step sizes the map can become chaotic. Deterministic gradient descent on a multiscale objective, with a step size that resolves only the macroscopic scale, converges to a statistical distribution (Kong & Tao, 2020), and stochastic optimizers at the edge of stability have fractal attractors whose dimension has been related to generalization (Tuci et al., 2026). On clock 1 the corresponding question is whether a forward pass keeps signals at a constant size, which Section 3.6 takes up, and on clock 3 it is the stability of a repetition loop, which Section 3.4 treats. Why training stays at the threshold instead of crossing it is explained with a reduced model in Section 4.4, and Experiment 2 measures the whole sequence on a transformer.

3.2 Basins, fractal boundaries and slow transients

A stable state is only useful if the system reaches it. Which state it reaches depends on where it starts, and on the way there it can linger near states that are not stable at all.

Basins and metastability. The basin of attraction of an attractor is the set of initial conditions that reach it. Basin boundaries can be fractal (McDonald et al., 1985). In a periodically forced oscillator, such as a ship rolling in beam seas, the basin of safe motion around the upright state erodes into fractal fingers as the wave amplitude grows, so that capsizing becomes sensitive to the initial state well before the safe attractor itself disappears (Thompson & Soliman, 1990; Thompson, 1997). The set of trainable hyperparameters of a neural network has the same kind of boundary: Sohl-Dickstein (2024) found that the boundary between step sizes that train and those that diverge is fractal, in the same way as the boundary of bounded orbits of the quadratic map that defines the Mandelbrot set. Figure 4 shows both cases side by side. For a forced double-well oscillator the two basins are separated by a smooth curve at small forcing and by interleaved, fractal bands at larger forcing. For a small network trained by gradient descent with one step size per layer, the boundary between training and divergence keeps its jagged structure when magnified 25 times. A metastable state is not an attractor but holds trajectories for a very long time, for example a cluster configuration of tokens that eventually merges (Geshkovski et al., 2024).

Slow transients and abrupt transitions. Near a saddle, or near the ghost of a fixed point that has just disappeared in a saddle-node bifurcation, a trajectory slows down for a long time and then moves on quickly. Training curves show both patterns. High-dimensional loss surfaces have far more saddles than local minima (Dauphin et al., 2014), and for deep linear networks the learning dynamics can be solved exactly: modes of the input-output map are learned one after another, separated by plateaus whose length depends on the initialization (Saxe et al., 2013). In recurrent networks, abrupt learning after a long plateau has been traced to the ghost of a saddle-node bifurcation (Dinc et al., 2025).

Abrupt transitions in what a model can do are the clock-2 counterpart of a bifurcation. Grokking, where test accuracy jumps long after the training loss has saturated, was first reported on algorithmic data (Power et al., 2022); inside the network the change is gradual, with circuit-level measures that improve steadily before the jump (Nanda et al., 2023), and a solvable two-layer model shows a first-order phase transition (Rubin et al., 2023). In language models, induction heads form during a short window of training, with a visible bump in the loss curve (Olsson et al., 2022), and across model scale abilities can appear suddenly (Wei et al., 2022), although part of the effect comes from discontinuous metrics (Schaeffer et al., 2023). Further explanations of these transitions are reviewed in Appendix E.

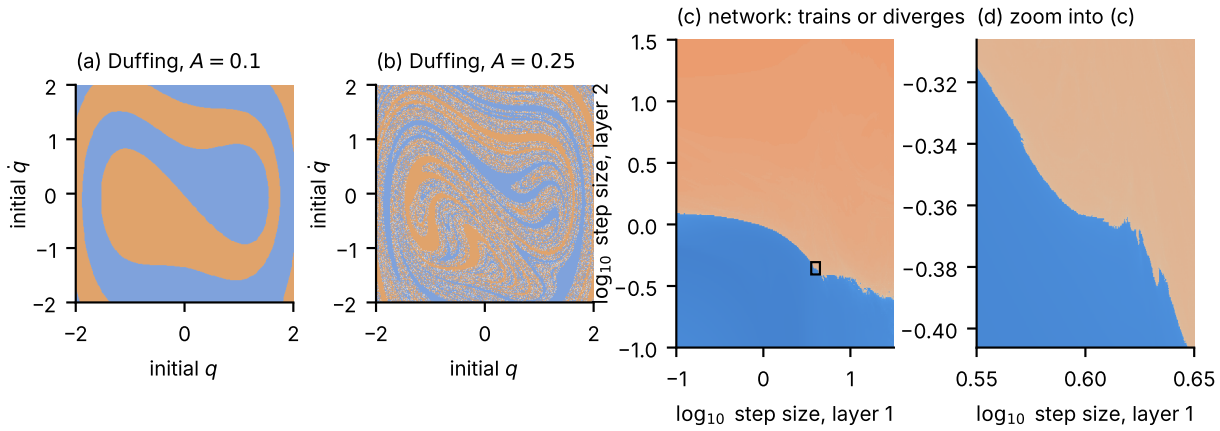


Figure 4: Basins of attraction in a mechanical system and in training. (a, b) Forced double-well Duffing oscillator $\ddot{q} + 0.25\dot{q} - q + q^3 = A\cos(1.2\tau)$: the attractor reached from each initial condition after 40 forcing periods (blue: right well; orange: left well; gray: motion that keeps visiting both wells). At $A = 0.1$ the basins are separated by a smooth boundary; at $A = 0.25$ the boundary is fractal and a region of cross-well motion appears. (c) A one-hidden-layer tanh network (width 16) trained by full-batch gradient descent for 500 steps on eight random points, with separate step sizes for the two layers, following Sohl-Dickstein (2024): blue runs train (darker: lower mean loss), orange runs diverge (lighter: later divergence). (d) Zoom into the box in (c).

3.3 Synchronization: tokens that cluster over depth

Section 3.2 was about one trajectory. The next concept is about many units that act on each other, and it is the one that organizes clock 1.

Synchronization and consensus. Weakly coupled oscillators can lock their phases. Bennett et al. (2002) reconstructed Huygens' 1665 observation that two pendulum clocks hanging from a common beam end up swinging in antiphase, the coupling being the small motion of the beam. Metronomes on a board that rolls freely synchronize in the same way (Pantaleone, 2002), and the Moon turns the same face to the Earth because tidal friction locked its rotation to its orbit (Goldreich & Peale, 1966). The Kuramoto model reduces many such systems to phases ψ_i with natural frequencies ω_i and coupling strength J,

$$\dot{\psi}_i = \omega_i + \frac{J}{n}\sum_{j=1}^{n}\sin(\psi_j - \psi_i), \tag{4}$$

and shows a transition to collective synchrony once J exceeds a critical value set by the spread of the ω_i (Kuramoto, 1975; Acebrón et al., 2005). Attention over depth has the same structure without the natural frequencies. Each token moves toward an average of the others, weighted by $e^{\beta\langle x_i, x_j\rangle}$, so tokens that are already aligned pull harder on each other (Geshkovski et al., 2023b). On the circle, with $x_i = (\cos\psi_i, \sin\psi_i)$, this gives $\dot{\psi}_i = Z_i^{-1}\sum_j e^{\beta\cos(\psi_j - \psi_i)}\sin(\psi_j - \psi_i)$, a Kuramoto model whose coupling depends on alignment (Li et al., 2026). For $\beta = 0$ all tokens synchronize into one cluster, and for large β several clusters form and persist for a long time before they merge (Section 4.6).

On clock 1. The transformer differs from a plain residual network because the update of token i depends on all tokens. Lu et al. (2019) read a transformer layer as a splitting scheme for a convection-diffusion equation of interacting particles. Sander et al. (2021b) showed that attention with doubly stochastic normalization is a Wasserstein gradient flow. Geshkovski et al. (2023a;b) studied a model in which tokens live on the unit

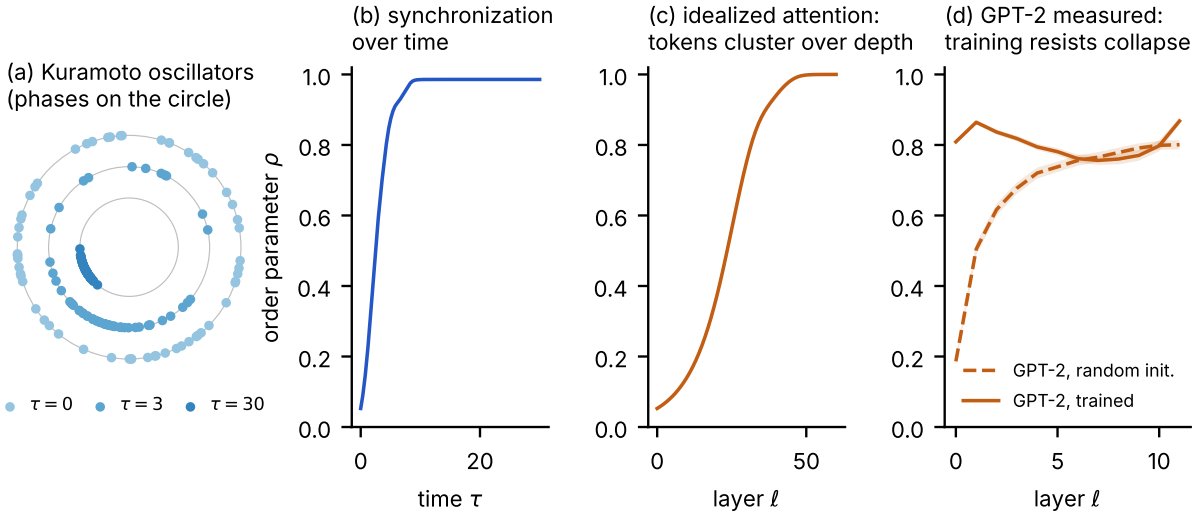


Figure 5: Depth as a dynamical system: synchronization of oscillators and clustering of tokens. (a) Kuramoto model equation 4 with $n = 60$, $J = 1.5$ and $\omega_i \sim \mathcal{N}(0, 0.3^2)$: phases at $\tau = 0$, 3 and 30 (outer to inner ring). (b) Its order parameter $\rho = |n^{-1} \sum_j e^{\mathrm{i}\psi_j}|$. (c) The attention dynamics on the circle with $\beta = 1$ and 60 tokens: the same order parameter, now over layers. (d) GPT-2, measured: order parameter of the residual stream entering each block, $\rho_\ell = \|n^{-1} \sum_i x_i^\ell / \|x_i^\ell\|\|$, for 20 prompts (mean ± one standard deviation). With random weights the tokens synchronize over depth as the idealized model predicts (ρ from 0.19 to 0.80). The trained model starts from aligned embeddings ($\rho = 0.81$) and stays between 0.76 and 0.87 over the twelve blocks.

sphere (a stand-in for layer normalization) and follow

$$\dot{x}_i = \Pi_{x_i}\Big(\frac{1}{Z_i}\sum_j e^{\beta\langle W_Q x_i, W_K x_j\rangle} W_V x_j\Big), \tag{5}$$

where Π_x projects onto the tangent space, W_Q, W_K, W_V are the query, key and value matrices, and β acts as an inverse temperature. They proved that, for broad choices of W_Q, W_K, W_V, the tokens cluster as time (depth) grows, in the simplest cases to a single point. Clustering is the particle-system version of an older observation: stacks of pure attention lose rank doubly exponentially with depth (Dong et al., 2021), and rank collapse at initialization slows training (Noci et al., 2022). Similar results hold for oversmoothing in attention-based graph networks, which Wu et al. (2023) analyze as nonlinear time-varying dynamical systems.

Real models have finite depth, so the long-time limit of equation 5 describes them only partly. Geshkovski et al. (2024) proved that particles stay near a configuration of several clusters for an exponentially long time before they merge, and a stack of finite depth spends its layers inside such a long-lived transient.

3.4 Noise-driven escape

Stable states and basins describe a system without noise. With noise, a system can leave a basin that it would never leave otherwise, and the time it takes is the subject of one of the oldest results of statistical physics.

Kramers' escape problem. Add noise of intensity D to a system resting at the bottom of a well of depth ΔU. The mean time to leave the well grows as

$$\bar{\tau} \simeq \frac{2\pi}{\omega_{\mathrm{w}}\,\omega_{\mathrm{b}}}\,\zeta\, e^{\Delta U/D} \tag{6}$$

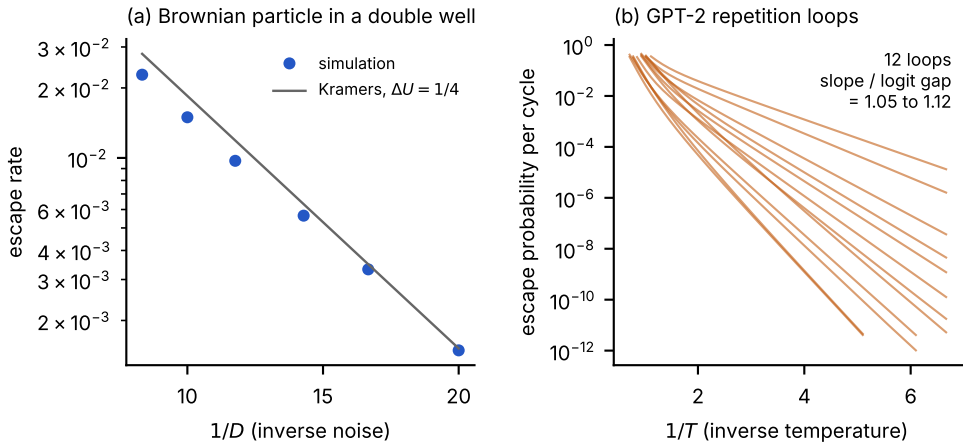


Figure 6: Noise-driven escape in a mechanical system and in a language model, on the same axes. (a) An overdamped particle, $dq = -U'(q)\,d\tau + \sqrt{2D}\,d\mathcal{W}$, in the double well $U(q) = (q^2-1)^2/4$ with barrier $\Delta U = 1/4$: inverse mean first-passage time from $q = -1$ to $q = +1$ (400 particles per noise level) against $1/D$, with Kramers' rate $(\sqrt{2}/2\pi)\,e^{-\Delta U/D}$. The fitted slope gives $\Delta U = 0.23$. (b) Twelve repetition loops of GPT-2 from Experiment 3: exact probability of leaving the loop during one cycle, computed from the stored logits, against $1/T$. At low temperature each curve is a straight line whose slope is the logit gap of the loop, as Proposition 1 predicts (ratio of slope to gap 1.05 to 1.12).

in the overdamped limit with friction coefficient ζ, where ω_{w} and ω_{b} are set by the curvature at the bottom of the well and at the top of the barrier (Kramers, 1940; Hänggi et al., 1990). The exponential factor is the Arrhenius law for the rate of chemical reactions, with temperature in the role of D (Arrhenius, 1889). In engineering the same question decides whether a ship capsizes in random waves or a structure snaps through to a buckled state. Escape from potential wells under deterministic periodic forcing is a related problem, where the escape threshold in forcing amplitude depends on the forcing frequency and on the shape of the well (Farid & Gendelman, 2019; Farid, 2022). In mechanics escape is usually the failure to avoid, whereas in text generation the well is a repetition loop and escape is what one wants. Figure 7 shows the two behaviors over time: a particle that rests in a well and hops, and GPT-2 sampled at a low temperature, which falls into repetition loops and leaves them.

On clock 2. Minibatch noise turns clock 2 into a stochastic process. Stochastic modified equations approximate SGD with step size η and batch size B by the SDE

$$d\theta = -\nabla\mathcal{L}(\theta)\,dt + \sqrt{\tfrac{\eta}{B}\,C(\theta)}\,d\mathcal{W}_t, \tag{7}$$

where C is the covariance of per-example gradients, $\mathcal{W}_t$ is a Wiener process, and time is measured in units of η (Li et al., 2015). Near a minimum with Hessian H the drift is linear, the process is of Ornstein-Uhlenbeck type, and its stationary covariance Σ solves the Lyapunov equation $H\Sigma + \Sigma H = (\eta/B)\,C$, so the ratio η/B acts as a temperature (Mandt et al., 2017; Smith & Le, 2017; Jastrzębski et al., 2017). Because SGD noise is anisotropic and state dependent, the stationary distribution is not a Gibbs distribution of the training loss, and the dynamics can converge to limit cycles instead of points (Chaudhari & Soatto, 2017).

Escape from a minimum is the most direct use of this theory. Xie et al. (2020) derived a Kramers-type rate for SGD and found that, unlike isotropic Langevin dynamics, SGD favors flat minima exponentially in the ratio of barrier height to Hessian eigenvalues, and anisotropic noise aligned with the curvature helps escape from sharp minima (Zhu et al., 2018). Because the noise of SGD is multiplicative, the escape rate can instead scale as a power of the loss ratio between saddle and minimum (Mori et al., 2021), and heavy-tailed gradient noise points toward Lévy-driven escape, dominated by single large jumps (Simsekli et al., 2019). The law that holds for language model pretraining has not been measured.

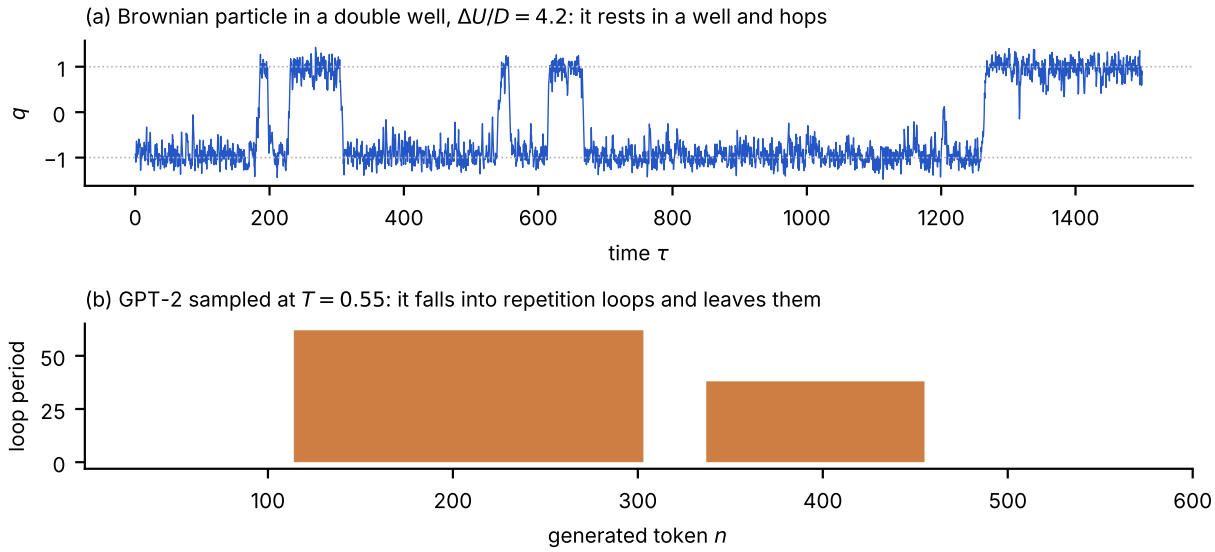


Figure 7: Escape over time. (a) Position of an overdamped particle in the double well $U(q) = (q^2 - 1)^2/4$ under noise of intensity $D = 0.06$: long rests near $q = \pm 1$ separated by quick hops. (b) GPT-2 continuing one prompt with plain temperature sampling at $T = 0.55$: at each generated token, the period of the repetition loop the text is in, where a stretch counts as a loop when it repeats at least three times (zero outside a loop). The text enters two loops, stays in each for more than a hundred tokens and leaves both.

On clock 3. With greedy or low-temperature decoding, models fall into loops that repeat a phrase or sentence until the length limit (Holtzman et al., 2019). At $T = 0$ the chain is deterministic and the loop is one of its periodic orbits. Xu et al. (2022) showed that loops reinforce themselves: the probability of repeating a sentence grows with the number of times it already appears in the context. Unlikelihood training (Welleck et al., 2019), truncation rules (Hewitt et al., 2022; Meister et al., 2022) and feedback control of perplexity (Basu et al., 2020) are partly attempts to keep the chain out of these attractors; Fu et al. (2020) give a theoretical analysis of why they form. Recent work studies the inside of the loop: how it shows up in residual-stream dynamics before it becomes visible in the output (Zhang et al., 2026b), and the conditions under which a loop "locks" so that escape becomes impossible in practice under a given sampler (Chen & Wang, 2026). Loops also appear at the level of whole texts. Repeated paraphrasing converges to period-two attractor cycles even with sampling noise (Wang et al., 2025). Reasoning models loop too, mostly at low temperature, which Pipis et al. (2025) trace to errors in learning hard steps of a task and to temporally correlated errors.

Section 4.1 shows that for repetition loops the correspondence with Kramers' problem is an exact identity, and derives when a loop locks.

3.5 Telling regimes apart from data

The previous subsections described the dynamics through models. In an experiment one usually has only a measured signal, and the same distinctions, between periodic motion, noise and chaos, must be read from it.

Four pictures of a signal. An experimentalist who sees only a signal classifies the motion with four standard pictures. The time history shows the motion directly. In the power spectrum a periodic motion gives sharp peaks at its frequency and harmonics, and noise or chaos gives a broad band. In the phase portrait a periodic orbit is a closed curve and noise fills a region. The recurrence plot marks the pairs of times (i, j) at which the state returns close to itself, $R_{ij} = 1$ if $\|u_i - u_j\| < \epsilon_R$, and a periodic orbit draws long lines parallel to the diagonal (Eckmann et al., 1987). Two numbers condense the last two pictures: the normalized entropy of the spectrum, $\mathcal{H}_s$, from 0 for a single line to 1 for white noise, and the determinism

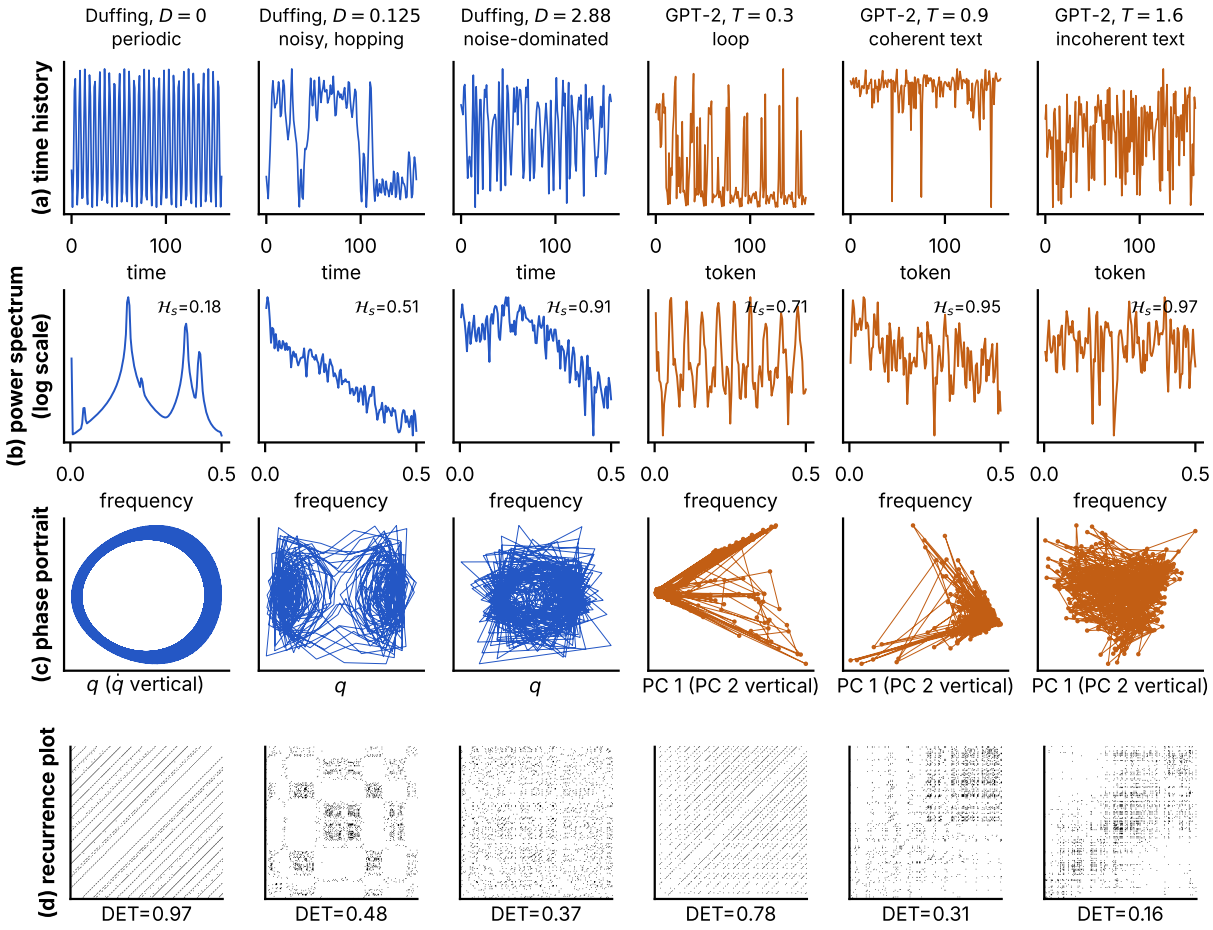


Figure 8: Generation as a dynamical system: the same four pictures for a mechanical oscillator and for a language model. Left: the forced double-well Duffing oscillator $\ddot{q}+0.25\dot{q}-q+q^3 = 0.1\cos(1.2\tau)+\sqrt{0.5D}\,\xi(\tau)$, sampled once per time unit, at noise intensities $D = 0$ (periodic orbit in one well), 0.125 (noisy orbits with hops between the wells) and 2.88 (noise-dominated). Right: GPT-2 continuing one prompt for 576 tokens at $T = 0.3$ (a loop of period 19), 0.9 (coherent text) and 1.6 (incoherent text); the state is the final hidden state of each token, scaled to unit norm and standardized per feature, after a transient of 64 tokens. (a) Time history of the first coordinate (q, or the first principal coordinate of the hidden state). (b) Its power spectrum (Welch), with the spectral entropy $\mathcal{H}_s$. (c) Phase portrait: $(q, \dot{q})$, or the first two principal coordinates. (d) Recurrence plot of the full state at a fixed recurrence rate of 5%, with the determinism DET.

DET, the fraction of recurrent points that lie on diagonal lines (Marwan et al., 2007). Figure 8 applies the four pictures to a forced double-well Duffing oscillator at three noise levels and to GPT-2 at three sampling temperatures. In both systems the quiet regime is a periodic orbit (for GPT-2, a repetition loop), with sharp spectral peaks and diagonal recurrence lines, and DET falls as the noise or the temperature grows. Over 10 noise seeds and 10 prompts, the median DET is 0.97, 0.48 and 0.42 for the Duffing oscillator and 0.67, 0.32 and 0.17 for GPT-2, and the median $\mathcal{H}_s$ is 0.18, 0.60 and 0.90 against 0.70, 0.95 and 0.95. The analogy has a limit that the pictures make plain: the spectra of coherent and of incoherent text are both close to flat, so for GPT-2 these two regimes differ in their recurrence structure and in the level of surprisal, not in the spectrum. Recurrence analysis of hidden states has recently been applied to the reasoning traces of DeepSeek-R1-Distill (Pham et al., 2026), and recurrence plots and correlation dimensions to chain-of-thought trajectories (Jaca et al., 2026).

On clock 3. At the level of statistics over many generations, temperature produces something close to a phase transition. Nakaishi et al. (2024) varied T in GPT-2 and found a low-temperature phase with repetitive structure, a high-temperature phase of incoherent text, and power-law correlations close to those of natural language at the transition point. Seen from clock 3, both phases are failures, and useful text lives in a window between them. The lower edge of the window is the loop escape of Section 3.4. The upper edge, where sampling noise spreads over so much of the vocabulary that the text loses coherence, has no dynamical definition yet, and Experiment 3 locates it only through proxies. Arnold et al. (2024) proposed statistical methods to detect such transitions automatically from output distributions. Jones (2026) found that in the continuous feedback version of generation, temperature and activation steering move the system between periodic orbits and chaos with positive finite-time Lyapunov exponents, with period-doubling scaling and periodic orbits proved by interval arithmetic. In chain-of-thought trajectories, Jaca et al. (2026) report sensitive dependence on initial conditions together with bounded evolution and fractal structure in hidden-state space. Sorting many generated trajectories into regimes (fixed point, periodic, irregular) without labels is a response-regime identification problem, for which unsupervised data-driven methods exist in mechanics (Farid, 2024).

3.6 Lyapunov exponents, order and chaos

The last concept measures how fast nearby trajectories separate. It is the quantity behind sensitivity to initial conditions, and it applies to all three clocks.

Lyapunov exponents. For a trajectory $u_0, u_1, \dots$ the finite-time Lyapunov exponent over K steps is

$$\Lambda_K(u_0) = \frac{1}{K} \log \sigma_{\max}\big(\mathrm{D}F(u_{K-1}) \cdots \mathrm{D}F(u_0)\big), \tag{8}$$

where $\sigma_{\max}$ is the largest singular value. Positive values mean that nearby trajectories separate exponentially, so that a perturbation of size δ reaches size one after about $\log(1/\delta)/\Lambda_K$ steps. Lorenz (1963) found this sensitivity in a three-variable truncation of atmospheric convection, and it gives weather forecasts a finite horizon however accurate the initial data. The limit $K \to \infty$, when it exists, is the largest Lyapunov exponent. Deep networks have a fixed and small number of steps, so only finite-time exponents make sense for clock 1 (Storm et al., 2023).

On clock 1. One line of work treats a randomly initialized deep network as a random dynamical system and asks whether small input differences grow or shrink with depth. Poole et al. (2016) found an order-to-chaos transition controlled by the weight and bias variances σ_w^2, σ_b^2. For a wide fully connected network with nonlinearity ϕ, the correlation between two nearby inputs is multiplied at each layer by

$$\chi_1 = \sigma_w^2 \, \mathbb{E}\big[\phi'(\sqrt{Q^*}\, G)^2\big], \tag{9}$$

where Q^* is the fixed point of the length map and G is a standard Gaussian variable. Inputs converge for $\chi_1 < 1$ (order) and separate for $\chi_1 > 1$ (chaos), so $\ln \chi_1$ plays the role of a Lyapunov exponent per layer; Schoenholz et al. (2016) showed that the depth over which signals propagate diverges at the transition, so networks are trainable to large depth only near it. The idea goes back to the chaotic phase of random recurrent networks (Sompolinsky et al., 1988). Requiring all singular values of the input-output Jacobian to be close to one (dynamical isometry) gives faster training still (Pennington et al., 2017) and allowed 10,000-layer convolutional networks to train (Xiao et al., 2018). For transformers, Cowsik et al. (2024) track the geometry of n token representations as a discrete-time system of interacting particles and derive conditions on the initialization that make deep transformers trainable.

For trained networks, Storm et al. (2023) computed finite-time Lyapunov exponents equation 8 of fully connected classifiers and found ridges of large exponents in input space that separate the classes, much like coherent structures in fluid flows. Our first experiment (Section 5.1) applies the same quantity to the residual stream of language models across training.

What a dynamical model of depth must reproduce. Interpretability research gives facts that any dynamical model of clock 1 has to reproduce. Predictions decoded from intermediate layers improve gradually, which suggests a trajectory that refines its estimate step by step (Belrose et al., 2023). Deleting or swapping middle layers hurts much less than touching the first or last ones, consistent with distinct stages of the trajectory (Lad et al., 2024). Token trajectories through depth stay close to a low-dimensional manifold (Sarfati et al., 2024). A few activations are orders of magnitude larger than the rest and nearly independent of the input (Sun et al., 2024), which together with attention sinks (Xiao et al., 2023) can be read as a near fixed point of the depth dynamics. Experiment 1 measures how training changes the stretching along this trajectory.

On clock 3. Training uses teacher forcing, while generation feeds the model its own outputs. Small errors therefore compound along clock 3, a problem known as exposure bias (Bengio et al., 2015; Wang & Sennrich, 2020). Errors snowball in multi-step reasoning (Dziri et al., 2023) and in hallucinated justifications (Zhang et al., 2023a), and next-token prediction can fail on planning tasks that require looking ahead (Bachmann & Nagarajan, 2024). In dynamical terms these are statements about the growth of perturbations along a trajectory of the closed-loop system: if the finite-time exponent of clock 3 is positive, an early error of size δ grows roughly as $\delta\, e^{\Lambda_K K}$ over the next K tokens. This exponent has not been measured for language models (**OP 3**).

Koopman linearization. Lyapunov exponents describe a nonlinear map through its linearization along one trajectory. A different, global linearization exists: a nonlinear map acts linearly on functions of the state, called observables, $h \mapsto h \circ F$ (Koopman, 1931). Eigenfunctions of this Koopman operator give coordinates in which the dynamics is linear, and finite approximations can be learned from data (Mezić, 2005). Dynamic mode decomposition, for instance, extracts the oscillation frequencies and spatial modes of a jet in crossflow directly from snapshots (Rowley et al., 2009). Related data-driven methods learn linear embeddings of nonlinear dynamics with neural networks (Lusch et al., 2017) or sparse governing equations from data (Brunton et al., 2015). Koopman methods have also been applied to the training dynamics of neural networks (Dogra & Redman, 2020; Redman et al., 2021).

4 Why the analogies hold: reduced-order models

Engineers often understand a complex system through a simple model that shares its governing terms. Sloshing in a partially filled liquid tank is a free-surface flow, yet its response regimes, including the strongly nonlinear ones with hydraulic impacts on the walls, are reproduced by an equivalent mechanical model in which a linear oscillator is coupled to a vibro-impact unit (Farid & Gendelman, 2017). Section 3 showed that the clocks and textbook physical systems behave alike. This section explains why, with reductions of the same kind. For each one we name the physical model and its governing equation (Figure 2), derive the part of the language model that reduces to it, and state where the reduction stops holding. Table 2 summarizes the six cases, and the subsections follow its order: three results on clock 3, two on clock 2 and one on clock 1.

4.1 Escape from a loop: the softmax is a Boltzmann distribution

Repetition loops are the case where the reduction is exact, so we start with the noise-driven escape of Section 3.4.

The softmax is a Boltzmann distribution. In statistical mechanics a system in contact with a heat bath occupies a state of energy E with probability proportional to $e^{-E/k_B T}$. The sampler equation 3 has exactly this form, with the energy of token y equal to minus its logit, $E_y = -z_y$, and $k_B = 1$. A token is a state, a logit is minus an energy, and the sampling temperature is a temperature in the physical sense. Suppose the chain has locked into a loop of period k with tokens $a_1, \dots, a_k$. As long as the sampler keeps choosing the loop token, the context is deterministic, so the logits z_i at the i-th step of the loop are fixed numbers. The loop token a_i has the highest logit, and every other token y is an exit whose energy is higher

Table 2: Reduced-order models of language-model behavior. The shared term is the part of the governing equation that produces the behavior in both systems.

Language model	Reduced model	Shared term	Status
Escape from a loop (clock 3)	Particle in a well under noise, Fig. 2(b)	Boltzmann factor: $e^{-\Delta/T}$ and $e^{-\Delta U/D}$	Exact identity (Prop. 1)
Loops that hold for a while (clock 3)	Particle in a deepening well, Fig. 2(d)	Barrier that grows with residence time	Fitted; $T_{1/2}$ within 9% to 25%
Regimes of generation (clock 3)	Noisy double-well Duffing oscillator, Fig. 2(a)	Ratio of barrier to noise	Same ordering, measured
Period two at the edge of stability (clock 2)	Explicit Euler step on a nonlinear spring	Step multiplier $1-\eta\lambda$ crossing -1, cubic term of the loss	Derived; multi-mode edge not covered
Loss spike (clock 2)	Ball in a valley whose stiffness changes along it, Fig. 11	Step multiplier $1-\eta\lambda$ below -1 where the valley is stiff	Exact for the two-parameter model; untested at scale
Clustering over depth (clock 1)	Kuramoto oscillators, Fig. 2(c)	Coupling $\sin(\psi_j-\psi_i)$	Exact for idealized attention; trained GPT-2 departs

by the logit gap $\Delta_i^y = z_i(a_i) - z_i(y) > 0$. The probability of staying in the loop at step i is

$$p_i(T) = \frac{1}{1+S_i(T)}, \qquad S_i(T) = \sum_{y \neq a_i} e^{-\Delta_i^y/T}, \tag{10}$$

and the probability of leaving during one cycle is $r(T) = 1 - \prod_{i=1}^{k} p_i(T)$ (Figure 2b). At low temperature the probability of leaving at step i is

$$1 - p_i(T) = \frac{S_i(T)}{1+S_i(T)} \;\approx\; m_i\, e^{-\Delta_i/T} \qquad (T \to 0), \tag{11}$$

where Δ_i is the smallest gap and m_i the number of tokens that attain it. This is the Arrhenius factor of Kramers' problem equation 6, with the logit gap in place of the barrier height ΔU and T in place of the noise intensity D (Figure 2b). The correspondence is an identity, which is why every loop in Figure 6(b) gives a straight line whose slope is its gap. Two differences remain. A Brownian particle diffuses up to the barrier, and the curvatures at the bottom and the top of the well set Kramers' prefactor; the sampler jumps in one step, so the prefactor is the number of exits m_i. In addition, the energies of a language model change as the context grows, which Section 4.2 treats.

Proposition 1 (Arrhenius law for loop escape)**.** *Let* $\Delta_{\min} = \min_i \min_{y \neq a_i} \Delta_i^y$ *and let* m *be the number of pairs* (i, y) *that attain it. Then* $r(T) \le \sum_i S_i(T)$ *for all* $T > 0$*, and*

$$r(T) = m\, e^{-\Delta_{\min}/T}\,\big(1 + o(1)\big) \quad \text{as } T \to 0, \qquad \text{so} \qquad \lim_{T\to 0} T \log r(T) = -\Delta_{\min}. \tag{12}$$

The proof is in Appendix C.

The logit gap therefore plays the role of the barrier height ΔU in Kramers' formula (Kramers, 1940), and T plays the role of noise intensity. Proposition 1 follows almost directly from the Gibbs form of the softmax. Chen & Wang (2026) use the same product of per-step continuation probabilities to ask when a sampler locks a loop exactly, with probability one, and show that plain temperature sampling never does. Here a loop locks in a weaker, probabilistic sense: the probability of never escaping is positive (Proposition 2), or escape takes longer than the generation (Section 5.3).

4.2 Loops that hold for a while: a well that deepens while it is occupied

The Boltzmann identity equation 11 holds at every step, but the energies change from cycle to cycle, because each completed cycle adds the same text to the context and the model grows more confident that it will

repeat it, the self-reinforcement reported by Xu et al. (2022). Figure 9(a) shows the smallest gap Δ_c of each cycle along three GPT-2 loops: it grows and then saturates. The physical counterpart is a particle in a well that deepens while it is occupied (Figure 2d). Rock friction behaves this way: the static friction between two surfaces grows with the time they have been in contact (Dieterich, 1972), so a block that has rested longer needs a larger force to slide.

To see what this growth does to escape, let Δ_c be the smallest gap during cycle c and $r_c(T)$ the escape probability during that cycle. The probability of never leaving is $\prod_{c\geq 1}(1-r_c(T))$, which is positive exactly when $\sum_c r_c(T) < \infty$ (given $r_c < 1$).

Proposition 2 (Deepening wells lock)**.** *If $\Delta_c \geq \Delta_1 + b\,(c-1)$ for some $b > 0$, then for every $T > 0$ the probability of never escaping is at least $1 - k(|\mathcal{V}|-1)\,e^{-\Delta_1/T}/(1-e^{-b/T})$ whenever this bound is positive, and it is positive for every $T > 0$. If instead the logits along the loop converge from cycle to cycle, so that $\Delta_c \to \Delta_\infty < \infty$, escape eventually happens with probability one, but the expected number of cycles before escape grows like $e^{\Delta_\infty/T}$ as $T \to 0$.*

The proof is in Appendix C.

Between linear growth and saturation lies a growth law with a sharp threshold in temperature.

Proposition 3 (Locking temperature)**.** *If $\Delta_c = \Delta_1 + \gamma \ln c$ for all $c \geq 1$ with $\gamma > 0$, the probability of never escaping is positive for $T < \gamma$, and escape happens with probability one for $T \geq \gamma$. For $T > \gamma$ the probability of staying in the loop for N cycles is $\exp\big(-\Theta(N^{1-\gamma/T})\big)$.*

The proof is in Appendix C.

Proposition 3 gives an infinite-horizon locking temperature $T_{\text{lock}} = \gamma$, below which survival decays to a positive constant and above which it decays as a stretched exponential. Linear growth locks at every temperature and saturating growth at none, so the three cases of Propositions 2 and 3 cover the possible outcomes for an unbounded generation. A real generation has a finite budget of N cycles. If the barrier has saturated at Δ_∞ and the escape probability per cycle is $m\,e^{-\Delta_\infty/T} \ll 1$, survival over N cycles equals one half at

$$T_{1/2}(N) \approx \frac{\Delta_\infty}{\ln\big(mN/\ln 2\big)}, \tag{13}$$

so the locking temperature observed in a saturating loop falls like $1/\ln N$ with the length of the generation. In practice logits are bounded, so the gaps cannot grow without limit. Experiment 3 (Section 5.3) measures Δ_c along real loops of GPT-2 and Pythia, tests Propositions 1 and 2 against the exact survival probabilities and against Monte Carlo sampling, and finds that the barriers saturate. A reduced model with a saturating barrier is therefore the right one for real loops.

A reduced model for real loops. The reduced model has two ingredients, a saturating barrier fitted to the gaps of each loop,

$$\Delta_c = \Delta_\infty - (\Delta_\infty - \Delta_1)\,e^{-(c-1)/c_*}, \qquad r_c(T) = 1 - \big(1 + e^{-\Delta_c/T}\big)^{-\nu}, \tag{14}$$

and an escape probability per cycle r_c for a single exit attempted ν times per cycle. The three numbers $\Delta_1, \Delta_\infty, c_*$ of each loop come from its greedy run alone, and ν plays the role of Kramers' prefactor; we use one value per model, chosen on a grid ($\nu = 3$ for GPT-2, $\nu = 4$ for Pythia-160M). Figure 9 compares the locking temperature $T_{1/2}(N)$ of this model with the full computation from all logits of all steps (Section 5.3). For horizons of 4 to 256 cycles the reduced model is within a median of 9% to 25% of the full value for individual loops, with correlations of 0.74 to 0.95 across loops, and it reproduces the slow fall of $T_{1/2}$ with N. For a single cycle it is off by 30% to 50%. For long horizons the reduced model gives $T_{1/2}(N) \approx \Delta_\infty/\ln(\nu N/\ln 2)$, the form of equation 13, since an escape within N cycles becomes likely once $\nu N e^{-\Delta_\infty/T}$ is of order one, so the locking temperature falls like $1/\ln N$.

4.3 One number orders the regimes of generation

The Boltzmann identity also accounts for the regimes of Section 3.5. If escape from a well is governed by the factor $e^{-\text{barrier}/\text{noise}}$, the motion of a system with wells is organized by one dimensionless number, the

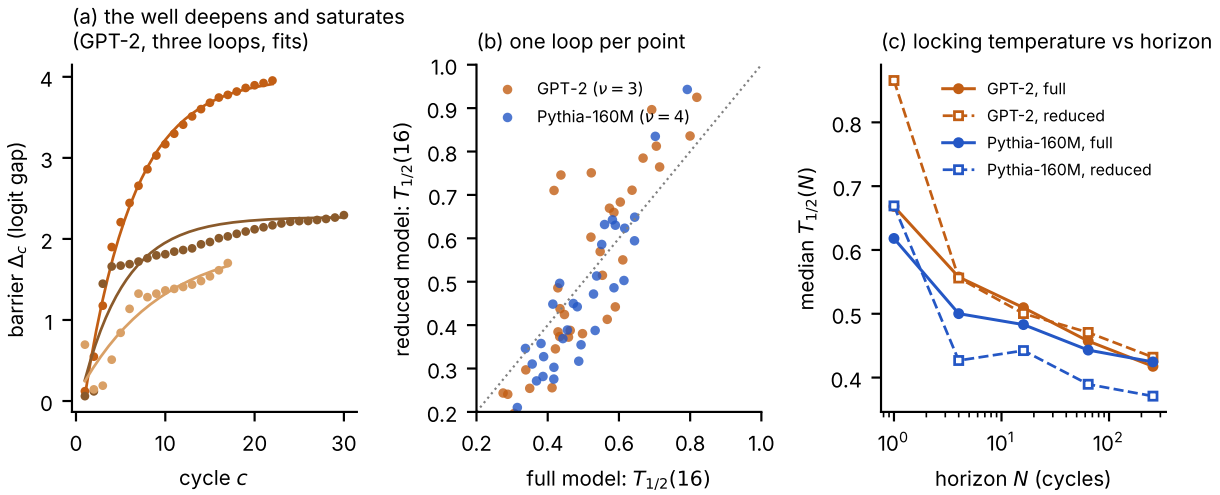


Figure 9: The reduced-order model of loop escape against the full computation. (a) Smallest logit gap of each cycle (dots) along three GPT-2 loops of at least 15 cycles, with the saturating fit of equation 14 (lines). (b) Locking temperature for a horizon of 16 cycles after a burn-in of two cycles, from the reduced model (three numbers per loop read from the greedy run, one attempt number ν per model) against the full computation from all logits, one point per loop. (c) Median locking temperature against the horizon for both models, full (filled) and reduced (open).

ratio of barrier height to noise intensity, whatever the other details of the system. The prediction is that the oscillator and the language model of Figure 8 leave their periodic orbits at the same value of this ratio, a value set by the length of the observation. For the double-well Duffing oscillator with unit mass, friction coefficient ζ and noise term $\sqrt{2\zeta D}\,\xi$, the unforced motion has the stationary density $\propto \exp\{-(q^2/2 + U(q))/D\}$, so the depth of a well in units of the noise is $\Delta U/D$, with $\Delta U = 1/4$ for $U = -q^2/2 + q^4/4$. If an escape has probability of order $e^{-\Delta U/D}$ per period, the oscillator is expected to leave a well within N periods once $\Delta U/D \lesssim \ln(N/\ln 2)$. Above this value it stays on a periodic orbit in one well, close to it the oscillator hops between wells, and far below it the noise dominates. For a loop of a language model the corresponding ratio is Δ/T, with Δ the barrier of the loop, and the same estimate gives $\Delta/T \lesssim \ln(\nu N/\ln 2)$, with N the number of cycles in the window. Figure 10 plots the determinism of the trajectory against this ratio for both systems. For the oscillator (512 samples, about 98 forcing periods, so $\ln(N/\ln 2) \approx 4.9$) DET rises from 0.38 to 0.78 as $\Delta U/D$ grows from 0.09 to 50, with its largest step between 3.1 and 5.6. For GPT-2 we sampled five prompts whose greedy runs loop, at eight temperatures from 0.3 to 1.4, and took Δ as the gap of the last cycles of the greedy loop. DET rises with Δ/T (correlation 0.75 with $\log(\Delta/T)$ over 40 runs), and the binned median climbs from 0.37 to 0.82 between $\Delta/T \approx 5.6$ and 12, just above the value 4.9 at which one escape is expected in the window. The scatter between prompts is large, and five prompts do not determine the shape of the curve, but both systems leave the periodic regime when the barrier falls to a few times the noise, at the value set by the length of the observation.

4.4 The edge of stability: an Euler step on a nonlinear spring

We now turn to clock 2. The edge of stability of Section 3.1, where training settles on a period-2 orbit at the threshold $2/\eta$, has a reduction that takes one line. Gradient descent $\theta_{t+1} = \theta_t - \eta\nabla\mathcal{L}(\theta_t)$ is the explicit Euler scheme with step η for the gradient flow $\dot\theta = -\nabla\mathcal{L}(\theta)$, the motion of an overdamped particle in the potential $\mathcal{L}$. Restrict it to the eigenvector v of the largest Hessian eigenvalue λ at a minimum θ^*, with $q = \langle\theta - \theta^*, v\rangle$ and $\mathcal{L}_v(q) = \mathcal{L}(\theta^* + qv)$. Then

$$q_{t+1} = q_t - \eta\,\mathcal{L}_v'(q_t), \qquad \mathcal{L}_v'(q) = \lambda q + \tfrac{1}{2}\mathcal{L}_v'''q^2 + \tfrac{1}{6}\mathcal{L}_v''''q^3 + \dots, \tag{15}$$

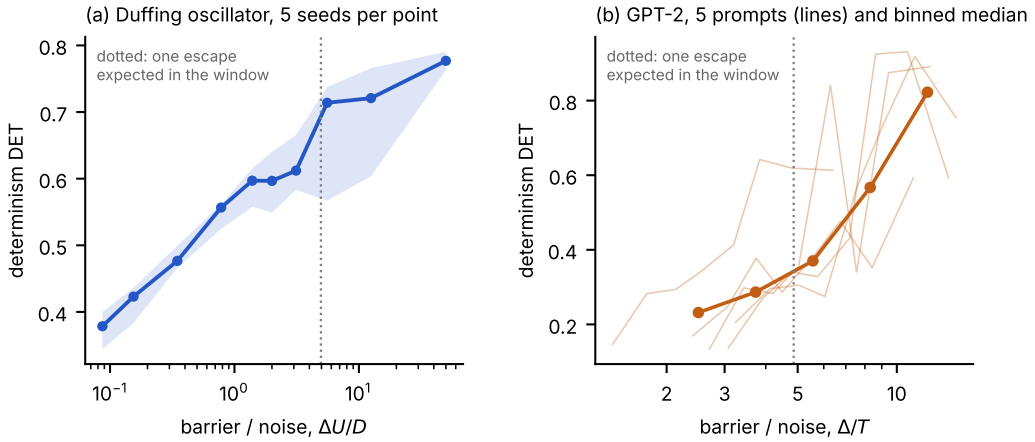


Figure 10: Determinism of the trajectory against the ratio of barrier height to noise intensity. (a) Forced double-well Duffing oscillator of Figure 8, ten noise levels, five seeds each (median and range), with $\Delta U = 1/4$. (b) GPT-2, five prompts whose greedy runs end in a loop, sampled at eight temperatures from 0.3 to 1.4; Δ is the median logit gap of the last three cycles of the greedy loop. Thin lines: individual prompts; thick line: median in bins of Δ/T. Dotted lines: the ratio at which one escape is expected within the observed window, $\ln(N/\ln 2)$ for the oscillator and $\ln(\nu N/\ln 2)$ with $\nu = 3$ for GPT-2.

and $-\mathcal{L}_v'(q)$ is the restoring force of a spring with linear, quadratic and cubic stiffness, the Duffing spring of Figure 2(a) when the quadratic term vanishes. The linear term alone gives $q_{t+1} = (1 - \eta\lambda)q_t$, whose multiplier crosses -1 at $\eta\lambda = 2$. The minimum then loses stability through a period-doubling bifurcation, the same instability that limits the time step of the explicit Euler scheme for a stiff spring. The nonlinear terms decide what follows. For the double well $\mathcal{L}_v(q) = (q^2 - 1)^2/4$ around $q = 1$, where $\lambda = 2$, the iterate settles on a period-2 orbit for $\eta\lambda$ between 2 and about 2.47, on a period-4 orbit from 2.48, on a period-8 orbit near 2.58, and on no orbit of period up to 64 from about 2.62 (Figure 3a,b). A period-2 orbit is what produces the spectral line at frequency 1/2.

A spring whose stiffness responds to the motion. A network adds a third ingredient: λ depends on θ, so the oscillation along v changes the curvature that drives it, and the motion and the stiffness are coupled. Damian et al. (2022) reduce this feedback to a two-variable map. With q_t the displacement along the top eigenvector v and λ_t the sharpness, linearized around a point where it equals $2/\eta$, the two evolve as

$$q_{t+1} = (1 - \eta\lambda_t)\, q_t, \qquad \lambda_{t+1} = \lambda_t + \eta\big(\alpha - \|\nabla\lambda_{\max}\|^2 q_t^2\big), \tag{16}$$

where $\alpha > 0$ measures progressive sharpening and the cubic term of the loss enters through the gradient of the sharpness, $\nabla\lambda_{\max}$. The first equation is the linear step of equation 15 with a stiffness that now moves. The oscillation grows while $\eta\lambda_t > 2$, and its square pulls λ_t back, so the pair cycles around the threshold instead of diverging. Mechanically, this is a mass in an elastic well whose walls soften when the mass swings hard, an oscillator whose stiffness is a slow variable driven by the amplitude of the motion. The feedback explains why the transformer of Experiment 2 shows a line at 1/2 (Figure 3c,f) across two decades of step size instead of a cascade to chaos at one step size: training keeps its own bifurcation parameter at the threshold. The one-direction reduction holds while one eigenvalue dominates. In our transformer two to six eigenvalues sit above $2/\eta$ (Section 5.2), which neither equation 15 nor the two-variable map describes. Other parts of this picture are conditions under which gradient descent converges while being locally unstable (Arora et al., 2022; Ahn et al., 2022), solvable minimal models with edge-of-stability dynamics (Zhu et al., 2022), convergence to stable period-two orbits beyond the threshold, a period-doubling bifurcation in the strict sense (Chen & Bruna, 2022), and the implicit penalty on the gradient norm that backward error analysis finds in the finite step (Barrett & Dherin, 2020).

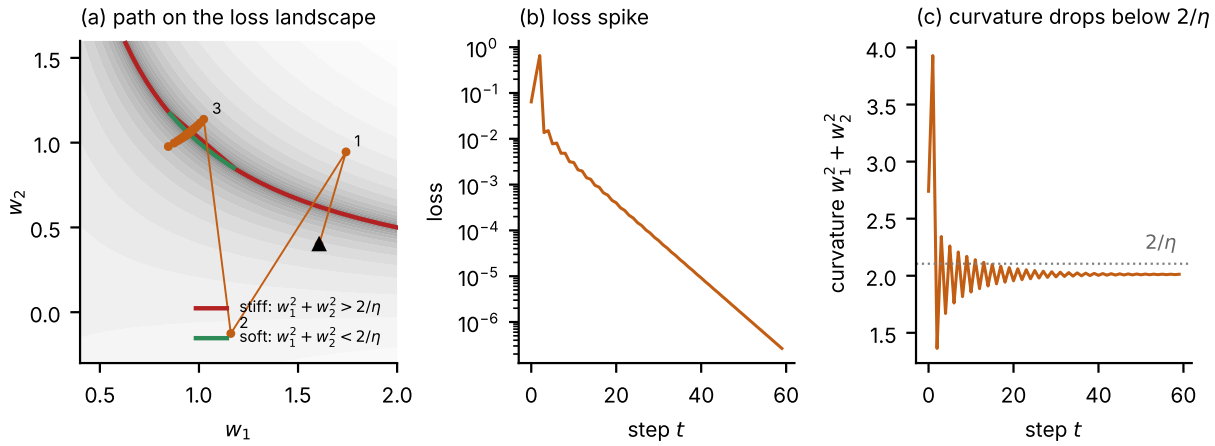


Figure 11: A loss spike in two parameters, seen as a ball thrown out of a stiff valley. Gradient descent on $(w_1 w_2 - 1)^2/2$ with $\eta = 0.95$, started near $(w_1, w_2) = (1.6, 0.4)$ where the curvature across the valley of minima, $w_1^2 + w_2^2 = 2.74$, exceeds $2/\eta = 2.11$. (a) Path of the iterate (triangle: start; numbers: the first three steps) on the loss landscape (gray: log loss); the valley of minima $w_1 w_2 = 1$ is red where it is too stiff for the step size and green where it is not. (b) The loss jumps from 0.064 to 0.66 and then decays. (c) The curvature oscillates and settles at 2.01, below $2/\eta$.

4.5 A loss spike: a ball thrown out of a stiff valley

The same instability, seen in a loss landscape whose stiffness changes from place to place, produces loss spikes. Large pretraining runs show such spikes. PaLM handled them by restarting from an earlier checkpoint and skipping data (Chowdhery et al., 2022). Wortsman et al. (2023) reproduced two instability mechanisms at small scale, growth of attention logits and divergence of the output logits, and studied how they depend on learning rate. Attention entropy collapse is another proposed cause (Zhai et al., 2023), as is the behavior of the Adam update at large scale (Molybog et al., 2023); conditions on the scale of embeddings and residual connections can prevent spikes (Takase et al., 2023). The smallest model with a loss spike has two parameters, and it shows the mechanism in a form that can be drawn.

In the catapult model of Lewkowycz et al. (2020) the product $w_1 w_2$ of two scalar weights is fit to a single target with step size η. The loss $(w_1 w_2 - 1)^2/2$ has a valley of minima along the hyperbola $w_1 w_2 = 1$, and the curvature across the valley at (w_1, w_2) is $w_1^2 + w_2^2$, large far out along the axes and smallest at $w_1 = w_2 = 1$ (Figure 11a). The mechanical picture is a ball moved by the explicit Euler steps of Section 4.4 in a channel whose walls are steep at one end and gentle at the other. Started on the steep part, where $\eta(w_1^2 + w_2^2) > 2$, each step overshoots the floor and the ball bounces from wall to wall with growing amplitude, the period-doubling instability of equation 15: the loss spikes. The bouncing also carries the ball along the channel toward the gentle end, and once it reaches the part where $\eta(w_1^2 + w_2^2) < 2$, the steps stop overshooting and the ball settles on the floor. With $\eta = 0.95$ the loss rises tenfold within two steps and the iterate settles where the curvature, 2.01, is below $2/\eta = 2.11$ (Figure 11b,c). Lewkowycz et al. (2020) call this the catapult phase; above a second threshold training diverges, and Kalra & Barkeshli (2023) mapped these early-training regimes as a function of learning rate, depth and width. Whether the spikes of large pretraining runs are catapults, crossings of the stability boundary or escapes from a basin is open (**OP 4**).

4.6 Clustering over depth: attention on the circle is a Kuramoto model

The last reduction concerns clock 1 and explains the synchronization of Section 3.3. In the transformer model of Geshkovski et al. (2023b), layer normalization keeps every token on the unit sphere, and self-attention

with identity query, key and value matrices moves token i by

$$\dot{x}_i = \Pi_{x_i}\Big(Z_i^{-1}\sum_j e^{\beta\langle x_i,x_j\rangle}x_j\Big), \qquad Z_i = \sum_j e^{\beta\langle x_i,x_j\rangle},$$

where Π_x projects onto the tangent space of the sphere at x. In two dimensions write $x_i = (\cos\psi_i, \sin\psi_i)$. The unit tangent at x_i is $(-\sin\psi_i, \cos\psi_i)$, its inner product with x_j is $\sin(\psi_j - \psi_i)$, and $\langle x_i, x_j\rangle = \cos(\psi_j - \psi_i)$, so

$$\dot{\psi}_i = Z_i^{-1}\sum_j e^{\beta\cos(\psi_j-\psi_i)}\sin(\psi_j - \psi_i). \tag{17}$$

For $\beta = 0$, $Z_i = n$ and equation 17 is the Kuramoto model equation 4 with $\omega_i = 0$ and $J = 1$, the model of identical metronomes on a common board, which synchronize from almost every initial condition (Figure 2c). In both systems each unit is pulled toward a weighted average of the others, and for $\beta > 0$ the pull is stronger between tokens that are already close, which lets several clusters form and persist (Section 3.3). The reduction accounts for Figure 5(c) and for GPT-2 with random weights in panel (d). Trained GPT-2 departs from it: its order parameter stays near 0.8 over all twelve blocks. The reduction drops the learned query, key and value maps, the MLP blocks and the finite depth. One candidate explanation is that these terms act like the spread of natural frequencies ω_i in the Kuramoto model, which opposes synchronization; we have not tested it.

5 Three experiments, one per clock

Sections 3 and 4 described the three clocks with the concepts of nonlinear dynamics and explained several correspondences with reduced models. This section tests three of them with new measurements on open models, one per clock. Experiment 1 asks how training changes the finite-time Lyapunov exponents of the forward pass (Section 3.6). Experiment 2 maps how gradient descent on a transformer moves through the regimes of Section 3.1 as the step size grows. Experiment 3 tests the escape theory of Sections 4.1 and 4.2 on the repetition loops of GPT-2 and Pythia, and follows the coherent window it implies across model size and into a reasoning model. Every run fits on a two-core CPU; most take under an hour, and the largest models take a few hours. Details are in Appendix D, and the code reproduces every number.

5.1 Experiment 1: the Lyapunov profile of the residual stream over training (clocks 1 and 2)

Question and setup. Finite-time Lyapunov exponents equation 8 describe how a forward pass stretches perturbations, layer by layer. How does this profile change as the model trains? This question couples clock 1 to clock 2 (**OP 1**). We use the public training checkpoints of Pythia (Biderman et al., 2023): eleven checkpoints of Pythia-70M (6 layers), from initialization to step 143,000, and six checkpoints of Pythia-160M (12 layers). Each block is treated as a map F_ℓ on the full residual stream of a 16-token context. For each block we estimate the largest singular value σ_ℓ of its Jacobian by power iteration and report the relative exponent

$$\tilde{\Lambda}_\ell = \log\sigma_\ell - \log\big(\|x^{\ell+1}\|/\|x^\ell\|\big), \tag{18}$$

which measures stretching beyond the growth of the residual norm. For the whole stack we report $\Lambda = \frac{1}{L}\log\sigma_{\max}(\mathrm{D}F_{0\to L})$, and the ratio $\kappa = \log\sigma_{\max}(\mathrm{D}F_{0\to L})/\sum_\ell \log\sigma_\ell$. Since $\sigma_{\max}$ of a product is at most the product of the σ_ℓ, $\kappa \le 1$ whenever $\sum_\ell \log\sigma_\ell > 0$, which holds throughout our measurements, and κ near one means that the most expanded direction of each layer is passed on to the next. Values are means over eight prompts, and κ is the ratio of the two means.

Results. Figure 12 shows three things. First, at initialization both models have the same profile, decreasing with depth ($\tilde{\Lambda}_\ell$ from 1.20 to 0.53 in Pythia-70M and from 1.19 to 0.39 in Pythia-160M), which reflects the shared initialization scheme. Second, training makes every layer more expansive. The mean of $\tilde{\Lambda}_\ell$ over layers rises from 0.75 to 2.32 in Pythia-70M and from 0.59 to 2.17 in Pythia-160M, and it is still rising at the last checkpoint. The rise is not monotone at the start, since at step 128 the deepest layers of Pythia-70M contract further (layer 6 drops from 0.53 to 0.22) before all layers expand. Third, the whole stack does not follow. In

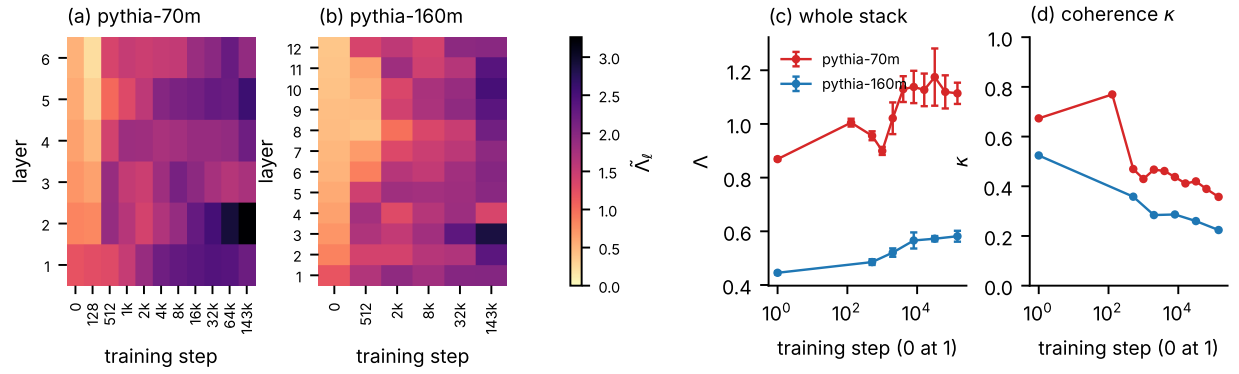


Figure 12: Experiment 1: finite-time Lyapunov exponents of the residual stream across public Pythia checkpoints. (a, b) Relative per-layer exponent $\tilde{\Lambda}_\ell$ (stretching of the top singular direction beyond the growth of the residual norm) for each layer and checkpoint. (c) Depth-averaged exponent of the whole stack, Λ; bars: standard deviation over prompts. (d) Coherence κ between per-layer and whole-stack stretching; values below one mean that the most expanded directions of consecutive layers are not aligned.

Pythia-70M, Λ goes from 0.87 to about 1.1 and stays between 1.11 and 1.18 from step 4,000 onward, which is under 3% of training. In Pythia-160M it moves from 0.45 to 0.58. As a result the coherence κ falls during training, from 0.67 to 0.36 in Pythia-70M and from 0.52 to 0.22 in Pythia-160M.

Reading. Training produces layers that are individually more sensitive to perturbations, while the expanding directions of consecutive layers become less aligned, so the forward pass as a whole stretches perturbations only moderately. The depth-averaged exponent settles early, while the per-layer exponents keep changing. This is consistent with the observation that the normalized singular value spectra of the weights stabilize early in pretraining (Zhang et al., 2026a), now seen through the Jacobians of the forward map. These are measurements on two small models with short contexts and a small prompt set. Whether the same pattern holds at scale, and whether κ predicts robustness to perturbations or the success of model editing, is part of **OP 1**.

5.2 Experiment 2: a learning-rate bifurcation diagram (clock 2)

Question and setup. How does the long-run behavior of gradient descent on a transformer change with the step size η? We train a one-block transformer (two heads, width 32) with full-batch gradient descent on next-token prediction of sequences from a fixed random Markov chain, for 56 step sizes between 0.02 and 4. Within one seed every step size starts from the same initialization and sees the same data, and we repeat the sweep for three seeds that change both. Full batches remove minibatch noise, so any irregularity comes from the map equation 2 itself. For each run we record the last 400 losses and the sharpness $\lambda_{\max}$ at the end.

Results. Figure 13 shows three regimes. In each seed, 48 of the 56 step sizes train. Across these 144 runs the final sharpness sits just above the stability threshold, with $\eta\lambda_{\max}/2$ having median 1.07 and range $[1.00, 1.21]$ across two decades of step size (panel b). This is the edge-of-stability behavior of Cohen et al. (2021), and in this model it holds over the entire trainable range we tested. None of the runs settles on a fixed point. Removing the slow downward trend of the loss shows what the iterate does instead (panels a and c). For $\eta < 0.05$ the fluctuations alternate in sign at every step, a period-2 orbit riding on the slow descent, and the median lag-1 autocorrelation of the detrended loss over the ten smallest step sizes is -0.99, -0.99 and -0.97 in the three seeds. As η grows the fluctuations become larger and less regular. The median lag-1 autocorrelation rises to between -0.95 and -0.92 for $0.05 \le \eta < 0.3$, between -0.74 and -0.68 for $0.3 \le \eta < 1$, and between -0.66 and -0.32 for $\eta \ge 1$. The edge of trainability depends on the seed. In seed

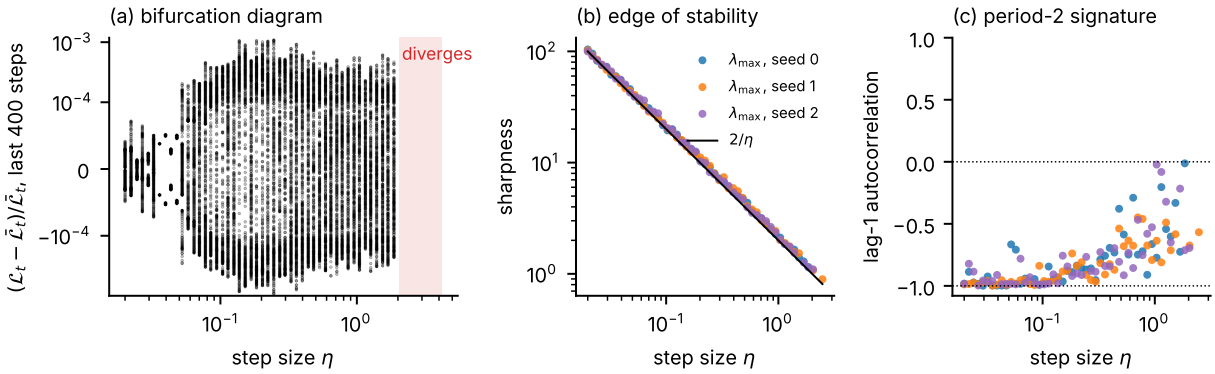


Figure 13: Experiment 2: full-batch gradient descent on a one-block transformer. (a) Bifurcation diagram for seed 0: relative deviation of the last 400 losses from their running mean (9-step window, symmetric-log axis) against the step size. Two branches at small η mark a period-2 orbit; the band at larger η is irregular oscillation; the shaded region diverges. (b) Sharpness at the end of training against $2/\eta$ for three seeds. (c) Lag-1 autocorrelation of the detrended loss for the same runs; -1 is a clean period-2 orbit.

0 every $\eta \geq 2.04$ diverges, while in seeds 1 and 2 divergent and trainable step sizes interleave between 1.5 and 2.5, and in seed 1 the largest trainable step size (2.47) lies above the smallest divergent one (1.68).

Reading. In dictionary terms, the trainable range of this model is a sequence from a period-2 orbit near the stability boundary, through increasingly irregular oscillation, to divergence. This is the picture that minimal models predict (Chen & Bruna, 2022; Zhu et al., 2022; Kong & Tao, 2020), here measured on a transformer with attention and layer normalization. The interleaving of trainable and divergent step sizes near the edge is consistent with the fractal trainability boundary reported by Sohl-Dickstein (2024), although a grid of 56 points cannot measure a fractal dimension. We do not claim a full period-doubling cascade, since with 400-step tails on top of a slowly drifting loss we cannot resolve periods above two.

Why the sharpness sits above $2/\eta$. In 26 trainable runs we continued gradient descent and computed, at consecutive iterates, the top six Hessian eigenpairs, the curvature along the step and the finite-time Lyapunov spectrum of the gradient-descent map (Appendix D.2). Two to six eigenvalues exceed $2/\eta$ at every iterate (median four), and the sharpness sits 5% to 14% above the threshold, yet the curvature averaged along the step equals $2/\eta$ to within 0.2% in every run, as the identity of Litman (2026) requires on average. The top eigenvectors at consecutive iterates are nearly orthogonal (median 86 degrees), so the top of the spectrum is a cluster whose members exchange order at every step, and the largest finite-time Lyapunov multiplier per two steps stays between 0.9998 and 1.019. The edge in this model is therefore a multi-mode regime with neutral or weakly chaotic dynamics, not a single stiff mode on a periodic orbit, and a warm-started power iteration, which averages over the alternating cluster, underestimates how far the sharpness sits above $2/\eta$ (**OP 10**).

5.3 Experiment 3: escape from repetition loops (clock 3)

Question and setup. Section 4.1 predicts that escape from a repetition loop follows an Arrhenius law with the logit gap as barrier (Proposition 1), and that loops lock if the barrier grows from cycle to cycle (Proposition 2). We test both on GPT-2 small (Radford et al., 2019) and on the final checkpoint of Pythia-160M (Biderman et al., 2023). From each of 40 prompts we decode greedily for 320 tokens and keep the runs that end in a loop. Because the context along a loop is deterministic, the logits stored during the greedy run give the exact probability of staying in the loop at any temperature, as also used by Chen & Wang (2026), and we check these exact values against Monte Carlo sampling.

Results. Greedy decoding ends in a loop for 37 of the 40 prompts in each model, with periods from 4 to 53 tokens (median 23) in GPT-2 and from 7 to 46 tokens (median 22) in Pythia-160M (Figure 14).

The barrier grows. The median smallest logit gap Δ_c rises from 0.36 in the second cycle of a loop to 2.33 in the sixth and 3.63 in the twentieth in GPT-2, and from 0.47 to 1.83 and then 2.77 by the tenth cycle in Pythia-160M (panel a). The barrier in cycle 6 is larger than in cycle 2 for 36 of 36 GPT-2 loops and 35 of 37 Pythia-160M loops. This is the self-reinforcement reported by Xu et al. (2022), measured here as the depth of a well.

Escape follows the Arrhenius law. For the first cycle after two cycles of burn-in, the escape probability per cycle approaches $e^{-\Delta_{\min}/T}$ at low temperature (panel b). Fitting $\log r$ against $1/T$ recovers the measured barrier with a correlation of 0.997 in both models and a median relative difference of 6.9% (22 GPT-2 loops) and 5.7% (21 Pythia-160M loops). Loops whose escape probability never falls below 10^{-2} on our temperature grid are excluded from the fit (panel c). The exact survival probabilities agree with Monte Carlo sampling. Across 24 comparisons per model (four loops, two temperatures, survival after 1, 2 and 4 cycles, 128 samples each), the largest absolute difference is 0.062 for GPT-2 and 0.068 for Pythia-160M, and every difference is within two binomial standard errors.

Loops lock below a fairly sharp temperature. The median probability of staying in a loop until the end of the 320-token horizon is 0.89 at $T = 0.31$ in both models, crosses 0.5 at $T \approx 0.53$ (GPT-2) and $T \approx 0.48$ (Pythia-160M), and falls to 0.06 and 0.015 at $T = 0.75$ (panel d). Free sampling gives a consistent picture. With three samples of 200 tokens per prompt and temperature, 68% (GPT-2) and 74% (Pythia-160M) of generations end in a loop at $T = 0.3$, 27% and 48% at $T = 0.5$, at most 6% at $T = 0.7$, and none at $T \geq 0.9$.

The barriers saturate, so locking depends on the horizon. Fitted from cycle 2 on, the barrier of most loops follows a saturating law, $\Delta_c = \Delta_\infty - (\Delta_\infty - \Delta_2)\, e^{-(c-2)/c_*}$, better than a logarithmic or a linear one (21 of 36 GPT-2 loops and 26 of 37 Pythia-160M loops by the corrected Akaike criterion), with saturation levels of median 4.2 and 3.6 reached within 3 to 4 cycles. The second case of Proposition 2 therefore applies: no loop locks in an unbounded generation, and the locking temperature is a finite-horizon quantity. Recomputed from the exact escape probabilities for horizons of 1 to 1024 cycles, the median $T_{1/2}$ falls from 0.67 to 0.40 in GPT-2 and from 0.62 to 0.37 in Pythia-160M, and equation 13 fits this decline with a largest deviation of 0.009 and 0.019. Appendix D.3 gives the fits and the convergence of the hidden state onto a periodic orbit, which is what makes the barriers saturate (Figure 16).

Coherent text lives in a narrow window between two failures. Raising T ends the loops but eventually produces incoherent text, so we map both failures with the same free generations (Figure 15). A generation counts as repetitive if its fraction of repeated 4-grams exceeds the 95th percentile of human continuations of 200 OpenWebText passages, and as incoherent if its mean negative log-likelihood under a separate reference model (Qwen2.5-0.5B) exceeds the 90th percentile of the human scores. Exact loops fall below one half at $T \approx 0.43$ (GPT-2) and $T \approx 0.45$ (Pythia-160M), loops and repetitive text together only at $T \approx 0.67$ and 0.72, and incoherent text rises above one half at $T \approx 0.88$ and 0.86. The coherent fraction peaks at $T = 0.8$, at 82% for GPT-2 and 71% for Pythia-160M, and is below 25% at $T = 0.6$ and at $T = 0.9$. Per prompt, the coherent range covers at most four of our twelve temperatures, typically two, and is empty for 2 of 40 prompts in GPT-2 and 5 of 40 in Pythia-160M. These proxies measure surprise and repetition, not meaning (Appendix D.3).

Larger models dig deeper wells. Repeating the loop analysis on Pythia-70M, 410M and 1B, which share training data and order with Pythia-160M (Biderman et al., 2023), the saturated barrier rises with size (medians 3.0, 3.6, 4.5 and 4.8 from 70M to 1B), and so does the temperature needed to leave a loop: the median $T_{1/2}$ for a horizon of 16 cycles is 0.43, 0.48, 0.59 and 0.64. Over a 14-fold increase in parameters the locking temperature rises by about half, while the coherent window of the larger models is wider (Figure 17).

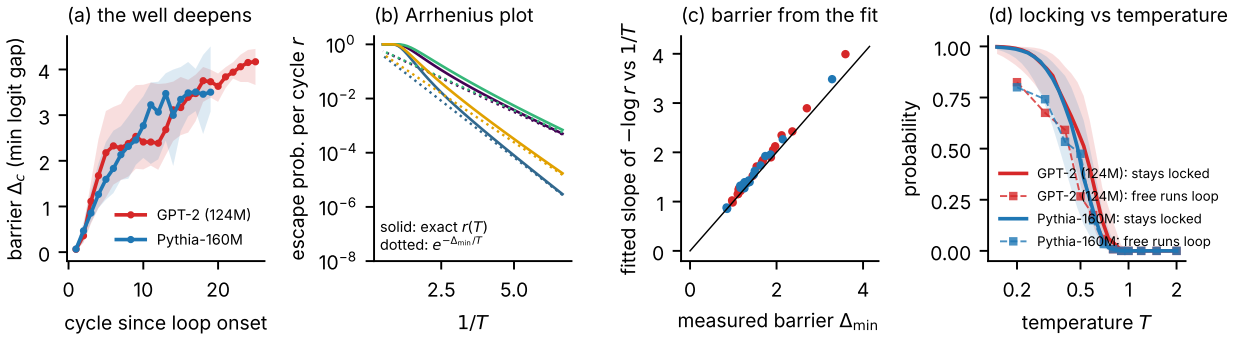


Figure 14: Experiment 3: escape from repetition loops. (a) Barrier Δ_c (smallest logit gap in cycle c) against cycles since loop onset; line: median over loops, band: interquartile range; only cycles reached by at least 10 loops are shown. (b) Exact escape probability per cycle for four GPT-2 loops (solid) and the Arrhenius term $e^{-\Delta_{\min}/T}$ (dotted). (c) Barrier recovered from the slope of $\log r$ against $1/T$ at low temperature, against the measured $\Delta_{\min}$; line: identity. (d) Solid: median probability of staying in the loop until the end of the 320-token horizon, after two cycles of burn-in (band: interquartile range); the number of remaining cycles differs between loops because their onsets differ. Dashed: fraction of free generations of 200 tokens that end in a loop (120 per temperature).

Reasoning models and the recommended temperature. Vendors of reasoning models prescribe a narrow sampling range: the DeepSeek-R1 model card asks for a temperature between 0.5 and 0.7 "to prevent endless repetitions or incoherent outputs" (DeepSeek-AI, 2025; DeepSeek-AI et al., 2025), and the Qwen3 card recommends $T = 0.6$ in thinking mode (Qwen Team, 2025; Yang et al., 2025). On the first 18 problems of GSM8K (Cobbe et al., 2021), Qwen3-0.6B in thinking mode with a budget of 1024 tokens reaches the answer in 0.28 of the traces at $T = 0.2$, 0.56 at 0.4, 0.61 at 0.6, and 0.22 at both 1.0 and 1.3 (Figure 18; Fisher exact test, p-value 0.09 for 0.6 against 0.2 and 0.04 against 1.0). The window is there, centered on the vendor's value, but its lower edge is a different failure: no trace loops exactly, and at $T = 0.2$ the model keeps returning to the same step with small rephrasings until 13 of 18 traces run out of budget. This is a recurrent orbit without a period, for which recurrence quantification (Eckmann et al., 1987; Marwan et al., 2007) is the natural tool (**OP 5**). Without thinking the model never loops, so the repetitive edge appears only in long traces. A single 0.6B model with a short budget tests the shape of the window, not the vendors' exact numbers.

Reading. The escape problem on clock 3 has the structure of Kramers' problem, with two differences that matter. The noise acts on a discrete choice through a Gibbs distribution, so the Arrhenius form follows directly from the softmax (Proposition 1) and needs no diffusion approximation. In addition, the barrier is a function of the residence time, so a loop becomes harder to leave the longer the model repeats it. Panel (a) also shows that the deepening is not uniform, and barriers often rise quickly over the first five cycles, pause, and rise again. Escape from a well whose depth grows with the time already spent in it is a non-stationary escape problem. Its mechanical analogue is a particle in a well that deepens as the particle settles, and how it scales beyond 1B parameters is open (**OP 5**).

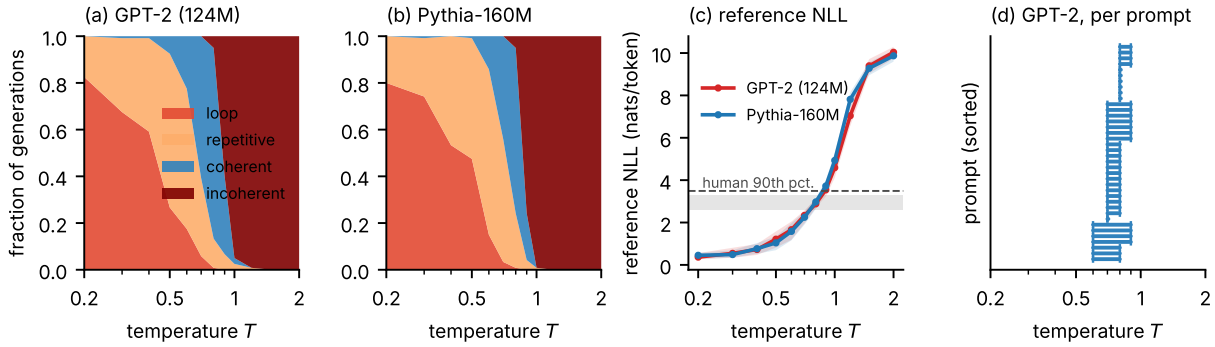


Figure 15: Experiment 3, the coherent window. (a, b) Fraction of free generations (three samples of 200 tokens for each of 40 prompts) that end in a loop, are repetitive (repeated 4-grams above the 95th percentile of human text), are incoherent (reference score above the 90th percentile of human text) or are coherent, against temperature. (c) Median reference score in nats per token (band: interquartile range); gray band: interquartile range of human text; dashed: the incoherence threshold. (d) Coherent temperature range of each GPT-2 prompt, sorted; two prompts with an empty range are not shown.

6 Open problems

The experiments answered some questions and opened others. Each problem below comes from an empty or weak cell of Table 1, from a result of Section 5, or from a place where two clocks meet, and each is stated with a first test. Problems **OP 9** and **OP 10** are conjectures that come from the mechanics of nonlinear oscillators; we include them because they make predictions that can be checked, and we do not claim evidence for them yet.

OP 1. Coupling of clocks 1 and 2. Which features of the Lyapunov profile of the forward pass settle before the loss does? In two small models the whole-stack exponent settles within 3% of training while the per-layer exponents keep growing (Experiment 1). The next test is the full Jacobian spectrum of every layer across the public checkpoints of a model family above a billion parameters.

OP 2. Periodic orbits in depth. Do models that reuse a block many times, such as looped and recurrent-depth models (Prairie et al., 2026), have periodic or quasi-periodic orbits in the residual stream, and do these orbits carry computation? Return maps of hidden states over many loop iterations would show them directly.

OP 3. The full Lyapunov spectrum of a language model. How many directions expand along a forward pass and along a generation, and what is the resulting attractor dimension? QR-based algorithms (Engelken et al., 2020; Vogt et al., 2020) make this feasible for models of GPT-2 size, and the exponent of clock 3 would turn exposure bias into a measured quantity.

OP 4. The type of a loss spike. Are the loss spikes of pretraining crossings of the stability boundary, period-doubling cascades, catapults (Lewkowycz et al., 2020) or escapes from a basin? Each type leaves a different signature in the curvature and in the update direction just before the spike, so open training logs and checkpoints can be classified by these signatures.

OP 5. Escape laws across samplers and scales. How do nucleus sampling and repetition penalties change the barrier of Proposition 1, and do the saturated barrier and the locking temperature keep growing with model size beyond 1B parameters? Reasoning models add the recurrent orbits of Experiment 3, which have no period: do they have an escape law of their own, and do recurrence plots of the trace predict when a model will run out of budget? A harder version asks whether the onset of a loop and the coherent window of a prompt can be read from the prompt and the weights before any text is generated.

OP 6. Fractal trainability of transformers. Does the fractal boundary between convergent and divergent hyperparameters (Sohl-Dickstein, 2024), reproduced for a small network in Figure 4, persist for transformers trained with Adam, and how does its dimension depend on depth? Fine grids over learning rate and warmup length give a first estimate.

OP 7. Metastability in generation. Is the length of a coherent stretch of generated text before drift a metastable residence time, exponentially long in some barrier, like the cluster states in depth (Geshkovski et al., 2024)? Survival curves of coherence against temperature would test this.

OP 8. Koopman modes of a forward pass. Can a linear operator fitted to residual-stream trajectories predict the effect of an intervention several layers ahead, and do its eigenfunctions align with known features? Dynamic mode decomposition on residual streams, compared with the tuned lens (Belrose et al., 2023), is a direct test.

OP 9. Targeted energy transfer (conjecture). In coupled mechanical oscillators, a nonlinear energy sink absorbs vibration energy from a primary system in one direction and irreversibly (Vakakis et al., 2009; Farid & Gendelman, 2016). We conjecture a similar one-way transfer of signal energy between spectral modes of the residual stream, with massive activations and attention sinks (Sun et al., 2024; Xiao et al., 2023) in the role of the sink. Tracking the energy of each singular mode of the layer Jacobians across depth and training would confirm or refute it.

OP 10. Escape under internal forcing (conjecture). At the edge of stability the iterate oscillates along stiff directions, and the other directions see this oscillation as periodic forcing. A periodically forced particle escapes from a potential well when the forcing amplitude crosses a threshold that depends on the forcing frequency (Farid & Gendelman, 2019; Farid, 2022), and we conjecture that some loss spikes are escapes of this kind. Gradient descent on a quartic well coupled to an oscillating direction is the simplest test. Experiment 2 adds a related question: self-stabilization (Damian et al., 2022) explains a single mode at the edge, but our transformer has several, which exchange order at every step, and their averaged dynamics is open.

OP 11. Early warning signals. Near a bifurcation, recovery from perturbations slows down and fluctuations grow. Do grokking, induction-head formation and loss spikes show rising autocorrelation and variance of the loss or of the gradient norm before they happen? Runs with a known transition step (Power et al., 2022; Olsson et al., 2022) allow a direct check.

OP 12. Control of clock 3. Can a controller built from the escape analysis keep generation coherent at a lower cost than repetition penalties, with Mirostat (Basu et al., 2020) and truncation sampling (Hewitt et al., 2022) as baselines? Three first tests on GPT-2 (Appendix D.3) fail in an instructive way. Heating the sampler when a loop is detected cuts looping generations at $T = 0.3$ from 68% to 6%, but 94% of the outputs remain repetitive. Rolling the text back to the first token of a loop and resampling there removes all exact loops at $T = 0.5$, but 91% of the outputs stay repetitive. A feedback controller on recurrence and surprisal settles near the center of the coherent window from any start, but its share of coherent texts (21% to 83%) does not beat the 82% of the best fixed temperature. All three act on the noise, while the trap is stored in the context. Controllers that act on the state, by editing the cache or by adding a steering vector to the residual stream (Turner et al., 2023; Zou et al., 2023), are the natural next step: in mechanical terms, a control force on the state instead of a change of the noise intensity.

7 Conclusion

A transformer language model is iterated in three ways: over layers, over optimizer steps and over generated tokens. Each iteration has been studied with tools from dynamical systems, mostly by different communities and under different names. We gave the three iterations one formal frame, reviewed the results on each, and summarized them in a dictionary that records which concepts have been proved, measured or only conjectured on which clock. In experiments on open models we found that training makes each layer of a Pythia model more expansive while the coherence of the whole forward pass falls, that gradient descent

on a small transformer stays at a multi-mode edge of stability across two decades of step size, and that repetition loops in GPT-2 and Pythia behave as wells whose depth grows while the model sits in them and then saturates, so that whether a loop locks depends on the length of the generation and on the size of the model, and coherent text lies in a narrow temperature window between repetition and incoherence. A small reasoning model has the same window, centered on its recommended temperature, and its low-temperature failure is recurrent rethinking, with no periodic loop.

A bridge between two fields can describe, explain and solve. This paper describes the three clocks in one vocabulary, shows the correspondences side by side, derives the reduced-order models behind them, and explains several behaviors that the machine learning literature had measured but not accounted for, from the Arrhenius escape from loops to the narrow window of coherent text. It does not yet solve a problem of language models that was unsolved before: our controllers for repetition find the right temperature but do not beat it. We think the third level is within reach, because many problems of language models have counterparts that dynamics has already solved, and the open problems of Section 6 point to where.

Many cells of the dictionary are empty, and several of the open problems sit where two clocks meet, such as the effect of training on the depth dynamics, the feedback of generated text into training, and the prediction of training instabilities with the tools used for escape problems in mechanics. Each of these questions has a counterpart that nonlinear dynamics has studied for decades on pendulums, ships and oscillator networks, and each can be tested with the weights and outputs of open models on a laptop. We hope that the shared vocabulary of this paper helps researchers from both communities recognize the same phenomenon when they meet it, and solve it together.

Code and data

Code to reproduce every figure and number, the raw results, and a maintained list of the papers in this review are available at `https://github.com/maorfarid1/three-clocks`. All experiments run on a CPU.

Broader impact statement

This is a review with small-scale experiments on public models, and we do not foresee direct negative consequences. A better account of repetition loops and of training instabilities could reduce wasted compute in pretraining and make generation more reliable.

Use of AI tools

An AI assistant (Claude, Anthropic) was used to search and verify the literature, draft and edit text, and write and run experiment code under the author's direction. All references were checked against arXiv or Crossref records, all numbers come from the released code, and the author is responsible for every claim in the paper.

References

Juan A. Acebrón, L. L. Bonilla, Conrad J. Pérez Vicente, Félix Ritort, and Renato Spigler. The Kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1), 2005. doi: 10.1103/RevModPhys.77.137.

Atish Agarwala, Fabian Pedregosa, and Jeffrey Pennington. Second-order regression models exhibit progressive sharpening to the edge of stability. arXiv preprint arXiv:2210.04860, 2022.

Andrea Agazzi, Giuseppe Bruno, Eloy Mosig García, Samuele Saviozzi, and Marco Romito. Stochastic Scaling Limits and Synchronization by Noise in Deep Transformer Models. *arXiv preprint arXiv:2604.26898*, 2026.

Kwangjun Ahn, Jingzhao Zhang, and Suvrit Sra. Understanding the unstable convergence of gradient descent. *arXiv preprint arXiv:2204.01050*, 2022. Published in International Conference on Machine Learning (ICML).

Kwangjun Ahn, Xiang Cheng, Hadi Daneshmand, and Suvrit Sra. Transformers learn to implement preconditioned gradient descent for in-context learning. *arXiv preprint arXiv:2306.00297*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? Investigations with linear models. *arXiv preprint arXiv:2211.15661*, 2022. Published in International Conference on Learning Representations (ICLR).

Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. Self-Consuming Generative Models Go MAD. *arXiv preprint arXiv:2307.01850*, 2023. Published in International Conference on Learning Representations (ICLR).

Julian Arnold, Flemming Holtorf, Frank Schäfer, and Niels Lörch. Phase Transitions in the Output Distribution of Large Language Models. *arXiv preprint arXiv:2405.17088*, 2024.

Sanjeev Arora, Zhiyuan Li, and Abhishek Panigrahi. Understanding Gradient Descent on Edge of Stability in Deep Learning. *arXiv preprint arXiv:2205.09745*, 2022.

Svante Arrhenius. Über die Reaktionsgeschwindigkeit bei der Inversion von Rohrzucker durch Säuren. *Zeitschrift für Physikalische Chemie*, 4U(1), 1889. doi: 10.1515/zpch-1889-0416.

Thomas Bachlechner, Bodhisattwa Prasad Majumder, Huanru Henry Mao, Garrison W. Cottrell, and Julian McAuley. ReZero is All You Need: Fast Convergence at Large Depth. *arXiv preprint arXiv:2003.04887*, 2020. Published in Conference on Uncertainty in Artificial Intelligence (UAI).

Gregor Bachmann and Vaishnavh Nagarajan. The pitfalls of next-token prediction. *arXiv preprint arXiv:2403.06963*, 2024. Published in International Conference on Machine Learning (ICML).

Krishnakumar Balasubramanian. Unbiased Gradients, Moving Stability Boundaries: Exact Mini-Batch Geometry in Linear Self-Attention. *arXiv preprint arXiv:2605.21292*, 2026.

David G. T. Barrett and Benoit Dherin. Implicit Gradient Regularization. *arXiv preprint arXiv:2009.11162*, 2020. Published in International Conference on Learning Representations (ICLR).

Sourya Basu, Govardana Sachitanandam Ramachandran, Nitish Shirish Keskar, and Lav R. Varshney. Mirostat: A Neural Text Decoding Algorithm that Directly Controls Perplexity. *arXiv preprint arXiv:2007.14966*, 2020.

Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned Lens. *arXiv preprint arXiv:2303.08112*, 2023.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. *arXiv preprint arXiv:1506.03099*, 2015. Published in Advances in Neural Information Processing Systems (NeurIPS).

Matthew Bennett, Michael F. Schatz, Heidi Rockwood, and Kurt Wiesenfeld. Huygens's clocks. *Proceedings of the Royal Society of London. Series A*, 458(2019), 2002. doi: 10.1098/rspa.2001.0888.

Raphaël Berthier. The Multiple Timescales of Gradient Descent on the Edge of Stability: A Perturbative Derivation of the Central Flow. arXiv preprint arXiv:2609.01034, 2026.

Quentin Bertrand, Avishek Joey Bose, Alexandre Duplessis, Marco Jiralerspong, and Gauthier Gidel. On the Stability of Iterative Retraining of Generative Models on their own Data. *arXiv preprint arXiv:2310.00429*, 2023. Published in International Conference on Learning Representations (ICLR).

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. *arXiv preprint arXiv:2304.01373*, 2023. Published in International Conference on Machine Learning (ICML).

Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data: Sparse identification of nonlinear dynamical systems. *arXiv preprint arXiv:1509.03580*, 2015. Published in Proceedings of the National Academy of Sciences.

Valérie Castin, Pierre Ablin, José Antonio Carrillo, and Gabriel Peyré. A Unified Perspective on the Dynamics of Deep Transformers. *arXiv preprint arXiv:2501.18322*, 2025.

Bo Chang, Minmin Chen, Eldad Haber, and Ed H. Chi. AntisymmetricRNN: A Dynamical System View on Recurrent Neural Networks. *arXiv preprint arXiv:1902.09689*, 2019. Published in International Conference on Learning Representations (ICLR).

Pratik Chaudhari and Stefano Soatto. Stochastic gradient descent performs variational inference, converges to limit cycles for deep networks. *arXiv preprint arXiv:1710.11029*, 2017. Published in Information Theory and Applications Workshop.

Angelica Chen, Ravid Shwartz-Ziv, Kyunghyun Cho, Matthew L. Leavitt, and Naomi Saphra. Sudden Drops in the Loss: Syntax Acquisition, Phase Transitions, and Simplicity Bias in MLMs. *arXiv preprint arXiv:2309.07311*, 2023. Published in International Conference on Learning Representations (ICLR).

Lei Chen and Joan Bruna. Beyond the Edge of Stability via Two-step Gradient Updates. *arXiv preprint arXiv:2206.04172*, 2022.

Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural Ordinary Differential Equations. *arXiv preprint arXiv:1806.07366*, 2018. Published in Advances in Neural Information Processing Systems (NeurIPS).

Yichuan Chen and Xin Wang. When Do Repetition Loops Lock? Truncation Sampling, Bounded Penalties, and a Blind Spot in DRY. *arXiv preprint arXiv:2610.24715*, 2026.

Lenaic Chizat and Francis Bach. On the Global Convergence of Gradient Descent for Over-parameterized Models using Optimal Transport. *arXiv preprint arXiv:1805.09545*, 2018. Published in Advances in Neural Information Processing Systems (NeurIPS).

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, Parker Schuh, Kensen Shi, Sasha Tsvyashchenko, Joshua Maynez, Abhishek Rao, Parker Barnes, Yi Tay, Noam Shazeer, Vinodkumar Prabhakaran, Emily Reif, Nan Du, Ben Hutchinson, Reiner Pope, James Bradbury, Jacob Austin, Michael Isard, Guy Gur-Ari, Pengcheng Yin, Toju Duke, Anselm Levskaya, Sanjay Ghemawat, Sunipa Dev, Henryk Michalewski, Xavier Garcia, Vedant Misra, Kevin Robinson, Liam Fedus, Denny Zhou, Daphne Ippolito, David Luan, Hyeontaek Lim, Barret Zoph, Alexander Spiridonov, Ryan Sepassi, David Dohan, Shivani Agrawal, Mark Omernick, Andrew M. Dai, Thanumalayan Sankaranarayana Pillai, Marie Pellat, Aitor Lewkowycz, Erica Moreira, Rewon Child, Oleksandr Polozov, Katherine Lee, Zongwei Zhou, Xuezhi Wang, Brennan Saeta, Mark Diaz, Orhan Firat, Michele Catasta, Jason Wei, Kathy Meier-Hellstern, Douglas Eck, Jeff Dean, Slav Petrov, and Noah Fiedel. PaLM: Scaling Language Modeling with Pathways. *arXiv preprint arXiv:2204.02311*, 2022. Published in Journal of Machine Learning Research.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, et al. Training Verifiers to Solve Math Word Problems. arXiv preprint arXiv:2110.14168, 2021.

Jeremy M. Cohen, Simran Kaur, Yuanzhi Li, J. Zico Kolter, and Ameet Talwalkar. Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability. *arXiv preprint arXiv:2103.00065*, 2021. Published in International Conference on Learning Representations (ICLR).

Jeremy M. Cohen, Behrooz Ghorbani, Shankar Krishnan, Naman Agarwal, Sourabh Medapati, Michal Badura, Daniel Suo, David Cardoze, Zachary Nado, George E. Dahl, and Justin Gilmer. Adaptive Gradient Methods at the Edge of Stability. *arXiv preprint arXiv:2207.14484*, 2022.

Jeremy M. Cohen, Alex Damian, Ameet Talwalkar, J. Zico Kolter, and Jason D. Lee. Understanding Optimization in Deep Learning with Central Flows. arXiv preprint arXiv:2410.24206, 2024.

Aditya Cowsik, Tamra Nebabu, Xiao-Liang Qi, and Surya Ganguli. Geometric Dynamics of Signal Propagation Predict Trainability of Transformers. *arXiv preprint arXiv:2403.02579*, 2024. Published in Physical Review E.

Alex Damian, Eshaan Nichani, and Jason D. Lee. Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability. *arXiv preprint arXiv:2209.15594*, 2022. Published in International Conference on Learning Representations (ICLR).

Yann Dauphin, Razvan Pascanu, Caglar Gulcehre, Kyunghyun Cho, Surya Ganguli, and Yoshua Bengio. Identifying and attacking the saddle point problem in high-dimensional non-convex optimization. *arXiv preprint arXiv:1406.2572*, 2014. Published in Advances in Neural Information Processing Systems (NeurIPS).

DeepSeek-AI. DeepSeek-R1 model card, usage recommendations. https://huggingface.co/deepseek-ai/DeepSeek-R1, accessed November 2026, 2025.

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645, 2025. doi: 10.1038/s41586-025-09422-z. arXiv:2501.12948.

James H. Dieterich. Time-dependent friction in rocks. *Journal of Geophysical Research*, 77(20), 1972. doi: 10.1029/JB077i020p03690.

Fatih Dinc, Ege Cirakman, Bariscan Kurtkaya, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer, and Hidenori Tanaka. A ghost mechanism: An analytical model of abrupt learning in recurrent networks. *arXiv preprint arXiv:2501.02378*, 2025. Published in Physical Review X.

Akshunna S. Dogra and William T Redman. Optimizing Neural Networks via Koopman Operator Theory. *arXiv preprint arXiv:2006.02361*, 2020. Published in Advances in Neural Information Processing Systems (NeurIPS).

Yihe Dong, Jean-Baptiste Cordonnier, and Andreas Loukas. Attention is Not All You Need: Pure Attention Loses Rank Doubly Exponentially with Depth. *arXiv preprint arXiv:2103.03404*, 2021. Published in International Conference on Machine Learning (ICML).

Mitia Duerinckx, Borjan Geshkovski, and Stefano Rossi. Kinetic theory for Transformers and the lost-in-the-middle phenomenon. *arXiv preprint arXiv:2605.09213*, 2026.

Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. Faith and Fate: Limits of Transformers on Compositionality. *arXiv preprint arXiv:2305.18654*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Weinan E. A Proposal on Machine Learning via Dynamical Systems. *Communications in Mathematics and Statistics*, 5(1), 2017. doi: 10.1007/s40304-017-0103-z.

Weinan E, Jiequn Han, and Qianxiao Li. A Mean-Field Optimal Control Formulation of Deep Learning. *arXiv preprint arXiv:1807.01083*, 2018. Published in Research in the Mathematical Sciences.

Jean-Pierre Eckmann, Sylvie Oliffson Kamphorst, and David Ruelle. Recurrence plots of dynamical systems. *Europhysics Letters*, 4(9), 1987. doi: 10.1209/0295-5075/4/9/006.

Rainer Engelken, Fred Wolf, and L. F. Abbott. Lyapunov spectra of chaotic recurrent neural networks. *arXiv preprint arXiv:2006.02427*, 2020. Published in Physical Review Research.

N. Benjamin Erichson, Omri Azencot, Alejandro Queiruga, Liam Hodgkinson, and Michael W. Mahoney. Lipschitz Recurrent Neural Networks. *arXiv preprint arXiv:2006.12070*, 2020. Published in International Conference on Learning Representations (ICLR).

Maor Farid. Escape dynamics of a particle from a purely nonlinear truncated quartic potential well under harmonic excitation. *Nonlinear Dynamics*, 2022. doi: 10.1007/s11071-022-07976-5.

Maor Farid. Unsupervised data-driven response regime exploration and identification for dynamical systems. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 2024. doi: 10.1063/5.0173938.

Maor Farid and Oleg V. Gendelman. Tuned pendulum as nonlinear energy sink for broad energy range. *Journal of Vibration and Control*, 2016. doi: 10.1177/1077546315578561.

Maor Farid and Oleg V. Gendelman. Response regimes in equivalent mechanical model of strongly nonlinear liquid sloshing. *International Journal of Non-Linear Mechanics*, 94, 2017. doi: 10.1016/j.ijnonlinmec.2017.04.006.

Maor Farid and Oleg V. Gendelman. Escape of a forced-damped particle from weakly nonlinear truncated potential well. *arXiv preprint arXiv:1910.08545*, 2019.

Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19, 1978. doi: 10.1007/BF01020332.

Jesseba Fernando and Grigori Guitchounts. Transformer Dynamics: A neuroscientific approach to interpretability of large language models. *arXiv preprint arXiv:2502.12131*, 2025.

Jesseba Fernando and Grigori Guitchounts. Dynamics of the Transformer Residual Stream: Coupling Spectral Geometry to Network Topology. *arXiv preprint arXiv:2605.14258*, 2026.

Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear Mode Connectivity and the Lottery Ticket Hypothesis. *arXiv preprint arXiv:1912.05671*, 2019. Published in International Conference on Machine Learning (ICML).

Zihao Fu, Wai Lam, Anthony Man-Cho So, and Bei Shi. A Theoretical Analysis of the Repetition Problem in Text Generation. *arXiv preprint arXiv:2012.14660*, 2020. Published in AAAI Conference on Artificial Intelligence.

Eric Gan. A Bifurcation Theory Framework for Gradient Descent on the Edge of Stability. arXiv preprint arXiv:2606.15551, 2026.

Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What Can Transformers Learn In-Context? A Case Study of Simple Function Classes. *arXiv preprint arXiv:2208.01066*, 2022. Published in Advances in Neural Information Processing Systems (NeurIPS).

Rong Ge, Furong Huang, Chi Jin, and Yang Yuan. Escaping From Saddle Points: Online Stochastic Gradient for Tensor Decomposition. *arXiv preprint arXiv:1503.02101*, 2015. Published in Conference on Learning Theory (COLT).

Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. The emergence of clusters in self-attention dynamics. *arXiv preprint arXiv:2305.05465*, 2023a. Published in Advances in Neural Information Processing Systems (NeurIPS).

Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. A mathematical perspective on Transformers. *arXiv preprint arXiv:2312.10794*, 2023b. Published in Bulletin of the American Mathematical Society.

Borjan Geshkovski, Hugo Koubbi, Yury Polyanskiy, and Philippe Rigollet. Dynamic metastability in the self-attention model. *arXiv preprint arXiv:2410.06833*, 2024.

Behrooz Ghorbani, Shankar Krishnan, and Ying Xiao. An Investigation into Neural Net Optimization via Hessian Eigenvalue Density. *arXiv preprint arXiv:1901.10159*, 2019. Published in International Conference on Machine Learning (ICML).

Justin Gilmer, Behrooz Ghorbani, Ankush Garg, Sneha Kudugunta, Behnam Neyshabur, David Cardoze, George Dahl, Zachary Nado, and Orhan Firat. A Loss Curvature Perspective on Training Instability in Deep Learning. *arXiv preprint arXiv:2110.04369*, 2021.

William Gilpin. Chaos as an interpretable benchmark for forecasting and data-driven modelling. *arXiv preprint arXiv:2110.05266*, 2021. Published in NeurIPS Datasets and Benchmarks Track.

William Gilpin. Model scale versus domain knowledge in statistical forecasting of chaotic systems. *arXiv preprint arXiv:2303.08011*, 2023. Published in Physical Review Research.

Peter Goldreich and Stanton Peale. Spin-orbit coupling in the solar system. *The Astronomical Journal*, 71:425, 1966. doi: 10.1086/109947.

Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large Language Models Are Zero-Shot Time Series Forecasters. *arXiv preprint arXiv:2310.07820*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Albert Gu and Tri Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv preprint arXiv:2312.00752*, 2023.

Albert Gu, Karan Goel, and Christopher Ré. Efficiently Modeling Long Sequences with Structured State Spaces. *arXiv preprint arXiv:2111.00396*, 2021. Published in International Conference on Learning Representations (ICLR).

Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient Descent Happens in a Tiny Subspace. *arXiv preprint arXiv:1812.04754*, 2018.

Eldad Haber and Lars Ruthotto. Stable Architectures for Deep Neural Networks. *arXiv preprint arXiv:1705.03341*, 2017. Inverse Problems, Volume 34, Number 1 Inverse Problems, Volume 34, Number 1, 2017.

Peter Hänggi, Peter Talkner, and Michal Borkovec. Reaction-rate theory: fifty years after Kramers. *Reviews of Modern Physics*, 62(2), 1990. doi: 10.1103/RevModPhys.62.251.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. *arXiv preprint arXiv:1512.03385*, 2015. Published in IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

John Hewitt, Christopher D. Manning, and Percy Liang. Truncation Sampling as Language Model Desmoothing. *arXiv preprint arXiv:2210.15191*, 2022. Published in Conference on Empirical Methods in Natural Language Processing (EMNLP).

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training Compute-Optimal Large Language Models. *arXiv preprint arXiv:2203.15556*, 2022. Published in Advances in Neural Information Processing Systems (NeurIPS).

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The Curious Case of Neural Text Degeneration. *arXiv preprint arXiv:1904.09751*, 2019. Published in International Conference on Learning Representations (ICLR).

Jesse Hoogland, George Wang, Matthew Farrugia-Roberts, Liam Carroll, Susan Wei, and Daniel Murfet. Loss Landscape Degeneracy and Stagewise Development in Transformers. *arXiv preprint arXiv:2402.02364*, 2024. Published in Transactions on Machine Learning Research.

Gregorio Jaca, Kristóf Benedek, and János Török. Chaos in reason: How chain-of-thought LLMs can look for an answer. *arXiv preprint arXiv:2607.27805*, 2026.

Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural Tangent Kernel: Convergence and Generalization in Neural Networks. *arXiv preprint arXiv:1806.07572*, 2018. Published in Advances in Neural Information Processing Systems (NeurIPS).

Stanisław Jastrzębski, Zachary Kenton, Devansh Arpit, Nicolas Ballas, Asja Fischer, Yoshua Bengio, and Amos Storkey. Three Factors Influencing Minima in SGD. *arXiv preprint arXiv:1711.04623*, 2017.

Chi Jin, Rong Ge, Praneeth Netrapalli, Sham M. Kakade, and Michael I. Jordan. How to Escape Saddle Points Efficiently. *arXiv preprint arXiv:1703.00887*, 2017. Published in International Conference on Machine Learning (ICML).

Timothy D. Jones. Chaos and bifurcations in continuous language-model feedback. *arXiv preprint arXiv:2610.31692*, 2026.

Dayal Singh Kalra and Maissam Barkeshli. Phase diagram of early training dynamics in deep neural networks: effect of the learning rate, depth, and width. *arXiv preprint arXiv:2302.12250*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Dayal Singh Kalra, Tianyu He, and Maissam Barkeshli. Universal Sharpness Dynamics in Neural Network Training: Fixed Point Analysis, Edge of Stability, and Route to Chaos. arXiv preprint arXiv:2311.02076, 2023.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361*, 2020.

Nikita Karagodin, Yury Polyanskiy, and Philippe Rigollet. Clustering in Causal Attention Masking. *arXiv preprint arXiv:2411.04990*, 2024. Published in Advances in Neural Information Processing Systems (NeurIPS).

Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. *arXiv preprint arXiv:2006.16236*, 2020. Published in International Conference on Machine Learning (ICML).

Nitish Shirish Keskar, Bryan McCann, Lav R. Varshney, Caiming Xiong, and Richard Socher. CTRL: A Conditional Transformer Language Model for Controllable Generation. arXiv preprint arXiv:1909.05858, 2019.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the Effects of RLHF on LLM Generalisation and Diversity. *arXiv preprint arXiv:2310.06452*, 2023. Published in International Conference on Learning Representations (ICLR).

Lingkai Kong and Molei Tao. Stochasticity of Deterministic Gradient Descent: Large Learning Rate for Multiscale Objective Function. *arXiv preprint arXiv:2002.06189*, 2020. Published in Advances in Neural Information Processing Systems (NeurIPS).

B. O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5), 1931. doi: 10.1073/pnas.17.5.315.

H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4), 1940. doi: 10.1016/S0031-8914(40)90098-2.

Vikram Krishnamurthy, Yiming Zhang, and Luke Snow. Stochastic Dynamics of Generative AI: From Autoregressive Models to Interacting LLM Networks and Neural SDEs. *arXiv preprint arXiv:2611.07065*, 2026.

Christian Kuehn and Jaeyoung Yoon. Spectral Selection in Symmetric Self-Attention Dynamics. *arXiv preprint arXiv:2604.26085*, 2026.

Tanishq Kumar, Blake Bordelon, Samuel J. Gershman, and Cengiz Pehlevan. Grokking as the Transition from Lazy to Rich Training Dynamics. *arXiv preprint arXiv:2310.06110*, 2023. Published in International Conference on Learning Representations (ICLR).

Daniel Kunin, Javier Sagastuy-Brena, Surya Ganguli, Daniel L. K. Yamins, and Hidenori Tanaka. Neural Mechanics: Symmetry and Broken Conservation Laws in Deep Learning Dynamics. *arXiv preprint arXiv:2012.04728*, 2020. Published in International Conference on Learning Representations (ICLR).

Yoshiki Kuramoto. Self-entrainment of a population of coupled non-linear oscillators. In *International Symposium on Mathematical Problems in Theoretical Physics*, volume 39 of *Lecture Notes in Physics*. Springer, 1975. doi: 10.1007/BFb0013365.

Vedang Lad, Jin Hwa Lee, Wes Gurnee, and Max Tegmark. The Remarkable Robustness of LLMs: Stages of Inference? *arXiv preprint arXiv:2406.19384*, 2024.

Itay Lavie, Kirsten Fischer, Andrey Lekov, Frederic Van Maele, Zohar Ringel, and Moritz Helias. Phase Transitions in Attention: A Bayesian Theory of Copy Head Emergence. *arXiv preprint arXiv:2606.12058*, 2026.

Aitor Lewkowycz, Yasaman Bahri, Ethan Dyer, Jascha Sohl-Dickstein, and Guy Gur-Ari. The large learning rate phase of deep learning: the catapult mechanism. *arXiv preprint arXiv:2003.02218*, 2020.

Qianxiao Li, Cheng Tai, and Weinan E. Stochastic modified equations and adaptive stochastic gradient algorithms. *arXiv preprint arXiv:1511.06251*, 2015. Published in International Conference on Machine Learning (ICML).

Sixu Li, Thomas Jacob Maranzatto, Jan Peszek, Trevor Teolis, Semih Akkoc, Konstantin Riedl, Sennur Ulukus, and Nicolás García Trillos. On the Diverse Dynamical Behaviors Arising in Deep Linear Transformers. *arXiv preprint arXiv:2607.18584*, 2026.

Elon Litman. The Origin of Edge of Stability. arXiv preprint arXiv:2604.20446, 2026.

Liyuan Liu, Xiaodong Liu, Jianfeng Gao, Weizhu Chen, and Jiawei Han. Understanding the Difficulty of Training Transformers. *arXiv preprint arXiv:2004.08249*, 2020. Published in Conference on Empirical Methods in Natural Language Processing (EMNLP).

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the Middle: How Language Models Use Long Contexts. *arXiv preprint arXiv:2307.03172*, 2023. Published in Transactions of the Association for Computational Linguistics.

Toni J. B. Liu, Nicolas Boullé, Raphaël Sarfati, and Christopher J. Earls. LLMs learn governing principles of dynamical systems, revealing an in-context neural scaling law. *arXiv preprint arXiv:2402.00795*, 2024. Published in Conference on Empirical Methods in Natural Language Processing (EMNLP).

Ziming Liu, Ouail Kitouni, Niklas Nolte, Eric J. Michaud, Max Tegmark, and Mike Williams. Towards Understanding Grokking: An Effective Theory of Representation Learning. *arXiv preprint arXiv:2205.10343*, 2022a. Published in Advances in Neural Information Processing Systems (NeurIPS).

Ziming Liu, Eric J. Michaud, and Max Tegmark. Omnigrok: Grokking Beyond Algorithmic Data. *arXiv preprint arXiv:2210.01117*, 2022b. Published in International Conference on Learning Representations (ICLR).

Edward N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20, 1963. doi: 10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2.

Yiping Lu, Aoxiao Zhong, Quanzheng Li, and Bin Dong. Beyond Finite Layer Neural Networks: Bridging Deep Architectures and Numerical Differential Equations. *arXiv preprint arXiv:1710.10121*, 2017. Published in International Conference on Machine Learning (ICML).

Yiping Lu, Zhuohan Li, Di He, Zhiqing Sun, Bin Dong, Tao Qin, Liwei Wang, and Tie-Yan Liu. Understanding and Improving Transformer From a Multi-Particle Dynamic System Point of View. *arXiv preprint arXiv:1906.02762*, 2019.

Bethany Lusch, J. Nathan Kutz, and Steven L. Brunton. Deep learning for universal linear embeddings of nonlinear dynamics. *arXiv preprint arXiv:1712.09707*, 2017. Published in Nature Communications.

Kaifeng Lyu, Jikai Jin, Zhiyuan Li, Simon S. Du, Jason D. Lee, and Wei Hu. Dichotomy of Early and Late Phase Implicit Biases Can Provably Induce Grokking. *arXiv preprint arXiv:2311.18817*, 2023. Published in International Conference on Learning Representations (ICLR).

Niru Maheswaranathan, Alex Williams, Matthew D. Golub, Surya Ganguli, and David Sussillo. Reverse engineering recurrent networks for sentiment classification reveals line attractor dynamics. *arXiv preprint arXiv:1906.10720*, 2019. Published in Advances in Neural Information Processing Systems (NeurIPS).

Stephan Mandt, Matthew D. Hoffman, and David M. Blei. Stochastic Gradient Descent as Approximate Bayesian Inference. *arXiv preprint arXiv:1704.04289*, 2017. Published in Journal of Machine Learning Research.

Thomas Jacob Maranzatto, Semih Akkoc, and Sennur Ulukus. The Role of Feed-Forward Layers in Transformer Dynamics. *arXiv preprint arXiv:2609.36230*, 2026.

Pierre Marion. Edge Flow: A Tractable and Predictive Continuous-Time Model for Gradient Descent at the Edge of Stability. arXiv preprint arXiv:2606.18080, 2026.

Norbert Marwan, M. Carmen Romano, Marco Thiel, and Jürgen Kurths. Recurrence plots for the analysis of complex systems. *Physics Reports*, 438, 2007. doi: 10.1016/j.physrep.2006.11.001.

Robert M. May. Simple mathematical models with very complicated dynamics. *Nature*, 261, 1976. doi: 10.1038/261459a0.

Steven W. McDonald, Celso Grebogi, Edward Ott, and James A. Yorke. Fractal basin boundaries. *Physica D: Nonlinear Phenomena*, 17(2), 1985. doi: 10.1016/0167-2789(85)90001-6.

Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A Mean Field View of the Landscape of Two-Layers Neural Networks. *arXiv preprint arXiv:1804.06561*, 2018. Published in Proceedings of the National Academy of Sciences.

Clara Meister, Tiago Pimentel, Gian Wiher, and Ryan Cotterell. Locally Typical Sampling. *arXiv preprint arXiv:2202.00666*, 2022. Published in Transactions of the Association for Computational Linguistics.

Igor Mezić. Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41, 2005. doi: 10.1007/s11071-005-2824-x.

Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The Quantization Model of Neural Scaling. *arXiv preprint arXiv:2303.13506*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Jonas M. Mikhaeil, Zahra Monfared, and Daniel Durstewitz. On the difficulty of learning chaotic dynamics with RNNs. *arXiv preprint arXiv:2110.07238*, 2021. Published in Advances in Neural Information Processing Systems (NeurIPS).

Igor Molybog, Peter Albert, Moya Chen, Zachary DeVito, David Esiobu, Naman Goyal, Punit Singh Koura, Sharan Narang, Andrew Poulton, Ruan Silva, Binh Tang, Diana Liskovich, Puxin Xu, Yuchen Zhang, Melanie Kambadur, Stephen Roller, and Susan Zhang. A Theory on Adam Instability in Large-Scale Machine Learning. *arXiv preprint arXiv:2304.09871*, 2023.

Takashi Mori, Liu Ziyin, Kangqiao Liu, and Masahito Ueda. Power-law escape rate of SGD. *arXiv preprint arXiv:2105.09557*, 2021. Published in International Conference on Machine Learning (ICML).

Rotem Mulayoff and Sebastian U. Stich. On the Stability of Nonlinear Dynamics in GD and SGD: Beyond Quadratic Potentials. arXiv preprint arXiv:2602.14789, 2026.

Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Phase transition in large language models and the criticality of natural languages. *arXiv preprint arXiv:2406.05335*, 2024.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023. Published in International Conference on Learning Representations (ICLR).

Lorenzo Noci, Sotiris Anagnostidis, Luca Biggio, Antonio Orvieto, Sidak Pal Singh, and Aurelien Lucchi. Signal Propagation in Transformers: Theoretical Perspectives and the Role of Rank Collapse. *arXiv preprint arXiv:2206.03126*, 2022. Published in Advances in Neural Information Processing Systems (NeurIPS).

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context Learning and Induction Heads. *arXiv preprint arXiv:2209.11895*, 2022.

Antonio Orvieto, Samuel L Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, and Soham De. Resurrecting Recurrent Neural Networks for Long Sequences. *arXiv preprint arXiv:2303.06349*, 2023. Published in International Conference on Machine Learning (ICML).

Mitchell Ostrow, Daoyuan Qian, Toni Liu, and Ila Fiete. A Mathematical Theory of the Jacobian Lens. *arXiv preprint arXiv:2610.31955*, 2026.

James Pantaleone. Synchronization of metronomes. *American Journal of Physics*, 70(10), 2002. doi: 10.1119/1.1501118.

Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of Neural Collapse during the terminal phase of deep learning training. *arXiv preprint arXiv:2008.08186*, 2020. Published in Proceedings of the National Academy of Sciences.

Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. On the difficulty of training Recurrent Neural Networks. *arXiv preprint arXiv:1211.5063*, 2012. Published in International Conference on Machine Learning (ICML).

Jaideep Pathak, Brian Hunt, Michelle Girvan, Zhixin Lu, and Edward Ott. Model-Free Prediction of Large Spatiotemporally Chaotic Systems from Data: A Reservoir Computing Approach. *Physical Review Letters*, 120(2): 024102, 2018. doi: 10.1103/PhysRevLett.120.024102.

Max Peeperkorn, Tom Kouwenhoven, Dan Brown, and Anna Jordanous. Is Temperature the Creativity Parameter of Large Language Models? *arXiv preprint arXiv:2405.00492*, 2024. Published in International Conference on Computational Creativity (ICCC).

Jeffrey Pennington, Samuel S. Schoenholz, and Surya Ganguli. Resurrecting the sigmoid in deep learning through dynamical isometry: theory and practice. *arXiv preprint arXiv:1711.04735*, 2017. Published in Advances in Neural Information Processing Systems (NeurIPS).

Quoc Tuan Pham, Mehdi Jafari, and Flora Salim. Is my model "mind blurting"? Interpreting the dynamics of reasoning tokens with recurrence quantification analysis (RQA). arXiv preprint arXiv:2602.06266, 2026.

Charilaos Pipis, Shivam Garg, Vasilis Kontonis, Vaishnavi Shrivastava, Akshay Krishnamurthy, and Dimitris Papailiopoulos. Wait, Wait, Wait... Why Do Reasoning Models Loop? arXiv preprint arXiv:2512.12895, 2025.

Ben Poole, Subhaneil Lahiri, Maithra Raghu, Jascha Sohl-Dickstein, and Surya Ganguli. Exponential expressivity in deep neural networks through transient chaos. *arXiv preprint arXiv:1606.05340*, 2016. Published in Advances in Neural Information Processing Systems (NeurIPS).

Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. *arXiv preprint arXiv:2201.02177*, 2022.

Hayden Prairie, Zachary Novack, Taylor Berg-Kirkpatrick, and Daniel Y. Fu. Parcae: Scaling Laws For Stable Looped Language Models. *arXiv preprint arXiv:2604.12946*, 2026.

Qwen Team. Qwen2.5 Technical Report. arXiv preprint arXiv:2412.15115, 2024.

Qwen Team. Qwen3-0.6B model card, best practices. https://huggingface.co/Qwen/Qwen3-0.6B, accessed November 2026, 2025.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language Models are Unsupervised Multitask Learners. OpenAI technical report, 2019.

Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A. Hamprecht, Yoshua Bengio, and Aaron Courville. On the Spectral Bias of Neural Networks. *arXiv preprint arXiv:1806.08734*, 2018. Published in International Conference on Machine Learning (ICML).

Sameera Ramasinghe, Ajanthan Thalaiyasingam, Hadi Mohaghegh Dolatabadi, Chamin Hewa Koneputugodage, Gil Avraham, Violetta Shevchenko, Yan Zuo, Karol Pajak, and Alexander Long. Taming Curvature: Architecture Warm-Up for Stable Transformer Training. *arXiv preprint arXiv:2606.16768*, 2026.

Gautam Reddy. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. *arXiv preprint arXiv:2312.03002*, 2023. Published in International Conference on Learning Representations (ICLR).

William T. Redman, Maria Fonoberova, Ryan Mohr, Ioannis G. Kevrekidis, and Igor Mezic. An Operator Theoretic View on Pruning Deep Neural Networks. *arXiv preprint arXiv:2110.14856*, 2021. Published in International Conference on Learning Representations (ICLR).

Matthew Renze and Erhan Guven. The Effect of Sampling Temperature on Problem Solving in Large Language Models. *arXiv preprint arXiv:2402.05201*, 2024. Published in Conference on Empirical Methods in Natural Language Processing (EMNLP).

Antônio H. Ribeiro, Koen Tiels, Luis A. Aguirre, and Thomas B. Schön. Beyond exploding and vanishing gradients: analysing RNN training using attractors and smoothness. *arXiv preprint arXiv:1906.08482*, 2019. Published in International Conference on Artificial Intelligence and Statistics (AISTATS).

Clarence W. Rowley, Igor Mezić, Shervin Bagheri, Philipp Schlatter, and Dan S. Henningson. Spectral analysis of nonlinear flows. *Journal of Fluid Mechanics*, 641, 2009. doi: 10.1017/S0022112009992059.

Noa Rubin, Inbar Seroussi, and Zohar Ringel. Grokking as a First Order Phase Transition in Two Layer Networks. *arXiv preprint arXiv:2310.03789*, 2023. Published in International Conference on Learning Representations (ICLR).

T. Konstantin Rusch and Siddhartha Mishra. Coupled Oscillatory Recurrent Neural Network (coRNN): An accurate and (gradient) stable architecture for learning long time dependencies. *arXiv preprint arXiv:2010.00951*, 2020. Published in International Conference on Learning Representations (ICLR).

Lars Ruthotto and Eldad Haber. Deep Neural Networks Motivated by Partial Differential Equations. *arXiv preprint arXiv:1804.04272*, 2018. Published in Journal of Mathematical Imaging and Vision.

Levent Sagun, Utku Evci, V. Ugur Guney, Yann Dauphin, and Leon Bottou. Empirical Analysis of the Hessian of Over-Parametrized Neural Networks. *arXiv preprint arXiv:1706.04454*, 2017. Published in International Conference on Learning Representations (ICLR).

Michael E. Sander, Pierre Ablin, Mathieu Blondel, and Gabriel Peyré. Momentum Residual Neural Networks. *arXiv preprint arXiv:2102.07870*, 2021a. Published in International Conference on Machine Learning (ICML).

Michael E. Sander, Pierre Ablin, Mathieu Blondel, and Gabriel Peyré. Sinkformers: Transformers with Doubly Stochastic Attention. *arXiv preprint arXiv:2110.11773*, 2021b. Published in International Conference on Artificial Intelligence and Statistics (AISTATS).

Michael E. Sander, Pierre Ablin, and Gabriel Peyré. Do Residual Neural Networks discretize Neural Ordinary Differential Equations? *arXiv preprint arXiv:2205.14612*, 2022. Published in Advances in Neural Information Processing Systems (NeurIPS).

Raphaël Sarfati, Toni J. B. Liu, Nicolas Boullé, and Christopher J. Earls. Lines of Thought in Large Language Models. *arXiv preprint arXiv:2410.01545*, 2024. Published in International Conference on Learning Representations (ICLR).

Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. *arXiv preprint arXiv:1312.6120*, 2013. Published in International Conference on Learning Representations (ICLR).

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are Emergent Abilities of Large Language Models a Mirage? *arXiv preprint arXiv:2304.15004*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear Transformers Are Secretly Fast Weight Programmers. *arXiv preprint arXiv:2102.11174*, 2021. Published in International Conference on Machine Learning (ICML).

Samuel S. Schoenholz, Justin Gilmer, Surya Ganguli, and Jascha Sohl-Dickstein. Deep Information Propagation. *arXiv preprint arXiv:1611.01232*, 2016. Published in International Conference on Learning Representations (ICLR).

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The Curse of Recursion: Training on Generated Data Makes Models Forget. *arXiv preprint arXiv:2305.17493*, 2023.

Umut Simsekli, Levent Sagun, and Mert Gurbuzbalaban. A Tail-Index Analysis of Stochastic Gradient Noise in Deep Neural Networks. *arXiv preprint arXiv:1901.06053*, 2019. Published in International Conference on Machine Learning (ICML).

Aaditya K. Singh, Stephanie C. Y. Chan, Ted Moskovitz, Erin Grant, Andrew M. Saxe, and Felix Hill. The Transient Nature of Emergent In-Context Learning in Transformers. *arXiv preprint arXiv:2311.08360*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Samuel L. Smith and Quoc V. Le. A Bayesian Perspective on Generalization and Stochastic Gradient Descent. *arXiv preprint arXiv:1710.06451*, 2017. Published in International Conference on Learning Representations (ICLR).

Jascha Sohl-Dickstein. The boundary of neural network trainability is fractal. *arXiv preprint arXiv:2402.06184*, 2024.

H. Sompolinsky, A. Crisanti, and H. J. Sommers. Chaos in Random Neural Networks. *Physical Review Letters*, 61 (3), 1988. doi: 10.1103/PhysRevLett.61.259.

Minhak Song and Chulhee Yun. Trajectory Alignment: Understanding the Edge of Stability Phenomenon via Bifurcation Theory. arXiv preprint arXiv:2307.04204, 2023.

L. Storm, H. Linander, J. Bec, K. Gustavsson, and B. Mehlig. Finite-time Lyapunov exponents of deep neural networks. *arXiv preprint arXiv:2306.12548*, 2023. Published in Physical Review Letters.

Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press, 2 edition, 2018. doi: 10.1201/9780429492563.

Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive Activations in Large Language Models. *arXiv preprint arXiv:2402.17762*, 2024.

David Sussillo and Omri Barak. Opening the Black Box: Low-Dimensional Dynamics in High-Dimensional Recurrent Neural Networks. *Neural Computation*, 25(3), 2013. doi: 10.1162/NECO_a_00409.

Sho Takase, Shun Kiyono, Sosuke Kobayashi, and Jun Suzuki. Spike No More: Stabilizing the Pre-training of Large Language Models. *arXiv preprint arXiv:2312.16903*, 2023.

Vimal Thilak, Etai Littwin, Shuangfei Zhai, Omid Saremi, Roni Paiss, and Joshua Susskind. The Slingshot Mechanism: An Empirical Study of Adaptive Optimizers and the Grokking Phenomenon. *arXiv preprint arXiv:2206.04817*, 2022.

J. M. T. Thompson. Designing against capsize in beam seas: Recent advances and new insights. *Applied Mechanics Reviews*, 50(5), 1997. doi: 10.1115/1.3101710.

J. M. T. Thompson and M. S. Soliman. Fractal control boundaries of driven oscillators and their relevance to safe engineering design. *Proceedings of the Royal Society of London. Series A*, 428(1874), 1990. doi: 10.1098/rspa.1990.0022.

Mario Tuci, Caner Korkmaz, Umut Şimşekli, and Tolga Birdal. Generalization at the Edge of Stability. *arXiv preprint arXiv:2604.19740*, 2026.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, et al. Steering language models with activation engineering. arXiv preprint arXiv:2308.10248, 2023.

Varun Vaidya. Learning-Induced Dynamical Transition in Recurrent Neural Networks. *arXiv preprint arXiv:2609.19288*, 2026.

Alexander F. Vakakis, Oleg V. Gendelman, Lawrence A. Bergman, D. Michael McFarland, Gaetan Kerschen, and Young Sup Lee. *Nonlinear Targeted Energy Transfer in Mechanical and Structural Systems*, volume 156 of *Solid Mechanics and Its Applications*. Springer, 2009. doi: 10.1007/978-1-4020-9130-8.

Lucrezia Valeriani, Diego Doimo, Francesca Cuturello, Alessandro Laio, Alessio Ansuini, and Alberto Cazzaniga. The geometry of hidden representations of large transformer models. *arXiv preprint arXiv:2302.00294*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. *arXiv preprint arXiv:1706.03762*, 2017. Published in Advances in Neural Information Processing Systems (NeurIPS).

Ryan Vogt, Maximilian Puelma Touzel, Eli Shlizerman, and Guillaume Lajoie. On Lyapunov Exponents for RNNs: Understanding Information Propagation Using Dynamical Systems Tools. *arXiv preprint arXiv:2006.14123*, 2020. Published in Frontiers in Applied Mathematics and Statistics.

Johannes von Oswald, Eyvind Niklasson, Ettore Randazzo, João Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, and Max Vladymyrov. Transformers learn in-context by gradient descent. *arXiv preprint arXiv:2212.07677*, 2022. Published in International Conference on Machine Learning (ICML).

Chaojun Wang and Rico Sennrich. On Exposure Bias, Hallucination and Domain Shift in Neural Machine Translation. *arXiv preprint arXiv:2005.03642*, 2020. Published in Annual Meeting of the Association for Computational Linguistics (ACL).

Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Dongdong Zhang, and Furu Wei. DeepNet: Scaling Transformers to 1,000 Layers. *arXiv preprint arXiv:2203.00555*, 2022. Published in IEEE Transactions on Pattern Analysis and Machine Intelligence.

Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. Unveiling Attractor Cycles in Large Language Models: A Dynamical Systems View of Successive Paraphrasing. *arXiv preprint arXiv:2502.15208*, 2025. Published in Annual Meeting of the Association for Computational Linguistics (ACL).

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and

William Fedus. Emergent Abilities of Large Language Models. *arXiv preprint arXiv:2206.07682*, 2022. Published in Transactions on Machine Learning Research.

Sean Welleck, Ilia Kulikov, Stephen Roller, Emily Dinan, Kyunghyun Cho, and Jason Weston. Neural Text Generation with Unlikelihood Training. *arXiv preprint arXiv:1908.04319*, 2019. Published in International Conference on Learning Representations (ICLR).

Kaiyue Wen, Zhiyuan Li, Jason Wang, David Hall, Percy Liang, and Tengyu Ma. Understanding Warmup-Stable-Decay Learning Rates: A River Valley Loss Landscape Perspective. *arXiv preprint arXiv:2410.05192*, 2024.

Mitchell Wortsman, Peter J. Liu, Lechao Xiao, Katie Everett, Alex Alemi, Ben Adlam, John D. Co-Reyes, Izzeddin Gur, Abhishek Kumar, Roman Novak, Jeffrey Pennington, Jascha Sohl-dickstein, Kelvin Xu, Jaehoon Lee, Justin Gilmer, and Simon Kornblith. Small-scale proxies for large-scale Transformer training instabilities. *arXiv preprint arXiv:2309.14322*, 2023. Published in International Conference on Learning Representations (ICLR).

Xinyi Wu, Amir Ajorlou, Zihui Wu, and Ali Jadbabaie. Demystifying Oversmoothing in Attention-Based Graph Neural Networks. *arXiv preprint arXiv:2305.16102*, 2023. Published in Advances in Neural Information Processing Systems (NeurIPS).

Xinyi Wu, Amir Ajorlou, Yifei Wang, Stefanie Jegelka, and Ali Jadbabaie. On the Role of Attention Masks and LayerNorm in Transformers. *arXiv preprint arXiv:2405.18781*, 2024. Published in Advances in Neural Information Processing Systems (NeurIPS).

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient Streaming Language Models with Attention Sinks. *arXiv preprint arXiv:2309.17453*, 2023. Published in International Conference on Learning Representations (ICLR).

Lechao Xiao, Yasaman Bahri, Jascha Sohl-Dickstein, Samuel S. Schoenholz, and Jeffrey Pennington. Dynamical Isometry and a Mean Field Theory of CNNs: How to Train 10,000-Layer Vanilla Convolutional Neural Networks. *arXiv preprint arXiv:1806.05393*, 2018. Published in International Conference on Machine Learning (ICML).

Zeke Xie, Issei Sato, and Masashi Sugiyama. A Diffusion Theory For Deep Learning Dynamics: Stochastic Gradient Descent Exponentially Favors Flat Minima. *arXiv preprint arXiv:2002.03495*, 2020.

Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On Layer Normalization in the Transformer Architecture. *arXiv preprint arXiv:2002.04745*, 2020. Published in International Conference on Machine Learning (ICML).

Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, Huayang Li, and Jian Li. Learning to Break the Loop: Analyzing and Mitigating Repetitions for Neural Text Generation. *arXiv preprint arXiv:2206.02369*, 2022. Published in Advances in Neural Information Processing Systems (NeurIPS).

Zhi-Qin John Xu, Yaoyu Zhang, Tao Luo, Yanyang Xiao, and Zheng Ma. Frequency Principle: Fourier Analysis Sheds Light on Deep Neural Networks. *arXiv preprint arXiv:1901.06523*, 2019. Published in Communications in Computational Physics.

An Yang, Anfeng Li, Baosong Yang, et al. Qwen3 Technical Report. arXiv preprint arXiv:2505.09388, 2025.

Greg Yang and Edward J. Hu. Feature Learning in Infinite-Width Neural Networks. *arXiv preprint arXiv:2011.14522*, 2020.

Shuangfei Zhai, Tatiana Likhomanenko, Etai Littwin, Dan Busbridge, Jason Ramapuram, Yizhe Zhang, Jiatao Gu, and Josh Susskind. Stabilizing Transformer Training by Preventing Attention Entropy Collapse. *arXiv preprint arXiv:2303.06296*, 2023. Published in International Conference on Machine Learning (ICML).

Hongtao Zhang, Wenjie Zhou, Chenxi Jia, Wei Chen, and Xueqi Cheng. The Stability of Singular Distribution: A Spectral Perspective on the Two-Phase Dynamics of Language Model Pre-training. *arXiv preprint arXiv:2605.26489*, 2026a.

Muru Zhang, Ofir Press, William Merrill, Alisa Liu, and Noah A. Smith. How Language Model Hallucinations Can Snowball. *arXiv preprint arXiv:2305.13534*, 2023a. Published in International Conference on Machine Learning (ICML).

Ruiqi Zhang, Spencer Frei, and Peter L. Bartlett. Trained Transformers Learn Linear Models In-Context. *arXiv preprint arXiv:2306.09927*, 2023b. Published in Journal of Machine Learning Research.

Shiyang Zhang, Aakash Patel, Syed A Rizvi, Nianchen Liu, Sizhuang He, Amin Karbasi, Emanuele Zappala, and David van Dijk. Intelligence at the Edge of Chaos. *arXiv preprint arXiv:2410.02536*, 2024. Published in International Conference on Learning Representations (ICLR).

Yuanhe Zhang, Xinyao Zhou, Haoran Gao, Yuyao Zhang, Zhenhong Zhou, Fanyu Meng, Li Sun, and Sen Su. Uncovering Uncontrolled Repetition through Residual Stream Dynamics. *arXiv preprint arXiv:2609.38802*, 2026b.

Xingyu Zhu, Zixuan Wang, Xiang Wang, Mo Zhou, and Rong Ge. Understanding Edge-of-Stability Training Dynamics with a Minimalist Example. *arXiv preprint arXiv:2210.03294*, 2022.

Zhanxing Zhu, Jingfeng Wu, Bing Yu, Lei Wu, and Jinwen Ma. The Anisotropic Noise in Stochastic Gradient Descent: Its Behavior of Escaping from Sharp Minima and Regularization Effects. *arXiv preprint arXiv:1803.00195*, 2018.

Andy Zou, Long Phan, Sarah Chen, James Campbell, et al. Representation engineering: A top-down approach to AI transparency. arXiv preprint arXiv:2310.01405, 2023.

A Glossary for readers from either field

The first list defines terms of machine learning for readers from nonlinear dynamics, and the second defines terms of nonlinear dynamics for readers from machine learning. Each definition is the minimum needed to follow the paper.

Terms from machine learning

Token, vocabulary. The unit of text a model reads and writes, usually a word or a piece of a word; the vocabulary $\mathcal{V}$ is the finite set of all tokens (about 50,000 for GPT-2).

Context. The sequence of tokens the model conditions on when it predicts the next one.

Embedding, hidden state, residual stream. Each token is mapped to a vector in $\mathbb{R}^d$; the vector of a token after block ℓ is its hidden state, and the sequence of these vectors through the blocks is called the residual stream.

Transformer block. One step of clock 1: an attention sublayer that mixes information across tokens, followed by a multilayer perceptron (MLP) that acts on each token separately, each added to the hidden state through a residual (skip) connection.

Attention. A weighted average of the hidden states of earlier tokens, with weights given by a softmax of inner products between learned projections (queries and keys); it is the coupling between tokens.

Layer normalization. Rescaling of each hidden state to zero mean and unit variance before a sublayer, which keeps the tokens near a sphere.

Logits, softmax. The logits z are the scores the model assigns to each token of the vocabulary; the softmax $e^{z_y} / \sum_{y'} e^{z_{y'}}$ turns them into probabilities.

Sampling temperature. The divisor T of the logits before the softmax; $T \to 0$ always picks the most likely token (greedy decoding), and large T makes all tokens nearly equally likely.

Top-k and nucleus (top-p) sampling. Truncation rules that sample only from the k most likely tokens, or from the smallest set whose probabilities sum to p.

Autoregressive generation. Producing text one token at a time, each sampled token being appended to the context for the next step.

Loss, cross-entropy. The training objective; for language models, the mean negative log-probability that the model assigns to the true next token.

Surprisal, perplexity. The surprisal of a token is $-\log p$ of that token; perplexity is the exponential of the mean surprisal.

Minibatch, SGD, learning rate. Stochastic gradient descent (SGD) estimates the gradient on a random subset of the data (a minibatch) and steps against it with step size η, the learning rate.

Adam, preconditioner. Adam rescales each coordinate of the gradient by running estimates of its first and second moments, which acts as a diagonal preconditioner.

Pretraining, checkpoint. Training a model on a large text corpus to predict the next token; a checkpoint is a saved copy of the weights at one training step.

Teacher forcing, exposure bias. During training the model always conditions on true text; during generation it conditions on its own samples, and the resulting mismatch is called exposure bias.

Sharpness. The largest eigenvalue $\lambda_{\max}$ of the Hessian of the loss.

Loss spike, grokking, induction head. A sudden, temporary rise of the training loss; a late jump in test accuracy long after the training loss has saturated; an attention circuit that copies a token that followed an earlier occurrence of the current token, which forms abruptly during training.

In-context learning. Learning a task from examples given in the prompt, without changing the weights.

Interpretability, tuned lens. The study of what the internal states of a network represent; the tuned lens decodes a next-token prediction from an intermediate layer.

Terms from nonlinear dynamics

Map, flow. A discrete-time system $u_{t+1} = F(u_t)$; a flow is its continuous-time counterpart $\dot{u} = f(u)$.

Fixed point, linear stability. A state with $F(u^*) = u^*$; it is linearly stable when every eigenvalue of the Jacobian $\mathrm{D}F(u^*)$ has modulus below one, so small perturbations decay.

Periodic orbit, limit cycle. A sequence of states that repeats with period k; an isolated attracting periodic orbit is a limit cycle.

Attractor, basin of attraction. A set that nearby trajectories approach; its basin is the set of initial states whose trajectories end on it.

Bifurcation. A qualitative change of the long-term behavior as a parameter crosses a critical value; in a period-doubling (flip) bifurcation an eigenvalue crosses -1 and a period-2 orbit appears, and in a saddle-node bifurcation two fixed points collide and vanish.

Ghost. The slow passage of trajectories through the region where a fixed point existed just before a saddle-node bifurcation.

Chaos, Lyapunov exponent. Chaos is bounded, aperiodic motion with sensitive dependence on initial conditions; the largest Lyapunov exponent is the mean exponential rate at which nearby trajectories separate, and a finite-time exponent measures it over a finite number of steps.

Fractal boundary. A boundary between basins with structure on every scale, so that arbitrarily small changes of the initial state can change the outcome.

Metastable state. A state that is not an attractor but holds trajectories for a very long time.

Duffing oscillator. A mass on a spring with a cubic term in its restoring force; with a negative linear stiffness its potential has two wells.

Kramers escape, Arrhenius law. The mean time for a noisy particle to leave a potential well of depth ΔU grows as $e^{\Delta U/D}$ with the noise intensity D; for chemical reactions this is the Arrhenius law with temperature in place of D.

Kuramoto model, order parameter. A model of oscillators that pull each other's phases together; the order parameter, the length of the mean of their phase vectors, measures how synchronized they are, from 0 to 1.

Slow-fast system, multiple scales. A system whose variables evolve on separated time scales; the method of multiple scales writes the solution as a function of a fast time and of slower times that differ by powers of a small parameter.

Reduced-order model. A low-dimensional model that keeps the term of the governing equations responsible for a behavior, such as an equivalent oscillator for a sloshing tank.

Time history, power spectrum, phase portrait, recurrence plot. Four standard pictures of a measured signal: the signal against time, its power at each frequency, the trajectory in the plane of two state variables, and the matrix of pairs of times at which the state returns close to itself.

Koopman operator. The linear operator that advances functions of the state (observables) along a nonlinear map.

B Notation

Each symbol has one meaning throughout the paper. The indices i and j count tokens (and i the steps within a loop); their range is given where they appear.

Symbol	Meaning
Language model and clock 1 (layer time)	
L	number of residual blocks (depth)
ℓ	block index; the layer time of clock 1
d	model width
$\mathcal{V}$, $\lvert\mathcal{V}\rvert$	vocabulary; its size
x_i^ℓ, x^ℓ	hidden state (residual stream) of token i after block ℓ; all tokens together
f_θ^ℓ	update added by block ℓ (attention and MLP)
F, F_ℓ	a generic map; the map of block ℓ
$\mathrm{D}F$	Jacobian of F
z, z_y	logits; logit of token y
W_U; W_Q, W_K, W_V	unembedding matrix; query, key and value matrices
β	inverse temperature of attention
Z_i	normalizer of the attention weights of token i
Π_x	projection onto the tangent space of the unit sphere at x
ϕ	activation function of a fully connected network (Section 3.6)
σ_w^2, σ_b^2	variances of the weights and biases at initialization
Q^*, G	fixed point of the length map; a standard Gaussian random variable
χ_1	factor by which one layer multiplies the distance between nearby inputs
Clock 2 (training time)	
t	optimizer step; the training time of clock 2
ε; τ_0, τ_1, τ_2	small parameter of a multiple-scales expansion ($\varepsilon = \eta$ in Section 2); fast and slow times $\tau_k = \varepsilon^k t$
θ, θ_t, θ^*	all trainable parameters; at step t; at a minimum
$\mathcal{L}$, $\mathcal{L}(\theta; \mathcal{B}_t)$	training loss; loss on the minibatch $\mathcal{B}_t$
$\mathcal{B}_t$	minibatch of training sequences used at step t; the sequence $\mathcal{B}_0, \mathcal{B}_1, \ldots$ is the data stream
B	batch size
η, η_t	learning rate (step size); at step t
P_t	preconditioner (identity for gradient descent)
g_t	gradient at step t
H	Hessian of the loss
λ, $\lambda_{\max}$	eigenvalue of the Hessian; the largest one (sharpness)
λ_t	sharpness at step t in the self-stabilization map equation 16
v	top eigenvector of the Hessian
$\mathcal{L}_v$	loss restricted to the line through θ^* along v
α	rate of progressive sharpening in equation 16
C, Σ	covariance of per-example gradients; stationary covariance of SGD near a minimum
$\mathcal{W}_t$	Wiener process
w_1, w_2	the two weights of the catapult model (Figure 11)
Clock 3 (generation time)	
n	number of tokens in the context; the generation time of clock 3
y_n, s_n	token at position n, an element of $\mathcal{V}$; context $s_n = (y_1, \ldots, y_n) \in \mathcal{V}^n$, whose length grows by one per step of clock 3
y	a token of the vocabulary
T, T_0	sampling temperature; base temperature of a controller
$\pi_T(\cdot \mid s_n)$	next-token distribution at temperature T, a probability vector with $\lvert\mathcal{V}\rvert$ entries
E, E_y, k_B	energy; energy of token y, $E_y = -z_y$; Boltzmann constant, set to 1
k	period of a loop in tokens; period of an orbit
a_i	i-th token of a loop
$p_i(T)$	probability that the sampler stays on the loop at step i
$S_i(T)$	sum of the Boltzmann factors of all exits at step i
Δ_i^y, Δ_i	logit gap between the loop token a_i and token y; smallest gap at step i

Symbol	Meaning
Δ, $\Delta_{\min}$	barrier of a loop: its smallest logit gap
m, m_i	number of exits that attain the smallest gap (of the cycle; at step i)
c	cycle index of a loop
Δ_c	smallest logit gap during cycle c
Δ_1, Δ_2, Δ_∞	barrier in cycles 1 and 2; its saturation level
b, γ	growth rate of the barrier under linear and logarithmic growth
c_*	number of cycles over which the barrier saturates
$r(T)$, $r_c(T)$	escape probability per cycle; during cycle c
r_*	target escape probability of the Arrhenius-guided controller
ν	number of escape attempts per cycle in the reduced model equation 14
N	horizon: number of cycles (or forcing periods) observed
$T_{1/2}(N)$	temperature at which a loop survives N cycles with probability one half
T_{lock}	temperature below which a loop is never left with positive probability
dist_c	distance between the hidden state and its value one period earlier, in cycle c
rep, $\mathcal{I}$	fraction of repeated 4-grams in a window; mean surprisal of the window
Physical and reduced models	
τ, $\bar{\tau}$	physical time; mean time to escape from a well
q, $\dot{q}$	displacement of an oscillator or particle (or along v); velocity
M, ζ	mass; friction coefficient
k_1, k_3	linear and cubic stiffness
A, Ω	amplitude and frequency of harmonic forcing
ξ, D	Gaussian white noise; its intensity
U, ΔU	potential energy; height of the barrier
ω_{w}, ω_{b}	frequencies set by the curvature at the bottom of the well and at the top of the barrier
ψ_i, ω_i	phase and natural frequency of oscillator i
J	coupling strength of the Kuramoto model
ρ, ρ_ℓ	order parameter of synchrony; of the residual stream entering block ℓ
u, μ	state of a generic map; parameter of the logistic map
Dynamical-systems tools	
Λ_K	finite-time Lyapunov exponent over K steps equation 8
Λ, $\tilde{\Lambda}_\ell$	exponent of the whole stack; relative exponent of block ℓ (Section 5.1)
$\sigma_{\max}$, σ_ℓ	largest singular value; largest singular value of the Jacobian of block ℓ
κ	coherence of stretching between blocks (Section 5.1)
δ	size of an initial perturbation
R_{ij}, ϵ_R	recurrence matrix; recurrence threshold
$\mathcal{H}_s$, DET	normalized spectral entropy; determinism of a recurrence plot
h	an observable acted on by the Koopman operator
I	identity matrix
O, o, Θ	asymptotic orders
$\mathcal{N}(0,\cdot)$	Gaussian distribution with mean zero and the stated variance

C Proofs

Proof of Proposition 1. Since $1-\prod_i p_i \le \sum_i (1-p_i)$ for $p_i \in [0,1]$, and $1-p_i = S_i/(1+S_i) \le S_i$, the bound follows. As $T \to 0$, each $S_i(T)$ is dominated by its smallest gaps, $S_i(T) = m_i e^{-\Delta_i/T}(1+o(1))$ with $\Delta_i = \min_y \Delta_i^y$, and $1-\prod_i p_i = \sum_i S_i + O(\max_i S_i^2)$. The terms with $\Delta_i = \Delta_{\min}$ dominate the sum. □

Proof of Proposition 2. By the bound in Proposition 1, $r_c \le k(|\mathcal{V}|-1)e^{-\Delta_c/T}$, which is summable under linear growth, and $\prod_c (1-r_c) \ge 1 - \sum_c r_c$. If the logits converge, then r_c tends to a positive limit and the second statement follows from the first part of Proposition 1. □

Proof of Proposition 3. At the step of cycle c that attains Δ_c, $1-p_i = S_i/(1+S_i) \ge \frac{1}{2}\min(1, e^{-\Delta_c/T})$, so $r_c \ge \frac{1}{2}\min(1, e^{-\Delta_c/T})$, and Proposition 1 gives $r_c \le k(|\mathcal{V}|-1)e^{-\Delta_c/T}$. For large c both bounds are constant multiples of $c^{-\gamma/T}$, so $\sum_c r_c$ converges exactly when $\gamma/T > 1$, and $\prod_c (1-r_c) > 0$ exactly when the sum

converges. For $T > \gamma$, $r_c \to 0$, so $-\log \prod_{c \le N}(1-r_c)$ is within constant factors of $\sum_{c \le N} c^{-\gamma/T} = \Theta(N^{1-\gamma/T})$. □

D Experimental details

All experiments run on a CPU with two cores and 7 GB of memory, using PyTorch and Hugging Face Transformers with the public checkpoints of GPT-2 small (Radford et al., 2019) and Pythia (Biderman et al., 2023). The repository contains one script per experiment, the exact command lines, the raw JSON outputs and the plotting scripts. The 40 prompts used in Experiments 1 and 3 are short neutral sentence openings written for this study (for example, "The weather today is cold and cloudy, with a chance of"), and the full list is in the repository.

D.1 Experiment 1: finite-time Lyapunov exponents

For each prompt we take the first n tokens (prompts shorter than n are repeated) and run the model once to record the residual stream $x^0, \dots, x^L$ before the final layer normalization. We then call each block of the Hugging Face implementation directly as a function $F_\ell : \mathbb{R}^{n \times d} \to \mathbb{R}^{n \times d}$, with rotary embeddings and an explicit causal mask, and check that chaining these calls reproduces the model's hidden states (the maximum relative deviation was zero in float32 for every checkpoint). The largest singular value of the Jacobian $\mathrm{D}F_\ell(x^\ell)$ is estimated by power iteration on $\mathrm{D}F_\ell^\top \mathrm{D}F_\ell$, using forward-mode (JVP) and reverse-mode (VJP) products from `torch.func`. The same procedure applied to the composition $F_{L-1} \circ \cdots \circ F_0$ gives the depth-averaged exponent $\frac{1}{L}\log \sigma_{\max}(\mathrm{D}F_{0 \to L})$. We use eight prompts, contexts of 16 tokens and 10 power iterations per estimate, for eleven Pythia-70M checkpoints (steps 0, 128, 512, 1k, 2k, 4k, 8k, 16k, 32k, 64k and 143k) and six Pythia-160M checkpoints (steps 0, 512, 2k, 8k, 32k and 143k).

D.2 Experiment 2: bifurcation diagram

Detail: Why the sharpness sits above $2/\eta$. In 26 trainable runs (three seeds, eight or nine step sizes each) we continued gradient descent for 200 to 400 steps and computed, at consecutive iterates, the top six Hessian eigenpairs by Lanczos iteration, the curvature along the step, a finite-time Lyapunov spectrum of the gradient-descent map and the leading multipliers of the two-step Jacobian by Arnoldi iteration (Appendix D.2). The sharpness at the iterates sits 5% to 14% above the threshold ($\eta\lambda_{\max}/2$ between 1.05 and 1.14, median 1.11), and two to six eigenvalues exceed $2/\eta$ at every iterate (median four). The curvature of the Hessian averaged over the segment between consecutive iterates, taken along the step, equals $2/\eta$ to within 0.2% in every run, as the telescoping identity of Litman (2026) requires on average. The stiff directions do not stay in place. The top eigenvectors at consecutive iterates are 74 to 88 degrees apart (median 86), so the top of the spectrum is a cluster whose members exchange order at every step, and the step is only partly aligned with the top eigenvector ($|\cos|$ between 0.08 and 0.42). Locally the two-step map expands, with a largest multiplier between 1.0002 and 1.126 that is often a slow complex pair near $+1$, yet along the trajectory the largest finite-time Lyapunov multiplier per two steps lies between 0.9998 and 1.019 while the loss stays bounded. In this model the edge is therefore a multi-mode regime with neutral or weakly chaotic dynamics, not a single stiff mode on a periodic orbit. A warm-started power iteration, which we used in a first pass, gives ratios of only 1.02 to 1.07, because it averages over the alternating cluster, so reported values of how far the sharpness sits above $2/\eta$ depend on the eigensolver whenever the top of the spectrum is degenerate. Why the oscillation stays bounded with several modes above the threshold is open (**OP 10**).

Data. A first-order Markov chain over $|\mathcal{V}| = 12$ symbols, with transition rows drawn from a Dirichlet distribution with concentration 0.3 (seed 0), generates 64 sequences of length 13, and the model predicts the next symbol at each of the 12 positions. *Model.* One pre-LN transformer block with causal attention (two heads, width 32, MLP width 64), learned positional embeddings and a linear readout, in float64. Attention is written out explicitly so that Hessian-vector products work on a CPU. *Training.* Full-batch gradient descent on the mean cross-entropy, from the same initialization for every step size within a seed. Seeds 0, 1 and 2 change both the Markov chain and the initialization. Each run lasts $\max(3000, 600/\eta)$ steps, capped at 30,000, so that all runs cover a comparable amount of gradient-flow time. A run is marked as diverged if the

loss exceeds 50 or becomes non-finite. *Measurements.* The last 400 losses of each run, and the top Hessian eigenvalue at the final iterate from 40 steps of power iteration on Hessian-vector products. The step sizes are 56 values spaced geometrically between 0.02 and 4.

Edge-of-stability geometry. For each seed and for nine step sizes spaced geometrically between 0.05 and 1.6, we train as above and continue gradient descent. The curvature along the step is measured over 200 further steps as $\langle g_t - g_{t+1}, g_t\rangle/(2\|g_t\|^2)$, which equals $\eta/2$ times the curvature of the segment-averaged Hessian along the step. Over a separate window of 400 steps we compute, every 50 steps at a pair of consecutive iterates, the top six Hessian eigenpairs by Lanczos iteration (`scipy eigsh` on Hessian-vector products, residuals near 10^{-13}), their mutual angles and the alignment of the step with the top eigenvector; a finite-time Lyapunov spectrum of six vectors with QR reorthogonalization at every step; and, at step 200, the eight largest multipliers of $(I - \eta H_{t+1})(I - \eta H_t)$ by Arnoldi iteration. Power iteration is not suitable for this two-step map, since thousands of flat directions have multipliers near one. The largest step size (1.6) diverged in seed 0, leaving 26 runs. Per-run values are in `results/eos_rotation/`.

D.3 Experiment 3: escape from repetition loops

Full results behind Section 5.3. The following paragraphs give the complete numbers summarized in the main text.

Detail: The barriers saturate, so locking depends on the horizon. For each loop with at least five cycles after the first, we fit Δ_c for $c \geq 2$ with the logarithmic law of Proposition 3, a linear law and a saturating law $\Delta_c = \Delta_\infty - (\Delta_\infty - \Delta_2)\, e^{-(c-2)/c_*}$, and compare the fits by the corrected Akaike criterion (Figure 16). The saturating law fits best for 21 of 36 GPT-2 loops and 26 of 37 Pythia-160M loops, the logarithmic law for 8 and 6, and the linear law for 7 and 5. The fitted saturation levels have medians 4.2 and 3.6, reached within a time constant of 3 to 4 cycles. The hidden state converges onto a periodic orbit at the same pace (Appendix D.3), and since the logits are a linear function of the final hidden state, this convergence is what makes the barriers saturate. On these models the second case of Proposition 2 applies, and no loop locks in an unbounded generation. The logarithmic law would not explain the measured locking temperatures either. Its rate γ has median 1.48 (GPT-2) and 1.15 (Pythia-160M), more than twice the measured $T_{1/2}$, and exceeds the loop's own $T_{1/2}$ in 35 of 35 and 32 of 35 loops (panel a). The locking temperature of Experiment 3 is therefore a finite-horizon quantity. We recompute it from the exact per-cycle escape probabilities for horizons of 1 to 1024 cycles after burn-in, repeating the last stored cycle beyond the end of the run, which matches the saturation. The median $T_{1/2}$ falls from 0.67 to 0.40 in GPT-2 and from 0.62 to 0.37 in Pythia-160M, and equation 13 fits this decline with a largest deviation of 0.009 and 0.019 (panel c). The fitted constants are effective values (Δ_∞ of 7.0 and 9.0, $\ln m$ of 10.6 and 15.7). They exceed the measured gaps because at $T \approx 0.5$ many competing tokens contribute to escape besides the nearest one, and the single-gap Arrhenius rate with $m = 1$ overestimates $T_{1/2}$ by a median of 37% (GPT-2) and 39% (Pythia-160M), although it ranks the loops correctly (correlation 0.89 and 0.92).

Detail: Coherent text lives in a narrow window between two failures. Raising T ends the loops but eventually produces incoherent text, so we map both failures with the same free generations (Figure 15). A generation counts as a loop by the test above. Otherwise it counts as repetitive if its fraction of repeated 4-grams exceeds the 95th percentile of human continuations of 200 OpenWebText passages (0.091), and as incoherent if its mean negative log-likelihood under a separate reference model (Qwen2.5-0.5B) exceeds the 90th percentile of the human scores (3.49 nats per token, against a human median of 2.92). The remaining generations count as coherent. Exact loops fall below one half at $T \approx 0.43$ (GPT-2) and $T \approx 0.45$ (Pythia-160M), but loops and repetitive text together fall below one half only at $T \approx 0.67$ and $T \approx 0.72$, and incoherent text rises above one half at $T \approx 0.88$ and $T \approx 0.86$. The coherent fraction peaks at $T = 0.8$, at 82% for GPT-2 and 71% for Pythia-160M, and is below 25% at $T = 0.6$ and at $T = 0.9$. Moving the incoherence threshold between the human median and the 95th percentile moves the peak between $T = 0.7$ and $T = 0.8$ and does not widen the window. For each prompt we also take the temperatures at which fewer than half of its samples loop and its median repetition and reference scores are within the human thresholds. This range covers at most four of our twelve temperatures, typically two, lies between 0.6 and 0.9, and is empty for 2 of 40 prompts in GPT-2 and 5 of 40 in Pythia-160M, so for these prompts no single temperature

avoids both failures. At the upper edge the entropy of the sampling distribution is 2.3 to 3.1 nats, and at $T = 2$ it reaches 10.3 nats, 95% of $\log|\mathcal{V}|$, so sampling there is spread over almost the whole vocabulary. These proxies measure surprise and repetition, not meaning, and a reference model of 0.5B parameters is a coarse judge of coherence. Figure 8 shows the three regimes of one prompt in the representations of Section 3.5: the loop is a periodic orbit with sharp spectral lines and long diagonal recurrence lines, and the determinism of the hidden-state trajectory falls from the loop to coherent and then to incoherent text.

Detail: Larger models dig deeper wells. We repeated the loop analysis on Pythia-70M, 410M and 1B, which share training data, tokenizer and data order with Pythia-160M (Biderman et al., 2023), so model size is the only variable (Figure 17). Greedy decoding loops for 33 to 37 of the 40 prompts at every size, the barrier grows from cycle 2 to cycle 6 in 34 of 36, 35 of 37, 33 of 33 and 33 of 33 loops from the smallest to the largest model, and the saturating law fits best for 22 to 26 loops per model. The saturated barrier rises with size, with medians 3.0, 3.6, 4.5 and 4.8 from 70M to 1B, and so does the temperature needed to leave a loop. The median $T_{1/2}$ for a horizon of 16 cycles is 0.43, 0.48, 0.59 and 0.64, and for 256 cycles it is 0.36, 0.42, 0.52 and 0.53. Over a 14-fold increase in parameters the locking temperature therefore rises by about half. Free generations of the 410M and 1B models (eight temperatures from 0.3 to 1.2, three samples per prompt) show a wider coherent window than at 160M. The coherent fraction exceeds one half only at $T = 0.8$ for Pythia-160M, and from $T = 0.7$ to $T = 0.9$ for 410M and 1B, with peaks of 81% and 82%. Within this family, larger models produce coherent text over a wider band of temperatures, while the exact loops they do enter are harder to leave.

Detail: Reasoning models and the recommended temperature. Vendors of reasoning models prescribe a narrow sampling range. The DeepSeek-R1 model card asks for a temperature between 0.5 and 0.7 "to prevent endless repetitions or incoherent outputs" (DeepSeek-AI, 2025; DeepSeek-AI et al., 2025), and the Qwen3 card recommends $T = 0.6$ with thinking, $T = 0.7$ without, and no greedy decoding (Qwen Team, 2025; Yang et al., 2025). We asked whether these values mark the same window between two failures. Qwen3-0.6B solved the first 18 problems of the GSM8K test set (Cobbe et al., 2021) in thinking mode with plain temperature sampling and a budget of 1024 tokens, one sample per problem and temperature (Figure 18). The fraction of traces whose last number equals the answer is 0.28 at $T = 0.2$, 0.56 at 0.4, 0.61 at 0.6, and 0.22 at both 1.0 and 1.3, and the fraction that also ends inside the budget follows the same shape (0.22, 0.28, 0.33, 0.06, 0.11). The best temperature on our grid is therefore the vendor's value, although with 18 problems per temperature the difference between $T = 0.6$ and $T = 0.2$ is at the edge of significance (Fisher exact test, p-value 0.09; 0.04 against $T = 1.0$). The two failures are not those of the loops above. No trace ends in an exact loop. At $T = 0.2$ the trace keeps returning to the same step with small rephrasings ("Wait, let me check again"), its median fraction of repeated 4-grams is 0.34 against 0.24 at $T = 0.6$, and 13 of 18 traces run out of budget. At $T = 1.3$ stray tokens from other scripts and broken phrases appear. In the language of Section 3.5, the low-temperature failure of this reasoning model is a recurrent orbit, not a periodic one, so recurrence quantification (Eckmann et al., 1987; Marwan et al., 2007) and not the exact escape probabilities is the natural tool (**OP 5**). Without thinking, the same model solves 0.42 to 0.48 of the first 20 problems at every temperature from 0.2 to 1.0, falling to 0.18 at 1.3, and never loops, so the repetitive edge appears only in the long thinking traces, consistent with the horizon dependence of the loops above. A single 0.6B model with a short budget tests the shape of the window and cannot confirm the vendors' exact numbers.

Loops converge onto periodic orbits in hidden-state space. Along each greedy loop we also record the final-layer hidden state x_n^L of the last token and compute the return distance dist_c, the mean of $\|x_n^L - x_{n-k}^L\|$ over the k steps of cycle c, divided by the mean norm of the states. If the loop is an attracting periodic orbit of the closed-loop map, $\mathrm{dist}_c \to 0$. The return distance falls between cycle 2 and cycle 6 in 33 of 36 GPT-2 loops and 35 of 35 Pythia-160M loops. Its median falls from 0.51 in the first cycle to 0.013 in the twelfth in GPT-2, and from 0.060 to 0.008 in Pythia-160M (Figure 19). It does not reach zero. Over the later cycles it stays near 0.01 in both models, because the context keeps growing, so the closed-loop map is not autonomous and the orbit keeps drifting slowly.

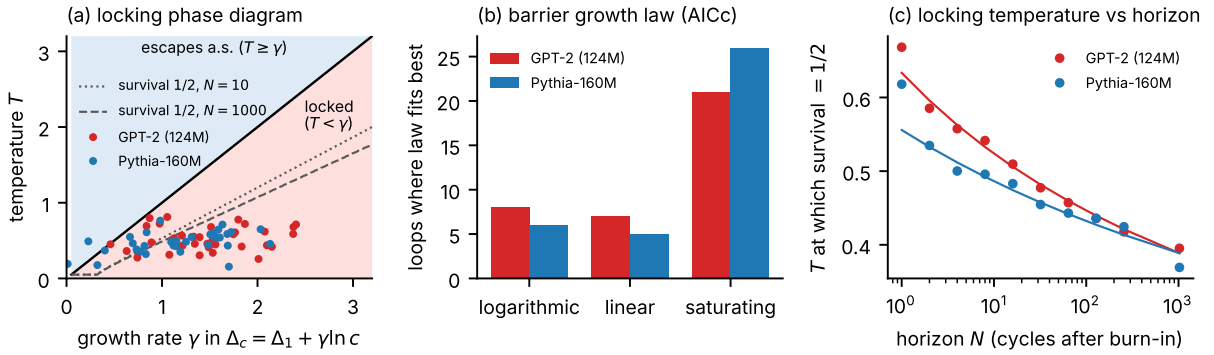


Figure 16: Experiment 3, locking. (a) Phase diagram for the logarithmic barrier $\Delta_c = \Delta_1 + \gamma \ln c$ (Proposition 3): loops lock with positive probability below $T = \gamma$ and escape almost surely above it. Dotted and dashed lines: temperature at which survival over $N = 10$ and $N = 1000$ cycles equals one half, for the single-gap rate with $m = 1$ and the median fitted Δ_1. Points: fitted γ and exact finite-horizon $T_{1/2}$ of each loop. (b) Barrier growth law preferred by the corrected Akaike criterion. (c) Median exact $T_{1/2}$ against the horizon N (points) and the fit of equation 13 (lines).

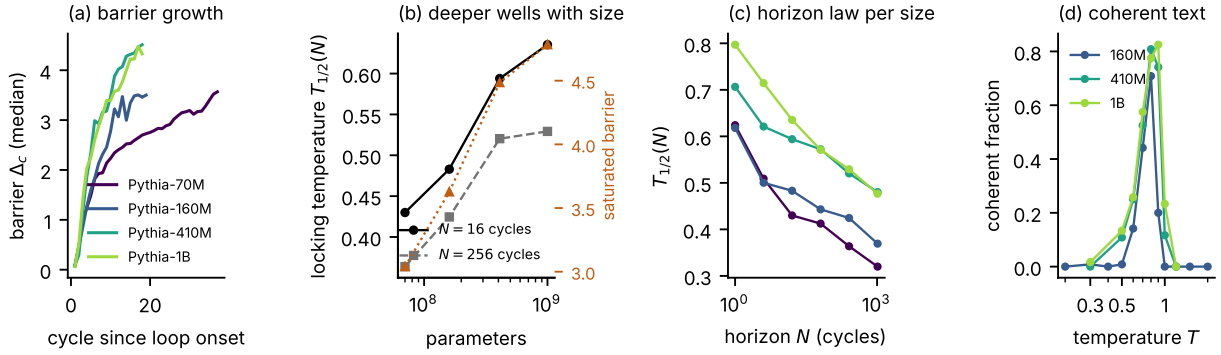


Figure 17: Experiment 3 across model size, Pythia-70M to Pythia-1B (same data and training order). (a) Median barrier Δ_c against the cycle since loop onset (cycles reached by at least 10 loops). (b) Median locking temperature $T_{1/2}(N)$ for horizons of 16 and 256 cycles (left axis) and median saturated barrier Δ_∞ from the saturating fits (right axis) against the number of parameters. (c) $T_{1/2}(N)$ against the horizon for each model. (d) Fraction of coherent free generations against temperature for the three larger models.

Loop detection. From each prompt we decode greedily for 320 tokens and store the full logit vector at each step. A loop of period $k \leq 64$ is detected when the last $3k$ tokens are k-periodic, and its onset is the earliest index from which the sequence stays k-periodic. *Stay probabilities.* Along the loop the context is deterministic, so the probability equation 10 of producing the loop token at temperature T follows from the stored logits, for 30 temperatures spaced geometrically between 0.15 and 2. From these we compute, for each cycle c, the barrier Δ_c (smallest gap between the loop token's logit and any other logit during the cycle) and the escape probability $r_c(T) = 1 - \prod_{t \in c} p_t(T)$. *Arrhenius fit.* For the first cycle after two completed cycles of burn-in, we fit $\log r$ against $1/T$ by least squares over the temperatures where $10^{-9} < r < 10^{-2}$, and compare the negative slope with $\Delta_{\min}$ of that cycle. *Monte Carlo check.* For the first four loops we sample 128 continuations at $T \in \{0.6, 1.0\}$ (nearest grid temperature) from the context after burn-in, record the first step whose token differs from the loop, and compare the fraction still in the loop after 1, 2 and 4 cycles with the exact survival probability. *Free sampling and coherence.* For each of 12 temperatures between 0.2 and 2 we sample three continuations of 200 tokens from every prompt with plain temperature sampling (no top-k or top-p), record whether each ends in a loop with at least three repetitions, the fraction

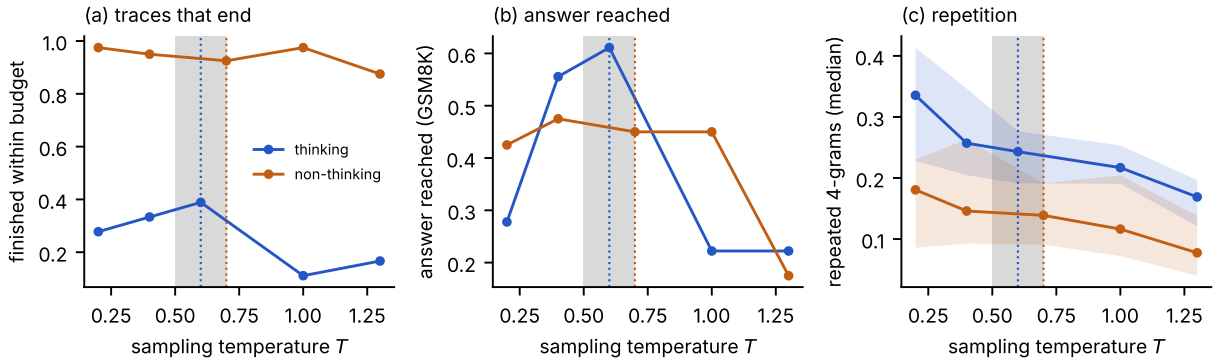


Figure 18: Experiment 3, reasoning model. Qwen3-0.6B on GSM8K test problems with plain temperature sampling, with thinking (18 problems, 1024-token budget) and without (20 problems, two samples each, 640 tokens). (a) Fraction of traces that end inside the budget. (b) Fraction whose last number equals the answer. (c) Median fraction of repeated 4-grams (band: interquartile range). Gray band: the range recommended for DeepSeek-R1; dotted lines: the Qwen3 recommendations for thinking (0.6) and non-thinking (0.7) mode. No trace ends in an exact loop.

of repeated 4-grams, and the mean entropy of the sampling distribution along the path. Each continuation is then scored by Qwen2.5-0.5B (Qwen Team, 2024) conditioned on its prompt, as the mean negative log-likelihood per reference token. The human baseline applies the same score to 200 passages of OpenWebText (the `stas/openwebtext-10k` sample), each split into its first 16 GPT-2 tokens as prompt and the next 200 as continuation. *Growth laws.* For each loop with at least five cycles after the first, we fit the barriers Δ_c, $c \geq 2$, with $\Delta_1 + \gamma \ln c$ and $\Delta_1 + b(c-1)$ by least squares and with $\Delta_\infty - (\Delta_\infty - \Delta_2)e^{-(c-2)/c_*}$ by nonlinear least squares, and pick the law with the smallest corrected Akaike criterion. The finite-horizon locking temperature $T_{1/2}(N)$ of a loop is the temperature, interpolated in $\log T$ on the 30-point grid, at which the exact probability of surviving N cycles after burn-in equals one half; beyond the stored cycles the escape probability of the last stored cycle is repeated. We fit equation 13 to the medians for $N \geq 4$. *Escape control.* The controller of **OP 12** samples at a base temperature T_0. When the generated tokens end in at least $\max(2, \lceil 8/k \rceil)$ repetitions of a block of $k \leq 64$ tokens, it computes the escape probability of the last cycle on a grid of temperatures from the 64 largest logits of each step, raises the temperature for the next k steps to the smallest value with escape probability r_* (capped at 1.5), and then returns to T_0. On the 40 prompts with three samples each, $T_0 = 0.3$ and $r_* = 0.9$ give 6% loops and 94% repetitive outputs (repeated 4-grams above the human 95th percentile), against 68% and 32% at constant $T = 0.3$. A repetition penalty of 1.2 (Keskar et al., 2019) removes loops and repetition but makes 73% of the outputs incoherent by our threshold. *Model size.* For Pythia-70M, 410M and 1B we run the greedy loop analysis without the Monte Carlo check, Pythia-1B in two halves of 20 prompts to fit in memory, and the free generations of 410M and 1B at eight temperatures in batches of 10 prompts, scoring them with the reference model in a separate pass. *Return distance.* We repeat the greedy runs while storing the final-layer hidden state (after the last layer norm) at each step. Figure 19 shows the median return distance and a projection of one GPT-2 loop on its first two principal components.

Reasoning model. We use Qwen3-0.6B in float32 with its own chat template, switching thinking on or off through the template flag, and append to each GSM8K question the instruction to reason step by step and put the final answer in `\boxed{}`. Sampling is plain temperature sampling with top-k and top-p disabled, in batches of 6 (thinking) or 10 (non-thinking) left-padded prompts, with a fixed seed per temperature, sample and batch. The answer is the content of the last `\boxed{}` if present and otherwise the last number in the trace, compared numerically with the reference answer. A trace counts as finished when it emits the end-of-turn token inside the budget. Loops are detected as in Experiment 3 with periods up to 256 tokens, and a loop must repeat over at least 30 tokens, which rules out runs such as the zeros of a number cut by the budget. The thinking runs take about 28 minutes per temperature on two CPU cores.

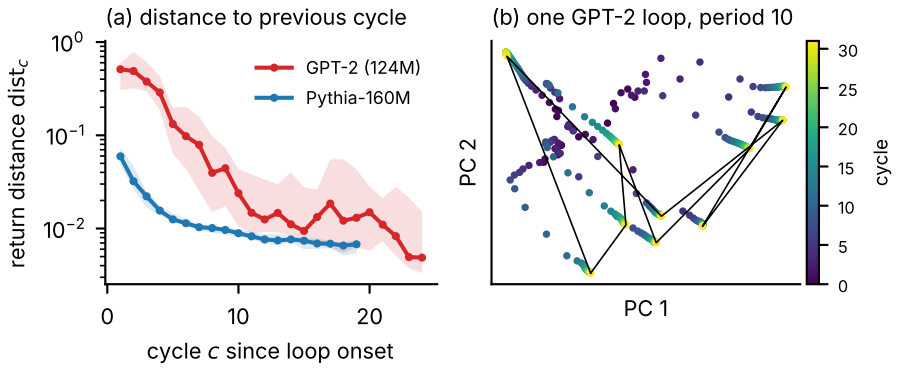


Figure 19: Experiment 3, loops as orbits. (a) Return distance dist_c (mean distance between the final hidden state and its value one period earlier, relative to the mean state norm) against the cycle; line: median over loops, band: interquartile range. (b) First two principal components of the final hidden states (each dimension standardized) along one GPT-2 loop of period 10 ("He was a man of great wealth and power."), colored by cycle. Each of the ten states of the cycle converges along its own track; black: the last cycle.

Paired figures and regime atlas. Figures 3, 6, 5 and 11 are produced by `paper/fig_pairs.py`, and Figure 8 by `exp3_escape/atlas.py`. The Duffing oscillator is integrated with the Euler-Maruyama scheme (step 0.025) after a transient of 100 time units and sampled once per time unit. For the recurrence plots the state is standardized and ϵ_R is chosen so that 5% of all pairs recur; DET counts diagonal lines of length at least 2 and excludes the main diagonal. The spectral entropy is the Shannon entropy of the normalized Welch spectrum (segments of 256 samples, zero frequency excluded) divided by its maximum. The statistics in Section 3 (`exp3_escape/atlas_stats.py`) repeat these measurements for noise seeds 0 to 9 and for the first 10 prompts of Experiment 3; at $T = 0.3$, 9 of the 10 GPT-2 runs end in an exact loop.

Controllers for clock 3. Rollback and kick (`exp3_escape/rollback.py`) samples at a base temperature T_0; when the generated tokens end in a loop it deletes the periodic tail back to its first token, forbids that token at that position and resamples there at T_0. The feedback controller (`exp3_escape/regime_control.py`) updates the temperature after every token from the last 32 tokens, $T \leftarrow \mathrm{clip}\big(T{+}0.05(\mathrm{rep}{-}0.02){-}0.01\max(0, \mathcal{I}{-}3),\ 0.4,\ 1.1\big)$, where rep is the fraction of 4-grams in the window that already occurred earlier in the generation and $\mathcal{I}$ is the mean surprisal of the window at $T = 1$ in nats. The gains were set by hand on three of the 40 prompts; without these three prompts the coherent fractions are 21%, 64% and 82%. Both use the 40 prompts, three samples and 200 tokens of Experiment 3b, and are scored with the same reference model and thresholds, so the constant-temperature rows of Experiment 3b are the baseline.

E Further related work

This appendix collects review material on each clock that supports Sections 3 and 4 but is not needed to follow the main argument.

E.1 Clock 1: Attention models closer to real decoders

Later work moved the model closer to real decoders. With causal masking the particle system is no longer a mean-field gradient flow, but clustering results still hold, and the first token, which attends only to itself, plays a special role (Karagodin et al., 2024). Wu et al. (2024) showed that the attention mask and layer normalization decide whether collapse happens at all. A mean-field limit with a Vlasov-type "Transformer PDE" covers several attention variants (Castin et al., 2025). Feed-forward layers can be read as a control that steers the particles (Maranzatto et al., 2026). Finite-width transformers with MLP blocks converge to a stochastic interacting particle system, in which noise can synchronize the tokens (Agazzi et al., 2026). For symmetric weights a spectral mode-selection mechanism decides the long-time state (Kuehn & Yoon, 2026),

and in two dimensions deep linear transformers reduce to Kuramoto-type models with a rich set of behaviors (Li et al., 2026). The kinetic theory of causal attention has also been used to explain why models use the middle of long contexts poorly (Duerinckx et al., 2026), the lost-in-the-middle effect of Liu et al. (2023).

Further measurements of the depth trajectory include the rise and fall of the intrinsic dimension of hidden representations across layers (Valeriani et al., 2023), smooth changes of individual units from layer to layer seen with tools from neural population dynamics (Fernando & Guitchounts, 2025), full Jacobian eigendecompositions of each layer in production-scale models (Fernando & Guitchounts, 2026), a theoretical analysis of methods that read features off the layer Jacobian (Ostrow et al., 2026), and looped models whose repeated block makes the iteration explicit and whose stability can be studied and scaled as such (Prairie et al., 2026).

E.2 Clock 1: Lessons from recurrent networks

Recurrent networks are the older case in which clocks 1 and 3 coincide: the same map is applied at every time step. Their analysis set many of the patterns used later for transformers. Fixed points and slow points of trained networks explain how they compute (Sussillo & Barak, 2013), and sentiment classifiers turn out to integrate evidence along a line attractor (Maheswaranathan et al., 2019). On the training side, exploding gradients were linked to bifurcations of the network dynamics (Pascanu et al., 2012; Ribeiro et al., 2019), and networks trained to reproduce chaotic dynamics must have diverging gradients (Mikhaeil et al., 2021). Lyapunov spectra quantify how much information a recurrent network keeps (Engelken et al., 2020; Vogt et al., 2020).

E.3 Clock 1: Explicit dynamical models of sequences

State-space models write the sequence dynamics in clock 3 explicitly, as a linear dynamical system in a hidden state, with eigenvalues placed to control memory and stability (Gu et al., 2021; Orvieto et al., 2023), or with input-dependent transitions (Gu & Dao, 2023). Linear attention has the same structure: it is a recurrent update of a matrix-valued state (Katharopoulos et al., 2020; Schlag et al., 2021). Recurrent networks built from coupled oscillators or constrained by Lipschitz bounds use the same design logic (Rusch & Mishra, 2020; Erichson et al., 2020). In these models the attractor and stability questions of this paper are explicit, since the eigenvalues of the state transition are parameters of the architecture.

E.4 Clock 2: Theory of the edge of stability

Marion (2026) proposes a three-variable flow for the center, direction and amplitude of the oscillation at the edge of stability. Song & Yun (2023) prove that in simple models trajectories align on a bifurcation diagram, Gan (2026) treats the onset of oscillation as a flip bifurcation of the normal dynamics near a manifold of minimizers, and Mulayoff & Stich (2026) give an exact criterion, in terms of third and fourth derivatives, for when oscillations near a minimum are stable. Why the sharpness rises in the first place, progressive sharpening, is proved only in reduced models (Agarwala et al., 2022), and the route from the edge to chaos at larger steps has been mapped in a two-layer linear model (Kalra et al., 2023).

E.5 Clock 2: Moving and controlling the stability boundary

Mini-batch sampling moves the stability boundary of gradient descent even when gradients are unbiased, as shown exactly for linear self-attention (Balasubramanian, 2026), and curvature control has been used to keep large transformers away from the boundary (Ramasinghe et al., 2026; Gilmer et al., 2021).

E.6 Clock 2: Plateaus, grokking and emergence

Noise or small perturbations let gradient methods leave strict saddles efficiently (Ge et al., 2015; Jin et al., 2017), but the time spent near a saddle can still dominate training. Spectral bias and the frequency principle describe an ordering similar to the plateaus of deep linear networks, with low-frequency components learned first (Rahaman et al., 2018; Xu et al., 2019), and learning can itself drive a recurrent network from chaotic to stable activity (Vaidya, 2026). Explanations of grokking include an effective theory of representation learning

with a phase diagram (Liu et al., 2022a), a dependence on weight norm (Liu et al., 2022b), a transition from lazy to rich training (Kumar et al., 2023) and two phases of implicit bias (Lyu et al., 2023); with adaptive optimizers, grokking coincides with cyclic phase changes and loss spikes (Thilak et al., 2022). In language models, syntactic attention structure appears abruptly and is followed by a drop in loss (Chen et al., 2023). In-context learning can emerge and later fade while training continues (Singh et al., 2023); minimal models explain the abruptness (Reddy, 2023), and Bayesian models of attention predict phase transitions in the emergence of copy heads (Lavie et al., 2026). Singular learning theory relates these stages to changes in the degeneracy of the loss landscape (Hoogland et al., 2024), and one model attributes emergence to discrete skills that are learned in order of frequency (Michaud et al., 2023).

E.7 Clock 2: Slow phases, landscape geometry and conserved quantities

Pretraining curves typically show a fast initial drop followed by a long slow phase. Zhang et al. (2026a) observe that the normalized singular value spectrum of the weight matrices stabilizes early and that this stabilization coincides with the slow phase. The river-valley picture explains why a constant learning rate followed by a fast decay works: the iterate moves along a valley floor while oscillating across it (Wen et al., 2024). The Hessian has a bulk of near-zero eigenvalues and a few outliers (Sagun et al., 2017; Ghorbani et al., 2019), and the gradient lies mostly in the subspace of the outliers (Gur-Ari et al., 2018). Two copies of a run that are split at step k and then see different minibatch noise end in the same linearly connected region only when k is past a short early phase; before that, the trajectory is sensitive to small perturbations (Frankle et al., 2019). Continuous symmetries of the architecture imply conservation laws for gradient flow, which finite steps and noise break in predictable ways (Kunin et al., 2020). At the end of classification training, features converge to a highly symmetric configuration (Papyan et al., 2020).

Infinite-width limits give two simpler dynamical systems. In the lazy limit, training is linear in function space (Jacot et al., 2018). In the mean-field limit, training two-layer networks is a Wasserstein gradient flow of particles (Mei et al., 2018; Chizat & Bach, 2018), and parametrizations exist that keep feature learning at infinite width (Yang & Hu, 2020). Scaling laws describe the slow phase at the level of loss averaged over the whole run (Kaplan et al., 2020; Hoffmann et al., 2022). Koopman operator methods have been used to speed up training and to explain magnitude pruning (Dogra & Redman, 2020; Redman et al., 2021).

E.8 Clock 3: Outer loops, when generation feeds training

The slow-fast separation of Section 2 breaks when generated text is used for training. Training repeatedly on a model's own outputs makes the tails of the distribution disappear (Shumailov et al., 2023; Alemohammad et al., 2023); whether the process stays near the original model depends on the fraction of real data in each round (Bertrand et al., 2023). Reinforcement learning from human feedback reduces output diversity (Kirk et al., 2023), which can be read as a contraction of the set of trajectories that clock 3 visits. These are dynamical systems over model versions, a fourth and much slower clock.

E.9 Clock 3: Language models as models of dynamical systems

A different line of work asks whether language models can forecast dynamical systems placed in their context. Pretrained models forecast time series zero-shot (Gruver et al., 2023) and extrapolate trajectories of physical systems with accuracy that improves with context length (Liu et al., 2024). Chaotic systems serve as a benchmark for such forecasters (Gilpin, 2021; 2023; Pathak et al., 2018), and training on data from cellular automata at the edge of chaos produces models with better downstream reasoning (Zhang et al., 2024). These results are about the content a model processes, not about the model as a dynamical system, but they share tools with this paper.